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Path of Corporate University to Enable Enterprises:
Based on the Best Corporate Universities in China

Author: Yi Zhou

Supervisor: Sarah Carr

New Zealand Otago University

ABSTRACT

Enterprises in the era of change are facing the severe challenge of knowledge management upgrading. Building corporate universities (CUs) has become the key strategic path for enterprises to cope with the challenge of knowledge management. This study takes 13 of China's CUs as cases, adopts qualitative and quantitative research methods to analyze the path mechanism of CUs enabling enterprise development, and explores how to connect multiple subjects, integrate multidimensional elements, improve organizational capacity, and obtain sustainable competitive advantages. Therefore, based on organizational learning theory, organizational capital theory, and knowledge-based view, this study proposes a path model of CUs' enabling behaviors in the context of China.

The research conclusions are as follows: (1) With the help of coding methods, this study identifies how CUs enable organizations and establishes a path model of CUs enabling enterprise development, including "linking strategy", "connecting businesses", "cultivating teams", "shaping organizations" and "inheriting culture". (2) This study proposes that CUs promote the development of organizational capabilities mainly reflected in the effective promotion of strategic capabilities, business capabilities, team motivation, organizational synergy capabilities, corporate cultural capabilities, and other five key elements, and with the help of qualitative analysis results and empirical analysis results to verify. (3) this study combines the path model and trend model into the integration model and analyzes the trend of CUs. Hence, by opening the black box of CUs' enabling path and trend, this study can further enrich the existing literature on CUs. Moreover, our conclusions can enlighten significance for the guiding practice of enterprises and CUs' managers.

Keywords: Corporate University; Enable; Case study; Coding method; Path analysis

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1 INTRODUCTION

Research in organizational development is an ever-evolving field that stems from social phenomena both domestically and internationally. Organization development refers to the broad application of behavioral science knowledge to plan, develop, improve, and strengthen strategies, structures, and processes that promote organizational effectiveness (Bushe & Marshak, 2009). In today's rapidly changing business environment marked by Variability, Uncertainty, Complexity, and Ambiguity (VUCA), as well as the emergence of novel organizational paradigms like borderless organizations, platform organizations, and agile organizations, staying ahead presents a formidable challenge for businesses (Whiteman, 1998). In this context, corporate universities emerge as indispensable tools, enabling businesses to navigate these complexities and maintain their competitive edge

Corporate universities are not just professional development units within organizations. They can serve a range of purposes to support organizational development. The service object of a corporate university is not only the internal employees of the enterprise but also extends to the outside of the enterprise. It is a strategic tool for the enterprise. In the actual practice of the Chinese scenario, most of the corporate universities have been dubbed "University", "College", "School of Management", "Business School", "Research Institute", "Training Institute" or "Development Institute" during the development process and other names, which have the function of providing talent training for the enterprise. However, there is a significant difference between a corporate university and traditional training centers. Traditional training centers aim to improve employees' work performance and cultivate employees' professional skills. In most organizations, the training method focuses on one-time or short-term skill teaching, which only develops the ability of employees to know and understand how to better perform their jobs (Liu Songbo and Wei Lili, 2008). As Roberts and Robertson (1992) described organization development as a practice for “enhancing individual development and improving organizational performance, through alteration of organizational members’ on-the-job behaviors” (p. 272). Following the organization development theory, Zhao Linjie et al (2008) indicated that the development of corporate universities reflects the shift from the orientation of training talents to the orientation of service organization development. This study believes that corporate

universities are the product of the evolution of corporate organizations, and are an upgrade and iteration of traditional corporate training centers.

The core purpose of corporate universities is to empower organizations by connecting multiple subjects (i.e., employees, customers, teams, and managers) to create value for customers. Corporate Universities integrate organizational elements (i.e., business, structure, process, mechanism) to enhance organizational vitality and enhance organizational capabilities, which in turn drive organizational performance, organizational vitality, and organizational capabilities. This research explores the paths the best corporate universities in China use to empower organizational development, investigates the role and mechanism of corporate universities in connecting multiple subjects and integrating multi-dimensional elements, to achieve the improvement and stimulation of organizational capabilities and vitality, and then achieve organizational performance.

This chapter will first introduce the research background of this research, that is, in the era of VUCA, corporate knowledge management is facing new challenges, and corporate universities have become a new carrier widely used in knowledge management around the world. The most critical empowerment path is the key measure to solve the challenge; therefore, based on the practice phenomenon of the corporate university in the context of China, this study proposes the research question of how does corporate university empower corporate development? Next, the significance of this research is discussed from a practical point of view; finally, the specific content of each chapter will be explained.

1.1 Research Background

For enterprises in the era of change, how to make limited resources produce greater performance is an important consideration. The limitation of resources in enterprises can stem from various factors such as budget constraints, economic fluctuations, market competition, and strategic decisions made by the organization's management. Limited resources might also be a result of prioritizing certain areas over others, optimizing operational efficiency, or adapting to changes in market demands and technological advancements. In the context of evolving business environments, understanding how to effectively utilize these limited

resources becomes crucial for enterprises to achieve optimal performance and sustain their competitiveness.

Training and development are tools they use. According to the research of the Training Industry, a global content research and development training organization, the market size of US corporate training in 2019 reached 169.4 billion US dollars, nearly 1,072.6 billion yuan, accounting for about 46% of the global market share (Training Industry, 2019). This report also predicts that the U.S. corporate training market will grow at a compound annual growth rate (CAGR) of more than 10% during the period 2019-2023 (Training Industry, 2019). Therefore, for the United States, the corporate training industry appears to be in a continuous upward cycle. The data from the "Analysis of the Market Status and Development Prospects of China's Enterprise Training Industry in 2020" shows that China's enterprise training market was about 734 billion yuan, and the average training investment was 2,474 yuan/person/year (51job.com, 2020). Compared with the United States, the overall and per capita training expenditure of Chinese enterprises is relatively limited. Effectively allocating limited resources poses a challenging dilemma for enterprises: should policymakers focus these resources on specific organizational units, or adhere to the principle of equal resource distribution across all entities? Additionally, enhancing the quality of enterprise human resources through education and training, transforming human capital into organizational capital, and establishing a sustainable cycle of learning-driven organizational development have garnered significant attention among researchers. However, within the context of this study, the specific relevance and organizational level of these issues remain unclear.

Meanwhile, in the past ten years, the Chinese government has issued a series of supporting policies, paying more and more attention to corporate universities as the carrier of vocational education, and will be used as an important indicator to measure the comprehensive competitiveness of the region. Therefore, the government is paying more and more attention to taking continuing education as an important indicator of regional comprehensive competitiveness, which highlight the importance of corporate universities in build competitive advantage (Francis et al., 2021). Entrepreneurs see the value of corporate universities in talent training, cultural inheritance, knowledge innovation, forward-looking research, change

management, and value chain integration, which aligns with the focus by government on continuous education.

Corporate University refers to "a carrier of lifelong learning for employees at all levels, as well as important customers and suppliers, helping them to improve job performance" (Meister, 1994). Since 1955, the Disney Company and 1956, when the world's first corporate university, the General Electric Company (GE) Croton School of Management, was officially established, corporate universities have continued to grow around the world. After that, well-known foreign corporate universities such as McDonald's Hamburg University (1961), Siemens School of Management (1997), and Apple University (2008) were successively established. As a learning organization, corporate universities are gradually being used by more and more companies. Successful practice shows that corporate university plays an important role in the process of organizational development, especially in the important links of organizational change - in other words, almost every well-known enterprise has a corporate university behind it (Meister, 1994; Allen, 2002).

Although compared with the practice of foreign corporate universities, Chinese corporate universities started relatively late, with a time difference of nearly 40 years, this period is also a stage of rapid economic development in China, which has been mirrored in the expansion of corporate universities. Chunlan College (established in 1997), was the first corporate university. After that, ZTE College (established in 2003), Huawei University (established in 2005), Taobao University (established in 2006), and Tencent University (established in 2014) were established successively, contributing to the development of enterprises. Especially in the context of China, the rapid development of e-commerce, intelligent manufacturing, infrastructure, and other industries has not only not lagged behind the development level of Western companies but has even become a global leader, which also provides a unique internal and external growth environment for corporate universities.

China's corporate universities have grown rapidly in recent years. According to the "In-depth Research and Analysis Report on the Construction and Operation of Chinese Corporate Universities" by the Prospective Industry Research Institute (2022), the number of corporate universities in China had exceeded 3,000 by the end of 2019, showing a leap-forward growth

characteristic different from that of Western corporate universities (see Figure 1-1). This report also concluded that the development momentum of domestic corporate universities has not weakened, and the number is still increasing, and it will be close to 10,000 in the next five years.

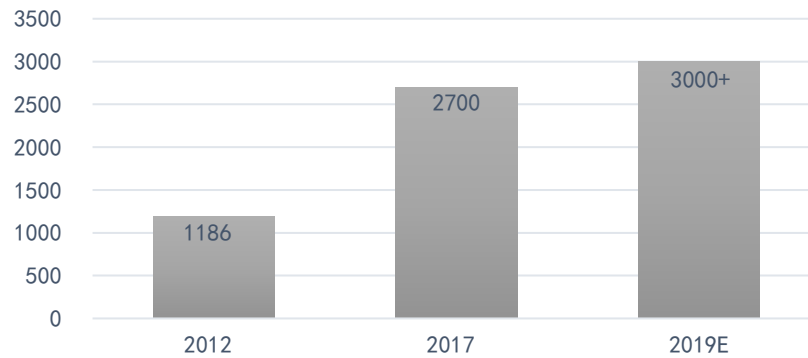


Figure 1-1 Statistical chart of the number of Chinese corporate universities

Chinese enterprises and their corporate universities are facing more challenges from the external environment (Chen Chunhua, 2018). With the advent of the VUCA era, international competition has been further intensified, Internet technology has continued to innovate, and consumer diversity has continued to increase. Over the past 20 years, China has experienced rapid digital development (Song et al., 2021; Wang et al., 2021), hence, Chinese companies are constantly catching up with international advanced companies, and even becoming leading companies in some industries. Taking advantage of growing digital resources and global challenges, many Chinese companies have established corporate universities with new development models. The successful experience of some corporate universities in China (e.g., Huawei University and Tencent University) shows that corporate universities can be key to meeting challenges such as talent development, human resource management, and strategy implementation. Therefore, it is necessary to combine the Chinese market environment and corporate characteristics, summarize and excavate the unique characteristics and excellent experience of Chinese corporate universities with best practices, explore the internal mechanism of their enabling enterprise development, and promote the best practices of corporate universities to support the development of Chinese organizations' competitive advantage.

1.2 Research Questions

With the combination of qualitative and quantitative research methods, this research uses the best corporate university in China as a typical case to analyze the research question: "How can corporate universities empower enterprises to develop and enable enterprises to build a good talent ecology to resist complex and uncertain business environment?" In a corporate university with high performance, which factors are more relevant in terms of strategy, talent, knowledge, resources, ecology, culture, brand, and other elements of organizational sustainable competitive advantage, and the corresponding enabling path mechanism is Content worth studying.

This research will analyze the effective practices of outstanding enterprises in the industry and outstanding corporate universities in terms of talent training and organizational development, and clarify the internal mechanism of the empowerment path of corporate universities by building a theoretical model. In addition, as an important participant in the global economy, China's rapid growth in economic strength and the rapid increase in the importance of Chinese enterprises have brought more research opportunities for domestic scholars. Previously, most of the research on corporate universities and empowerment was carried out by Western scholars, but now the number and level of Chinese corporate universities are increasing, can research results generated from the paradigm of Western scholars be successfully extended to China? Based on the analysis of the typical practices and key measures of the best practice corporate universities in China, this research hopes to enrich the existing research on corporate universities, especially for corporate administrators and enterprises University workers provide useful ideas for reference.

The researcher has participated in the selection activity called "China Corporate university Ranking List" for 13 years, which is a selection activity initiated by the well-known university Shanghai Jiaotong University as an impartial third party since 2012. This selection activity ranked the top corporate universities and gave them the award of "China Outstanding Corporate University" every year. After consultation with the organizers of the selection activity, they have endorsed this research starting in 2019.

The researcher, with the endorsement of the sponsor of the ranking event, has accumulated a large amount of up-to-date detailed information and data on corporate universities. By the end of 2020, according to incomplete statistics, there are more than 2,000 large corporate universities in China, more than half of which have communicated with the researcher and more than 100 leading corporate universities have collaborated with the researcher. In addition, as the deputy dean of the School of Overseas Education, at Shanghai Jiao Tong University, the researcher has participated in organizing the follow-up research of Chinese corporate universities for more than ten years and has provided training and consulting services for hundreds of enterprises, with profound practical experience. Therefore, this study has accumulated sufficient primary data in the field of corporate university research.

1.3 Research Purpose and Value

Aiming to provide a better understanding of how corporate universities enable organization development, this study tries to help corporate universities to be better at what they do by. This research has important practical significance for both the enterprise itself and the corporate university. For enterprises whose primary purpose is to make profits, organizational performance is the top priority of business operations. A corporate university can prove to be an important carrier for enterprises to dynamically coordinate and configure intangible assets (e.g., human resources) and tangible assets (e.g., training platforms) from the perspective of ability and vitality, and then play an important role in promoting organizational performance. Long-term organizational development is one of the main goals of many enterprises. Future organizational development must be the pursuit of both ability and vitality. The management of organizational capabilities and vitality should be approached comprehensively and systematically, instead of concentrating solely on individual factors.

Analyzing the key elements of the corporate university, helps corporate managers to reach a consensus on talent construction and organizational development and promotes the precise investment of corporate resources and corporate budgets in talent training. From a practical point of view, this study can describe the organizational empowerment mechanism in a more in-depth manner. By building a path model for the corporate university to empower enterprise

development, it can improve the ability of the corporate university to promote organizational performance and provide useful ideas for corporate managers.

1.4 Research Content

Based on the analysis of the challenges brought by current environmental changes to organizational development, this research studies the enabling path of the best corporate universities in China for corporate development. This research endeavors to analyze the core of organizational empowerment by examining how it links multiple subjects, including employees, customers, teams, and managers, to create value for customers. This connection relies on the integration of various organizational elements such as business strategies, structure, processes, and mechanisms. The study aims to enhance organizational vitality and improve capabilities, thereby driving organizational performance and fostering comprehensive organizational development. On this basis, this thesis will present the analysis of the path model of the role of corporate universities in connecting multiple subjects and integrating multi-dimensional elements and examines how to effectively improve employee capabilities and stimulate employee vitality through the practice of corporate universities, build organizational capabilities and organizational vitality, and then achieve Organizational Performance. The content of each chapter is arranged as follows:

Chapter 1 is the introduction. This chapter expounds on the research background, research questions, and research significance, and defines the concepts of "corporate university" and "empowerment" in this study based on the classic literature. and technical route of corporate university empowering enterprise development.

Chapter 2 is the literature review. This chapter begins with an overview of organizational learning theory as the theoretical basis for the research. Other major theories, including Organizational capital theory and the knowledge-based view, are reviewed and analyzed to provide an appropriate framework for the research. This study uses bibliometric software to sort out the English literature in the field of corporate university research. Starting from two main research statuses, “corporate university and knowledge management” and “corporate

university and organizational empowerment”, it analyzes the current literature on corporate university empowerment research.

Chapter 3 is the research design. This study mainly adopts a combination of qualitative and quantitative research methodologies such as textual analysis, case study, benchmarking, questionnaire, and coding. Based on the three-level coding method, with the help of qualitative analysis software NVivo 12 Coding, summarizing and analyzing the differentiated paths of the best corporate universities to empower corporate development. To improve the validity and reliability of the research conclusions, this study collected first-hand data using the questionnaire method and semi-structured interview method, and then empirically tested the viewpoints of the theoretical model obtained by the coding analysis.

Chapter 4 is case analysis and text analysis. Based on 13 corporate universities that have won the "Best Corporate University Practice in China" award, this research identifies key events from the aspects of mission and functional orientation, training objects, organizational structure, management mechanism, and excellent performance, and provides first-hand information on corporate universities. The text analysis with second-hand data preliminarily analyzes the key factors of corporate university empowering enterprise development and provides a reference for further detailed research.

Chapter 5 contains two analytical components of qualitative and quantitative data, using a mixed analytical approach. First, the quantitative section provides a fundamental analysis of the corporate university empowerment path. The sample for this study is the best practice of Chinese corporate universities as discussed in Chapter 4. The original data comes from the work summary and interview materials compiled and filled out by the work team of the case corporate university. Secondly, by analyzing the original data with the help of the three-level coding of the programmatic basic paradigm, a preliminarily constructed corporate university authorization path model in the Chinese context. Finally, quantitative analysis was carried out with the help of the questionnaire to verify the reliability and validity of the three-level coding.

Chapter 6 constructs the theoretical model of the enterprise authorized by the corporate university in the context of China. In order to avoid corporate universities only sharing successful experiences without revealing deficiencies, the first subsection conducts a forward-

looking trend study on corporate universities to further improve and enhance the empowerment capabilities of corporate universities. The second part constructs the path mechanism model of the best corporate university to help the development of enterprises in the context of China. It also provides a management tool for other corporate universities or similar specialized agencies within a business to understand how improvements can be made to incorporate these aspects into their current operations.

Chapter 7 is the conclusion and prospects. This chapter sorts out the main research conclusions put forward research contributions from the theoretical and practical levels, expands the research perspectives of existing organizational learning and knowledge management theories, and provides experience for the practical practice of corporate universities. In addition, in the future outlook, the next research direction and ideas are proposed.

2 THEORETICAL BASIS AND LITERATURE REVIEW

This chapter first reviews the theoretical basis of corporate university research represented by organizational learning theory, organizational capital theory, and knowledge-based view. Furthermore, this chapter introduces the literature of the corporate university, also including the relations between organizational capacity, organizational learning, knowledge management, enabling behaviors.

2.1 Basic Concepts and Related literature

Corporate university: Fundamentally speaking, a corporate university is a product of the evolution of corporate organizations, an upgrade, and iteration of traditional corporate training centers, and is fundamentally different from traditional corporate training centers. Li Lin et al. (2015, p. 81) suggested that a "Corporate University is a vehicle for lifelong learning, which helps employees at all levels, as well as key customers and suppliers, to improve their performance." This study also lists some scholars' definitions of 'corporate university' in Table 2-1.

Table 2-1 Corporate university's definitions

Source	Definition of the corporate university
Meister (1994)	The corporate university is a lifelong learning process for employees at all levels, as well as key customers and suppliers, to improve their work performance.
Gilley & Maycunich (2000)	The corporate university conducts strategic human resource development, and its strategic development level is marked by whether the project can maximize organizational performance, create an organizational culture of continuous learning, and implement and manage organization-wide change initiative activities.
Prince (2001)	Corporate university becomes laboratories for the creation or transformation of individual learning into organizational knowledge.
Heinz (2001)	Being part of large businesses or companies, corporate universities take into consideration long-term strategic goals that fit within the parent organization and help them drive them to certain essential resources
Elkeles & Phillip (2007)	The biggest characteristic of a corporate university is its strategy, which helps the organization realize its strategic value.

Enabling (or Empowerment): Empowerment refers to the mechanism and management of organizations to fully release their rights, maximize the personal potential of employees, empower others, and provide needed supportive behaviors through various services (Menon, 1999). At present, the scope of application of "empowerment" in China has expanded, it

includes more empowerment content, and the empowerment objects have also expanded from individuals to organizations. For example, organizational empowerment refers to improving organizational capabilities. To improve employees' internal work motivation and efficiency, organizations or leaders delegate power to employees (Luo Jianhe and Nie Wei, 2010). The powers here include empowering employees to make decisions and actions autonomously at work, empowering employees to improve their abilities, and accessing information. From a microscopic perspective, psychological empowerment refers to employees' self-efficacy and intrinsic work motivation based on organizational empowerment behaviors (Zhang Yan et al., 2006), improve the psychological empowerment of employees, and help to stimulate the vitality of employees.

Knowledge management's objective is to identify and leverage the collective knowledge in an organization to help organizations compete and survive. An enabling lever is e-learning, the creation and distribution of organizational knowledge through the online delivery of information, communication, education, and training (Wild, 2002). Recently, enterprises offer promising digital opportunities to empower knowledge workers in their production environment by leveraging their knowledge processes, decision-making skills and social interaction practices (Hannola et al., 2018).

Organizational development: Roberts & Robertson (1992) described organizational development as a practice for “enhancing individual development and improving organizational performance, through alteration of organizational members’ on-the-job behaviors” (p. 272). Joseph (2004) believes that organizational development refers to the process of an organization adapting to changes in the environment and ultimately realizing self-renewal, self-growth, self-improvement, and self-development through a series of behavioral changes. Brown and Isaacs (2005) said that organizational development “includes a series of planned behavioral science intervention activities to help find improved ways of working together toward individual and organizational goals (p. 3)”.

Organizational capability: The connotation of organizational capability: creating value for customers, surpassing competitors, sustainable development, and is deeply rooted in organizations rather than individuals (Tomer, 1987; McDowell et al., 2018). The rapid and

sustainable growth of an enterprise must rely on the strength of the entire organization. Besides, Youndt et al (2004) indicated that organizational capital exists in databases, patents, manuals, structures, and systems. institutionalized knowledge and codified experience in and through databases, patents, manuals, systems, and processes. Although organizational capability enhancement may generate higher revenues, it may also incur higher operating costs (Lun et al., 2016). Some literature indicated that the positive influence of organizational capability on performance outcomes. Hindasah and Nuryakin (2021) shows that organizational capability positively and significantly affects enterprises' financial performance.

Organizational performance: Williams (1998) proposed that performance management is a system that combines organizational performance management with employee performance management. Organizational performance is the evaluation of organizational efficiency in combination with organizational goals. It includes not only the assessment of the current situation of the enterprise but also the assessment of the future potential of the enterprise. The measurement of organizational performance can be divided into some dimensions, including productivity, retention, accounting returns, growth, and market returns (Combs et al., 2006). The existing research generally believes that there is a close relationship between organizational performance and human resource management (Sun et al., 2007; Van De Voorde et al., 2012).

2.2 Theoretical Perspective of Corporate Universities

Corporate university is the key knowledge management department in the enterprise. Many studies indicate that it is necessary to analyze corporate university along the main line of knowledge management theory. But among the plethora of definitions of knowledge management is that its objective is to identify and leverage the collective knowledge in an organization to help organizations compete and survive. Based on these research consensuses, this section includes three theories that are closely related to corporate universities. First, classical organizational learning theory focuses on how companies acquire resources and capabilities through learning, which provides theoretical support for the motivation of companies to establish and operate corporate universities (Gavetti & Levinthal, 2000). Second, the organizational capital theory goes a step further than the organizational learning theory,

arguing that organizations can build their own capital through learning, which is closely related to the sustainable competitive advantage of enterprises (Francis et al., 2021). Third, the knowledge spillover and absorption behavior in the above-mentioned process is the content explained by the knowledge-based view.

2.2.1 *Organizational Learning Theory Perspective and Corporate Universities*

Organizational learning theory was introduced by Argyris and Schon in "Organizational Learning" published in 1978 (Argyris & Schon, 1978), that organizational learning is a kind of learning in the social relationship formed by the organizational system. Its theoretical viewpoints include the connotation of organizational learning and the influencing factors and results of organizational learning.

From the connotation of organizational learning, the current research consensus is that corporate university is the collective learning of organizational members: Dodgson (1991) believes that organizational learning includes the learning process and learning results; Zangwill and Kantor (1998) believe that organizational learning is to deal with complex and complex learning. Berends et al. (2003) believe that organizational learning is a process that develops over time, interacting and maintaining the past, present, and future of the organization at the same time connect. From the analysis level of organizational learning, according to Bontis (2002), it can be divided into three levels: individual, group, and organization. Among them, individual learning is the direct motivation of organizational learning theory, and individual learning is the basic element of organizational learning; group learning is the process of collective learning, which depends on the accumulation of learning experience of individuals in the organization; the learning experience of individuals and groups can be transferred into the process and structure of the organization, and then have an impact on the organization's future learning activities (Shrivastava, 1983).

According to the research of Argyris and Schon (1978), from the perspective of the types of organizational learning, it can be divided into single-loop learning, double-loop learning and relearning: (1) Single-loop learning is a single-direction learning method based on a stimulus-response mechanism (Argyris and Schon, 1978). A corporate university can improve an organization's ability to accomplish its stated goals using a single-loop learning approach

without major changes to the organization's strategy, the most common being top-down training for employees. (2) Dual-loop learning to deal with complex environments. Organizations can not only change their behaviors but also modify their mental models. Corporate universities use dual-loop learning to re-evaluate organizational goals and the nature of organizational culture, including changing organizational culture and organizational learning. (3) Relearning is the ability to establish a new operating mode based on the above. For example, corporate universities can iterate learning methods, especially in the digital economy, using a variety of digital methods to help employees and enterprises interact and communicate.

Existing research on organizational learning theory focuses on the influencing factors and results of organizational learning, as well as the intermediary mechanism that acts as a "bridge". Early research focused more on the effect of organizational learning, that is, organizational learning can play a positive role in organizations and employees. For example, organizational learning has a positive impact on employee job performance. Chen Guoquan (2002) based on the survey data of 201 enterprises in China, empirically demonstrated that there is a significant positive correlation between learning ability and organizational performance in the process of discovery, invention, selection, implementation, promotion, feedback, and knowledge. Chen Guoquan and Liu Wei (2017) used 213 samples from Chinese enterprises and found that the external environment and internal structural characteristics of the organization played a moderating role in the relationship between organizational learning and organizational performance at the same time. As the number of studies examining organizational learning outcomes has grown, research has also discussed which factors play a mediating role. For example, Gavetti and Levinthal (2000) found that previous behaviors promote the generation of learned behaviors, and insights generated in the process have an impact on subsequent behaviors. Chen Guoquan (2017) analyzes the relationship among organizational learning, innovation and learning from a theoretical perspective, and believes that organizational learning has a good role in promoting organizational innovation, which in turn promotes the improvement of organizational performance. Recent studies have shown that more scholars have paid attention to the improvement measures of organizational learning, that is, what factors can promote efficient organizational learning. For example, Chen Guoquan and Wu Fan (2018)

used a systematic viewpoint to construct a theoretical model of learning transfer, arguing that human factors, problem factors, the interaction between people and problems, and the moderating effect of the environment will jointly affect the effect of learning transfer. Zhao Linjie et al. (2008) took China's auto industry as an example to demonstrate that there is a positive correlation between factors such as learning organization characteristics, organizational learning system, inter-organizational relationships and organizational learning ability and inter-organizational learning effect.

Existing studies have pointed out that organizational learning theory is a key perspective to analyze the function of a corporate university (Chen Yunqi & Xu Yusen, 2018): Corporate university encourages different departments to actively exchange information and experience through the establishment of learning exchange platform, and promotes the interaction and collaboration among members of various departments; Some corporate universities have even established learning platforms across organizational boundaries to promote internal and external knowledge exchange.

Organizational learning theory can be used as a reference for this study because the corporate university is the key carrier to promoting organizational development using learning. Since organizational learning revolves around the acquisition of information and knowledge, the subject ("who will organize learning") refers not only to individuals but also to teams and the entire organization. Existing research suggests that the effectiveness of organizational learning is not the sum of the learning effectiveness of individuals in the organization, it can far exceed the effectiveness of individual learning (Shrivastava, 1983). Successful businesses are those that fully develop their employees' commitment to the business and their capabilities at every level. For traditional companies, transforming into a learning enterprise represents a huge challenge. To achieve this, companies need to encourage managers and other employees to actively try new ideas, and the results of the experiments are fed back and applied at the broader level of the company. At the same time, Senge also proposed five skills that can help companies transform and upgrade to learning organizations: systematic thinking, self-learning, breaking the "spiritual model", establishing a common outlook, and team learning. For example, for systematically building a learning organization, the key to success is to have a suitable corporate

culture. If learning is not supported by the overall corporate culture, it will not be well executed. The corporate university may be an effective carrier that can use learning methods to promote the formation of systematic thinking at the organizational level and break down organizational boundaries to establish a common vision.

2.2.2 *Organizational Capital Theory Perspective and Corporate Universities*

Based on human capital, Tomer (1987) defined organizational capital (Organization capital) as "a human capital that is reflected in organizational relationships, organizational members, and organizational information collection, and has the attributes of improving organizational functions." Youndt et al. (2004) believe that organizational capital exists in databases, patents, manuals, structures, and systems, institutionalized knowledge and codified experience in and through databases, patents, manuals, systems, and processes. For the purposes of this thesis, it is this definition of organizational capital from Youndt et al (2004) us used to understand corporate universities. Organizational capital is one of the important dimensions of enterprise intellectual capital that the firm directly owns and remains within the company even when employees (human and social capital) leave the organization (Subramaniam & Youndt, 2005). Prior studies have indicated that organizational capital is closely related to the sustainable competitive advantage of enterprises (Francis et al., 2021). For example, McDowell et al. (2018) believe that organizational capital is the combination of human skills and physical capital in an organization to meet organizational knowledge for production and product needs. George (2015) believes that organizational capital is the knowledge of technology, publications, organizational management, and procedures that employees remain in the enterprise after work. Subramaniam and Youndt (2005) believe that the essence of organizational capital is the accumulation of knowledge, that is, enterprise-specific knowledge capital that is produced along with organizational products. Therefore, existing studies have agreed that organizational capital, human capital, and social capital are listed as three prominent aspects of organizational intellectual capital (Kaplan & Norton, 2004; Subramaniam & Youndt, 2005). Following those views, this study uses the system-based perspective from Youndt et al. (2004) to understand corporate universities' practices.

In addition, existing studies have shown that the improvement of various intellectual capital, such as organizational capital, human capital, social capital, can affect the absorptive capacity of the enterprise through human resource practices, thereby improving the innovation performance of the organization (Soo et al., 2016). For example, McDowell et al. (2018) used a sample of small firms to show the positive relationship between intellectual capital and organizational performance. Li and Wu (2021) explored organization capital and its spillover effects among Chinese manufacturing firms and found organization capital generates positive spillover in China when geographical proximity is considered.

The theory of organizational capital can guide the development of corporate universities. Organizational capital can make the assets of a group of people or an organization more efficient than separate (Chen et al., 2019). The accumulation of organizational capital and the choice of the path are directly related to the question of how to obtain and improve the ability of corporate university and affect the choice of the path of corporate university enabling enterprise development. Most studies believe that there are multiple output channels for the positive impact of organizational capital on enterprise formation, but the existing research has not reached a consensus on the specific mechanism. For example, Chen and Inklaar (2016) argue that investment in organizational capital not only increases the productivity of investing firms but also spills over to other firms. Specific to the enterprise, organizational capital can effectively promote the innovation ability of the enterprise. For example, Subramaniam and Youndt (2005), through a longitudinal study of 93 organizations, found that human capital, organizational capital and social capital and their interrelationships selectively affect incremental innovation capabilities and breakthrough innovation capabilities, where organizational capital is positive Influence incremental innovation capabilities. Tronconi and Vittucci (2011) took European firms as a sample and found that organizational capital has a positive impact on firm performance. Francis et al. (2021) found that a firm's organizational capital has a positive impact on innovation, and when firms with more organizational capital have a greater number of patents, their patents are cited more often. As a unique form of educational arrangement, a corporate university focuses on building the intellectual capital of

the enterprise (Singh et al., 2019), which can achieve the goals of the organization by building the intellectual capital of the employees.

This study integrates the theory of organizational capital into the management practice and theoretical research of corporate universities, which can further identify the dimensions of organizational capital in corporate universities, and explore what organizational capital is needed to effectively empower corporate development and provide a reference for effectively cultivating organizational capital.

2.2.3 Knowledge-Based View and Corporate Universities

Knowledge-based view theory suggests that knowledge, which can be defined as the information generated by individuals by encoding and abstracting large amounts of situational and complex data into specialized forms in individual members of the organization, with unique and inimitable characteristics, is a vital resource for enterprises to form competitiveness (Grant, 1996). Therefore, the essence of enterprise knowledge is the integration of individual professional knowledge, and enterprise knowledge management is a strategy that consciously promotes the transfer of the right knowledge to the right person at the right time, and finally promotes the development of the enterprise.

The existing knowledge-based view research believes that internal and external knowledge can affect the breadth and depth of knowledge learning in enterprises (Lane et al., 2006; Zahra & George, 2002). Zhou and Li (2012) integrated existing research perspectives and analyzed how the existing knowledge base (i.e., knowledge breadth and depth) interacts with knowledge integration mechanisms (i.e., external market knowledge acquisition and internal knowledge sharing) to influence breakthrough innovation, pointing out that Firms with a greater breadth of knowledge are more likely to achieve breakthrough innovations with internal knowledge sharing rather than market knowledge acquisition. Côte-Real et al. (2017) based on the knowledge-based view, revealed the positive impact of big data methods at the enterprise level on the business value chain of enterprises. Scuotto et al. (2017) analyzed the positive impact of knowledge dynamics of SMEs in determining the choice of establishing formal and informal open innovation models and the innovativeness of service firms. Mahdi et al. (2019) argue that acquiring knowledge and using it efficiently is the only way to have a sustainable competitive

advantage in the marketplace, and using a sample of colleges and universities from educational settings, demonstrate how knowledge management processes can create sustainable competitive advantage.

Knowledge management capability, which has been defined as the degree to which a company creates, shares and utilizes knowledge resources among various functional departments, (Dabic et al., 2020) is an important bridge connecting organizational development and internal and external factors. On the one hand, the existing literature develops a knowledge-based perspective that focuses on external knowledge transfer into the organization, that is, either the internal and external knowledge processes of a firm need an integrative knowledge management capability to successfully implement innovation strategy (Taghizadeh et al., 2021; Nickerson & Zenger, 2004). For example, studying the role of external knowledge sources in identifying strategic opportunities, the realization of opportunities usually involves important interactions with external knowledge sources, especially organizational design can facilitate the interaction of enterprises with external resources (Foss et al., 2013). Based on the internal perspective of the organization, the mediating role of knowledge management capability in the relationship between strategic human resource practice and innovation performance is demonstrated (Chen & Huang, 2009). Other scholars have used the data of 404 small and medium-sized enterprises to find that the knowledge value and knowledge creation practices of top managers affect open innovation, which in turn affects organizational performance (Singh et al., 2021); or demonstrate that knowledge-oriented leadership plays an important role in enterprise knowledge management positive effect in (Donate & Sánchez de Pablo, 2015).

On the other hand, existing research expands the boundaries of knowledge management to the outside of the organization, arguing that companies can establish strategic alliances based on knowledge perspectives. Grant and Baden-fuller (2004) used the knowledge-based view to analyze the causes of inter-firm alliances, arguing that the main advantage of alliances over firms and markets lies in "exploring" knowledge rather than "acquiring" knowledge, compared with previous learning-based views, based on the knowledge-based view, the analysis of alliance knowledge acquisition is more rigorous. The research of Kale and Singh (2007) demonstrated that the alliance learning process includes the expression, coding, sharing, and

internalization of alliance management knowledge, and it is positively related to the overall alliance success of the enterprise. In addition, in the VUCA era, some scholars have applied the knowledge-based view to the research on the digital business value of enterprises. There are also studies demonstrating that big data analytics improves organizational agility by influencing knowledge management and its processes (Côte-Real et al., 2017), or analyzing the problem of identifying lead users in open innovation communities from a knowledge-based perspective (Shan Xiaohong et al., 2021).

The theoretical support of this study lies in the fact that the enterprise university can solve the practical problems of knowledge management inside the enterprise and knowledge diffusion outside the enterprise. Corporate universities undertake the important function of corporate knowledge management (Chen Yunqi and Xu Yusen, 2018), which needs to effectively link internal and external knowledge to improve employees' knowledge ability, thereby promoting the continuous accumulation of corporate knowledge. In addition, the knowledge-based view believes that the knowledge management of an organization is no longer limited to the organization, but can establish a close knowledge cooperative relationship with external suppliers, customers, and universities. From the actual practice of corporate university, the viewpoint of knowledge-based view can provide useful guidance for corporate university knowledge management activities.

2.3 Research on Corporate Universities

2.3.1 Research Overview

Since GE established its first corporate university in 1956, research related to corporate universities has continued to evolve. An earlier piece of literature on corporate universities is "*Business Segments and Corporate Relations with US Universities*" published by Michael (1981), which analyzes the relationship between corporations and universities. The first article explicitly about corporate universities was Buchbinder and Newson (1985)'s paper titled "*Corporate University Linkages and the Scientific-Technical Revolution*", based on research in Canada, which analyzed that universities were facing a lack of funds and that corporations had an unprecedented demand for new technologies. The question of interest, that the integration of

enterprises and universities can provide new productivity for enterprises. This article kicked off the study of the corporate university, and for the next four decades, the existing literature focused on the concept of the corporate university, the function of knowledge management, and driving organizational development.

This study collated literature published in the corporate university field from 1985-2020. We use software A to visualize the key words, author cooperative network, and clustering of literature over the past decades (see Appendix 1). The first paper on "Corporate University" appeared in 1985, which was the beginning of research in this field. In this study, all the literatures with the types of "Articles" and "Review" were extracted from the Web of Science Core Collection database, and 120 articles with the subject (i.e., title, keywords and abstract) containing "corporate university" were obtained (search formula: TS = "corporate university "). As shown in Figure 2-1, there has been little interest in this area of research until the last decade since the first studies of corporate universities emerged in 1985. In 2013, publications in the field exceeded 10 for the first time, then declined, peaking at 15 in 2019. From the perspective of management practice, the world's first corporate university, the General Electric Company (GE) Croton School of Management, was officially established in 1956, and corporate universities entered a period of rapid development in the 1980s. Although China's corporate universities started relatively late, the first corporate university was Motorola University established in 1993, followed by Chunlan College established in 1997, Hewlett-Packard School of Business established in 2001, and Shanghai Jiaotong University-General Motors established in 2008. Automobile Research Institute". It can be seen that there is a certain disconnect between the literature research of corporate universities and the actual management practice, and the industry management practice precedes the management theory research. This also shows that there is a large gap in research in this field, and high-quality theoretical research is urgently needed to guide practice.

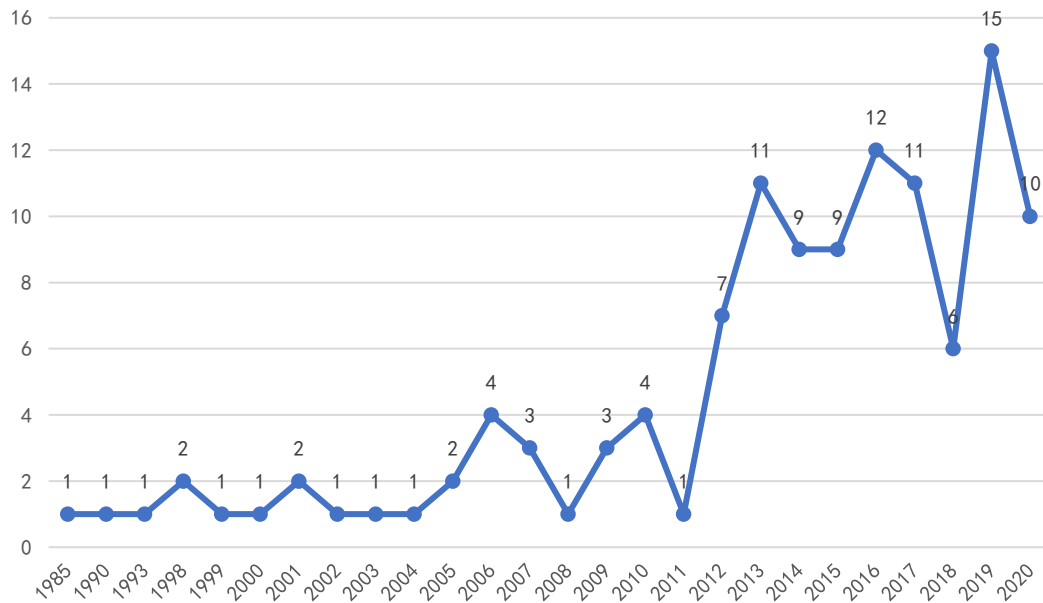


Figure 2-1 Number of papers per year in the field of "Corporate University"

Organizations need to constantly adapt themselves to changing and increasingly complex environments. The "Corporate University" puts "intelligent learning" at the forefront, and its strategy is to ensure consistency between the organization and the environment, which requires both speed and direction of learning (Rademakers, 1988). The number of papers published in the field of corporate universities mainly came from the United States ($n=33$, 27.5%), Canada ($n=19$, 15.83%) and Australia ($n=17$, 14.17%). Islamabad University, Rotterdam, York University in Canada and the University of Quebec are among the top publishers. The same rule can also be found in business practice. In terms of types, the articles in the field of corporate universities are dominated by education (38 articles, 31.67%), management (18 articles, 15.00%), and management (7 articles, 5.83%). A corporate university has the dual attributes of corporate management and teaching and training, and many types coexist. Therefore, by searching domestic and foreign literature, we can see that foreign scholars' research on corporate universities is earlier than Chinese scholars. Prior research on corporate universities is not only analyzing the definition, functions and characteristics of corporate universities, but also analyzing the antecedents and influencing mechanisms of corporate universities. This research will continue to be explained in more detail in the following sections.

2.3.2 Concept and Characteristics of Corporate Universities

Corporate university is a lifelong development for employees at all levels, as well as important customers and suppliers. The carrier of learning will help them improve their work performance (Meister, 1994). This consensus has been recognized by many researchers. Allen (2002) defines a corporate university as an educational entity. As a strategic tool, the responsibility of corporate university is to cultivate the knowledge and wisdom of individuals or organizations by implementing learning activities to assist the parent company to achieve its mission. Hence, based on existing literature, this study summarizes the development characteristics of the three stages of the corporate university, including stage of germination, stage of growth and stage of maturity. Table 2-2 shows that CUs' functions are constantly extending and expanding.

Table 2-2 Corporate university's evolution

Stage	Characteristics of corporate universities
Stage of germination (1920s to 1940s)	• Focus on internal staff, the scope has expanded; • The training content focuses on skill operation; • The training method is relatively simple, mainly teaching courses; • The lecturer adopts the mode of internal and external combination and has not paid attention to the construction of an internal lecturer team
Stage of growth (1940s to 1980s)	• The strategic functions of corporate universities have been strengthened • The service object is mainly internal, but the system and individuation have been enhanced • The training curriculum is linked to the corresponding competency standards • The training forms are diversified and a variety of effective learning methods have been developed and adopted
Stage of maturity (1980s to present)	• The organizational operation system of the corporate university has been improved, and the organizational independence has been enhanced • The business system is more related to performance, competency standards, employee career development, and value chain users, and more systematic • Knowledge management has been emphasized and applied • Information technology has been widely used in the construction and operation of corporate universities • The function of corporate university tends to generalize

The core feature of corporate university is reflected by its strategic value. Prince (2001) believes that the core functions of corporate universities include learning and training, knowledge management and strategic linkages. Elkeles and Phillips (2007) believe that the most important feature of corporate university is its strategic nature, which can connect business departments with the top-level strategy of the enterprise, help business departments to clarify

the overall strategic plan of the enterprise, and then help the entire enterprise to realize strategic value. Wu Feng (2014) believes that corporate universities are the product of the evolution of corporate organizations, and are the upgrade and iteration of traditional corporate training centers, whilst being essentially different from traditional corporate training centers. Gilley and Maycunich (2000) believe that the core function of corporate university is the development of strategic human resources, and the level of strategic development indicates whether it can maximize organizational performance, create an organizational culture of continuous learning, and implement and manage organizational-wide change initiatives. Heinz (2001) argues that as part of a large enterprise or company, corporate universities consider long-term strategic goals that fit the parent company and help them drive certain necessary resources.

Since the first domestic corporate university in China was established in 1997 — nearly forty years after the establishment of the first corporate university in the West, research on corporate universities in China also started relatively late. The earliest research appeared in the business world. Ma Jinhong (2001) took Ericsson China Institute as an example to analyze the function and role of corporate university in competitive strategy, and believed that the function of corporate university includes three points: (1) Design, develop and provide Learning programs to support the core strategy of the enterprise; (2) to promote the cultural construction with the help of the corporate university; (3) to strengthen the core competence of the organization through the curriculum design of the corporate university. Those functions are consisting with western corporate university.

Liu Wei and Xiang Gang (2001) published the paper " *On the Sustainable Development, Continuous Innovation and Continuous Learning Mechanism Construction of Enterprises*" proposing that modern excellent enterprises should take sustainable development as the primary goal, and the establishment of continuous learning mechanism should be It is an important foundation for enterprises to achieve sustainable development. For large-scale enterprises with far-sightedness, the establishment of corporate universities is a better feasible model for enterprises to promote the construction of continuous learning mechanism. With the continuous development of Chinese enterprises and the escalating demand for talent management, more enterprises have begun to establish corporate universities and use corporate universities as a

learning platform to promote their own development. For example, from 2009 to 2013, the number of Chinese corporate universities showed a sudden growth trend. On average, more than 200 corporate universities were born every year. By 2013, the total number of Chinese corporate universities exceeded 2,000 (Li Lin et al., 2015), it has exceeded 3,000 in 2020. Luo Jianhe and Nie Wei (2010) pointed out that corporate university is an advanced form of corporate education development by sorting out the development of Chinese corporate universities at that time, and believed that Chinese corporate universities still have broad development prospects, but the follow-up development should not only develop their own resources, form a unique personality, and make effective use of resources from external collaborators. After that, a part of representative studies has made a clear concept definition and characterization of China's corporate universities. The author of the book "Corporate University Code" provides a clear concept for the corporate university in the Chinese context: "Corporate university is an internal enterprise organization that serves corporate strategy and takes organizational knowledge operation and knowledge education as its core content (Li Lin et al., 2015, p. 81)".

Combining with the research consensus that "corporate universities are learning organizations" (Wu Feng, 2014; Li Lin et al., 2015), this study believes that corporate universities are the evolutionary products of corporate organizations, that is, the upgrading and iteration of traditional corporate training centers. A new model of education and training. Corporate University is funded by enterprises, with senior managers of enterprises, first-class business school professors and full-time and part-time trainers as teachers, with the help of classroom training, action learning, case studies, actual combat simulation, online learning and other teaching methods, to achieve the cultivation of internal talents within the enterprise , promote business development, form the core competitiveness of the organization, and ultimately empower the entire industry chain ecology.

2.3.3 Knowledge Management Research in Corporate Universities

Currently, there is a consensus in the academic community regarding the use of knowledge management perspective to analyze the activities of corporate universities. Most scholars believe that employing the concept of knowledge management is instrumental in gaining a

deeper understanding of the role and functions of corporate universities, viewing them as vehicles for creating and transforming individual knowledge into organizational knowledge (Prince, 2001; Scarso, 2016). From this perspective, corporate universities aggregate and integrate various internal and external resources, facilitating the promotion, consolidation, and dissemination of knowledge within the organization, thereby fulfilling a crucial function in knowledge management (Scarso, 2016; Singh et al., 2019).

Existing literature identifies key characteristics of knowledge management in corporate universities, which include: ① aligning the educational objectives of corporate universities with the strategic needs of the organization; ② closely aligning employees' earnings with the learning objectives of their activities; ③ expanding the scope of learning to encompass key clients and suppliers; ④ managing corporate universities as commercial entities, allowing for economic pursuits, and developing corporate university alliances; ⑤ ensuring accurate representation and assessment of the value created through knowledge management (Abel & Li, 2012; Chen et al., 2019; Nagy & Burch, 2009). Therefore, from the perspective of knowledge management, corporate universities are not merely places for imparting knowledge; they are also creators, disseminators, and preservers of knowledge (Chen et al., 2019). Specifically, corporate universities provide the following support to drive organizational development:

(1) At the strategic level of knowledge management, scholars have delved into how corporate universities become pivotal forces in strategic execution. Meister (1994) pointed out that corporate universities serve not only internal employees but also extend their services externally, positioning themselves as strategic tools for organizations. For instance, Antonelli's (2013) research highlighted the crucial role of corporate universities in conveying corporate strategies and social responsibility perspectives. Antonelli used the example of six corporate universities in the European energy industry, emphasizing that these institutions have surpassed traditional training models. Additionally, Scarso's (2016) study emphasized the role of corporate universities as strategic educational platforms, providing employees with training and development oriented towards strategic goals. These studies indicate that corporate universities

play essential roles in strategic planning and transmission, empowering organizations in the strategic domain.

(2) At the operational level of knowledge management, insights from organizational learning theory and the knowledge-based view have been valuable. These theories posit that organizational learning outcomes surpass individual learning effects, encompassing knowledge transfer, knowledge creation, and application incubation within a company (Shrivastava, 1983; Grant, 1996; Zhou & Li, 2012). Corporate universities, as key platforms for knowledge transmission, deeply engage in business activities by transferring knowledge to operational departments and facilitating the application of new knowledge. This profound integration of knowledge management activities into business operations substantially supports corporate universities in empowering operations, thereby facilitating comprehensive business development. Corporate universities employ methods such as training, sharing, and research to introduce external knowledge, innovate internal knowledge, assimilate explicit knowledge, transform tacit knowledge, and utilize it. In practical terms, corporate universities often invite external experts to provide specialized training for employees in different business departments, enhancing the flow of knowledge within the organization and increasing the success rate of knowledge integration (Li et al., 2015).

(3) At the team level of knowledge management, Gilley and Maycunich (2000) assert that a core function of corporate universities is strategic human resource development, with teams being the smallest units of internal human resource management within organizations. In the actual practices of corporate universities, most institutions nurture internal trainers to provide training services for business teams. Knowledge transfer is considered a key approach to strengthening the internal trainer workforce, knowledge creation results from collaborative efforts between internal trainers and business teams, and knowledge application involves business teams transforming knowledge into practical technologies and products. Throughout this process, the end-to-end knowledge management activities of corporate universities not only enhance employees' professional skills but also promote knowledge sharing and collaboration within teams.

(4) At the organizational level of knowledge management, existing literature has focused on how corporate university knowledge management activities evolve within and outside organizations. Prince and Beaver (2001) regard corporate universities as tools and carriers of knowledge management, emphasizing their characteristics, where their mission is to serve as laboratories where employees create knowledge and transform individual learning into organizational knowledge. Similarly, Chen et al. (2021), using a sample of 34 Chinese corporate universities and employing fuzzy set qualitative comparative analysis, found that internal expertise and the clarification of internal knowledge processes play crucial roles in enhancing knowledge within corporate universities. Given that the most crucial intangible resource for businesses is knowledge—a key investment ensuring long-term competitive advantage—corporate universities, through internal knowledge management, functional items, and transmission activities, assist in building more flexible and adaptive organizational structures, promoting sustained organizational development.

(5) At the cultural level of knowledge management, considering the cultivation of corporate culture is a complex and vital topic. While no study has directly revealed the relationship between corporate universities and the cultivation of corporate culture (Gilley & Maycunich, 2000), existing literature suggests that corporate universities can help employees better understand and internalize the company's core values through knowledge management activities (Antonelli, 2013). Thus, corporate universities provide support for fostering a positive corporate culture atmosphere.

Besides emphasizing the importance of knowledge management functions, some scholars have highlighted the differences in knowledge management activities among corporate universities of different natures (Antonelli, 2013) and the disparities in corporate universities driven by different dominant knowledge management functions (Chen et al., 2021). This insight inspires further research to explore the differentiated behaviors of corporate universities in knowledge management activities based on their diverse natures. However, the majority of current studies still focus on the knowledge management functions of corporate universities (Abel & Li, 2012; Singh et al., 2019), rather than analyzing how corporate universities leverage knowledge management to empower enterprises from multiple dimensions. Therefore, it is

necessary to construct a comprehensive theoretical framework to analyze the pathways and mechanisms through which corporate universities empower enterprise development by utilizing knowledge management functions.

2.3.4 Research on Enabling Behavior in Corporate Universities

Empower/Enable refers to the empowerment of individuals or organizations, for example, managers decentralize power to allow employees to have more authority on work (Zhang & Kathryn, 2010). The views expressed in existing literature on empowerment and empowerment can be classified into two levels: the individual and the organization. Based on the individual perspective, empowerment is to give energy to others through changes in words and deeds, attitudes, and environment, and maximize personal potential (Seibert et al., 2004). In the field of management, Adler et al. (1996) first proposed that empowerment is a management tool for managers to make employees better control their tasks. Specific to the inside of the enterprise, some studies have analyzed the empowerment behavior of individuals in the enterprise. Spreitzer (1995) believes that empowerment is the internal process of improving employees' work beliefs and enhancing their self-efficacy through giving them responsibility. Self-efficacy is an individual's confidence in his ability to complete work tasks, for example, entrepreneurs can achieve self-empowerment as institutional entrepreneurs through social status (Battiana, 2006), and the impact of leadership empowerment on leadership effectiveness (Zhang Jianping, 2021). Another example is that employees' psychological empowerment is based on the perceived self-efficacy and intrinsic work motivation of organizational empowerment behaviors (Zhang Yan et al., 2006; Lei Qiaoling, 2006).

With the development of psychological empowerment theory, more research has turned to exploring the empowerment behavior among different individuals within an enterprise, focusing on the mechanism through which leaders' "top-down" empowerment has a positive effect on employees: Deng Yulin (2018) found that leaders' empowerment behavior can significantly predict voice behavior, and voice efficacy plays a complete mediating role between leader empowerment and employee voice behavior. Hou Man et al. (2021) used leader-employee matching. Their data demonstrated that leadership empowerment and employee innovation performance are significantly positively correlated, and tacit knowledge sharing

plays a mediating role between the two Empowerment leadership behavior has a positive impact on employees' leadership identity with supervisors whilst leadership identity plays a mediating role between empowering leadership behavior and employee organizational citizenship behavior (Lang Yi and Wang Hui, 2016). With the continuous development of authorization forms, some scholars have found that senior managers of enterprises can achieve corporate goals more efficiently when they implement other positive management activities in combination with empowerment activities for employees than authorization alone (Feng Jiao et al., 2019; Chin et al., 2019; Yang Jiaoping et al., 2021) , such as sharing information, providing various resource support and development platforms, accommodating employees, and developing relevant work skills of employees.

Through these empowerment activities, on the one hand, the work level of employees can be improved, and the confidence of employees to deal with tasks and difficulties at work can be increased (Parker et al., 2010). It can also enhance employees' sense of belonging to the organization, improve employees' psychological safety, enhance employees' intrinsic motivation to work, and reduce employees' risk perception of change behaviors, thereby making Their willingness to engage in forward-looking behaviors such as change will be stronger (Li Lin et al., 2015).

Based on the organization's point of view, organizational empowerment refers to the organization's top-down, left-to-right release of rights, through various services, to maximize employees' personal potential, empower others, and provide the mechanism and support needed (Luo Zhongwei et al., 2017). Organizations can stimulate employees' passion for innovation through empowerment mechanisms such as capability support, benefit sharing, self-driving, and common growth (Qi Xugao, 2013). Therefore, the core of organizational empowerment lies in connecting multiple subjects, including employees, customers, teams, and managers to create value for customers. (Seibert et al., 2004; Zhang & Kathryn, 2010). This connection needs to rely on the integration of organizational elements (such as business, structure, process, and mechanism), with the aim of improving organizational vitality, organizational capabilities, and then drive organizational performance. With the continuous development of the digital economy, a new generation of technological innovation has also affected the form of organizational

empowerment. As far as current management practices are concerned, the empowerment management of an organization is always compatible with knowledge management in the digital age (Zhang Cuijuan et al., 2020). For example, existing literature has studied the impact of new empowerment methods at the organizational level on organizations and individuals within organizations: digital technology empowers enterprises to form organizational resilience (Shan Yu et al., 2017) , and the impact of empowerment on enterprise service innovation behavior (Shan Yu et al., 2017; Mao Qian et al., 2021) , and information technology empowers enterprises to continuously optimize their production and business processes (Jin Lulu and Zhang Pei, 2019) . In addition, empowerment subjects are no longer limited to traditional corporate subjects, and empowerment subjects represented by platform-based organizations have emerged. Existing research has analyzed the empowerment behavior of the platform-based enterprise Haier Group (a well-known manufacturing enterprise in China, which has established several B2B platforms) by means of coding methods, and found that it specifically includes structural empowerment (information rights, opportunities rights, and resource rights), leadership empowerment (teaching experience, solving problems and improving ability), psychological empowerment (control, influence and efficacy), resource empowerment (resource identification, resource acquisition and resource utilization) and cultural empowerment (Interactive culture, sharing culture and innovation culture) (Zhou Wenhui et al., 2018). Some scholars have also found that the resource empowerment behavior of platform-based enterprises stimulates employees' innovation passion by increasing employees' innovative self-efficacy (Yang Jiaoping et al., 2021).

Corporate universities have experienced rapid growth in recent years, evolving into vital knowledge construction and sharing institutions (Nagy & Burch, 2009). For instance, studies such as Chin et al.'s (2019) work have explored how corporate universities drive environmental protection-oriented social innovation. By encouraging employees to engage in innovative and environmental education programs, these universities foster a sense of commitment to the organization's norms, subsequently enhancing both innovation and environmental conservation efforts.

However, existing research primarily delves into empowerment behavior at the individual or organizational level, leaving a gap in understanding the specific role of corporate universities in this process. This gap is especially noticeable in emerging markets. According to Yang Guoan (2010), organizational capability, manifested through the collective efforts of all members, stands as the cornerstone of an organization's competitiveness. Scholars emphasize the need for corporate universities to establish a robust capability support system. This system plays a pivotal role in enabling corporate development, a point underscored by scholars like Xu Yusen and Chen Yunqi (2018). Addressing this area of corporate university empowerment in emerging markets remains a key research focus.

2.4 Corporate Universities Enable Enterprises Development

After searching domestic and foreign corporate university research, the existing literature has described the emergence, development, rationality of existence and the significance of corporate university systematically and in detail. The emergence of corporate university stems from the integration and development of human resources by enterprises, and it is a learning organization aimed at enterprise employees' learning, development, cultural input and ability enhancement. Scholars at home and abroad generally believe that corporate universities are important places for employees to continue their professional learning and an important carrier for their lifelong learning. However, the existing research on the future development trend of corporate universities in emerging markets still needs to be further explored. Based on this, this study relies on the theoretical evolution path of "Organizational learning theory → Organizational capital theory → Knowledge-based view" to provide profound theoretical support for researching the pathways of corporate university empowerment, as illustrated in Figure 2-2.

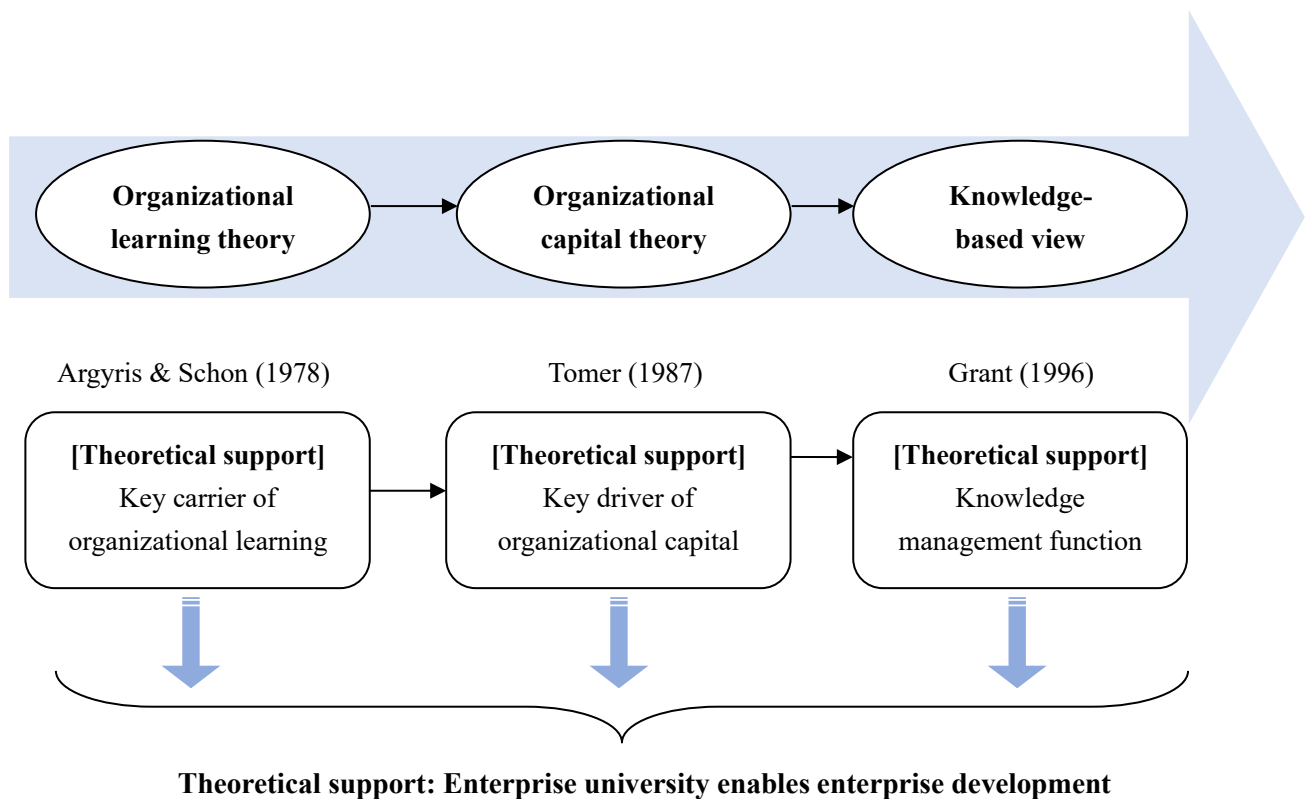


Figure 2-2 Supporting theory diagram

First, organizational learning theory provides a theoretical foundation for understanding the intrinsic traits, nature, and functions of corporate universities. Rooted in the collective learning foundation of organizational members (Argyris & Schon, 1978), corporate universities are viewed as platforms through which organizations offer lifelong learning opportunities to both internal and external stakeholders, including employees, suppliers, and clients. Under the framework of organizational learning theory, the enterprise university is considered to have the following three important core functions. (1) Corporate universities serve as pivotal hubs for organizational learning (Meister, 1994). One of the primary objectives behind establishing corporate universities is to provide continuous learning opportunities for internal employees (Prince, 2001). By offering training programs, knowledge dissemination, and skill development services, corporate universities enable employees to continually enhance their knowledge and skills.

(2) Corporate universities have also extended learning opportunities to suppliers and clients. Several outstanding corporate universities have initiated knowledge sharing with external stakeholders. This external knowledge dissemination not only strengthens

collaboration between the company and its ecosystem members (Scarso, 2016) but also fosters collaborative development within the entire industry chain. Serving as learning platforms, corporate universities bridge knowledge gaps between enterprises and external partners, propelling innovation and progress within the entire industry ecosystem.

(3) Corporate universities foster deep collaborations through strategic alliances with internal and external stakeholders (Chen et al., 2019). These collaborations go beyond mere knowledge transfer and encompass strategic planning and business synergy. As vital platforms for organizational learning, corporate universities promote alignment of strategic objectives among the company and its ecosystem members. This alignment, based on a shared strategic vision, collectively enhances the overall competitiveness of the ecosystem where the corporate university resides.

Second, organizational capital theory emphasizes the crucial role of corporate universities in empowering enterprise development. Organizational capital stands as a key dimension of intellectual capital within a business, closely linked to sustainable competitive advantage (Francis et al., 2021). Within this theoretical framework, corporate universities are perceived not only as practical venues for organizational learning but also as vital tools and platforms for enhancing organizational capital. (1) Corporate universities serve as platforms for internal learning and knowledge transfer, enhancing the quality of organizational human capital. Within corporate universities, employees gain knowledge, training, and practical experiences from both internal and external experts. This internal knowledge sharing facilitates the accumulation of internal knowledge capital. By nurturing talents with innovative thinking and leadership skills, corporate universities effectively elevate the organization's level of organizational learning, thereby enhancing intellectual capital and the quality of human resources (Subramaniam, 2005).

(2) Corporate universities nurture and develop internal talents, providing a continuous talent pool for the organization. The training provided by corporate universities encompasses not only technical skills but also soft skills such as leadership, teamwork, and innovation capabilities—key components of organizational capital. By consistently cultivating internal

talents, corporate universities ensure a robust talent reservoir, providing essential support for the organization's long-term development.

(3) Corporate universities serve as platforms where external knowledge combines with internal knowledge. These institutions not only offer learning opportunities for internal employees but also invite external experts and scholars for lectures and seminars, introducing the latest external knowledge and technologies. This fusion of internal and external knowledge enhances the organization's internal knowledge system, fostering innovation and improving the organization's knowledge capital. Consequently, it strengthens the organization's competitiveness (Tronconi & Vittucci, 2011; Singh et al., 2019).

Third, the knowledge-based view reveals specific pathways and mechanisms through which corporate universities empower enterprise development. Organizations can absorb knowledge from internal and external sources, storing it within individual members of the organization in specialized forms. Through specific mechanisms, this knowledge is transformed into the organization's capabilities (Grant, 1996). Consequently, leveraging the knowledge-based view, this study reveals, from a micro perspective, how corporate universities conduct knowledge management activities to ultimately empower organizations. It provides concrete pathways and mechanisms for empowering enterprises: (1) Corporate universities empower enterprise development by integrating internal and external knowledge. In practical terms, corporate universities attract experts and scholars from different fields through various activities such as training sessions, seminars, and lectures, promoting the flow and exchange of knowledge. This accumulation and transmission of knowledge not only enrich the organization's internal knowledge system but also provide learning opportunities for employees, encouraging them to continuously update their knowledge and enhance their professional skills. Through the interaction between internal employees and external experts, corporate universities create an "ecosystem" or ecosystem based on knowledge sharing.

(2) Corporate universities empower continuous enterprise development by transforming knowledge into capabilities. Considering the emphasis of the knowledge-based view on converting knowledge into organizational capabilities, corporate universities help internal employees apply acquired knowledge to practical work, thereby forming the core

competitiveness of teams and organizations internally. Simultaneously, corporate universities encourage internal and external clients to integrate acquired knowledge into practical work, creatively solving problems and encouraging innovative breakthroughs beyond organizational boundaries. The knowledge management functions of corporate universities enhance the competitiveness of enterprises and provide the impetus for their sustained development.

Therefore, combined with the existing research on corporate universities and the above theoretical perspectives, as well as new management phenomena emerging in the Chinese context, this study believes that existing corporate university research is still insufficient for the following content. First, the strategic positioning of corporate universities in emerging markets remains open for discussion. As a carrier with the dual attributes of an educational entity and a strategic tool, the corporate university itself is an existence that constantly breaks through the limitations of traditional definitions. When the strategic function of the corporate university is extensible, the organizational structure and strategic positioning of the corporate university are also more diversified, and these need to be examined with the help of the real case of the corporate university. Different from mature corporate universities in developed countries with decades of development, corporate universities in emerging markets represented by China are developing rapidly, showing a series of different evolutionary characteristics in a unique institutional environment. Jin Weilin and Wu Huabin (2019) analyzed the practice of Haier University (belonging to the Chinese enterprise Haier) in order to undertake the Group's construction of an ecological brand strategy in the Internet of Things era, such as building a non-linear talent learning interaction platform and establishing various stages of maker experiments room. Second, the existing research on the boundaries of corporate university knowledge management is no longer limited to the internal enterprise. With the continuous deepening of management practice, the most typical phenomenon is that some corporate universities in China have expanded their service targets, which is different from the original corporate universities that only serve their own employees, such as Trinity University, Dongfeng Nissan University and other corporate universities not only Carry out knowledge management for internal employees of the enterprise, as well as knowledge management activities for external stakeholders represented by suppliers and sellers. Third, in the context of

the digital age, there is still a lack of a systematic and integrated theoretical analysis framework for corporate universities to empower corporate development. For example, Liang Jiaming (2020) pointed out that digital means facilitate the establishment and management of corporate universities, and corporate universities need to establish digital learning platforms to enrich training and knowledge management means. Existing studies still mostly focus on the analysis of traditional training and learning platforms, and the analysis of digital platforms, new media platforms, and online and offline integration platforms used in the practice of corporate universities is still insufficient, especially the urgent need to establish local enterprises that reflect the Chinese situation. A systematic analysis framework for university empowerment pathways.

In summary, this study, based on the theoretical framework of "organizational learning theory → organizational capital theory → knowledge-based view," delved into how top-tier corporate universities in China provide support for organizational development. We explored the roles and mechanisms of corporate universities in connecting various stakeholders and integrating multifaceted elements. This research not only reviewed organizational learning theory, organizational capital theory, and knowledge management theory but also offered profound insights into the roles and functions of corporate universities in the context of China.

3 RESEARCH DESIGN

This chapter explains how the research plan is carried out, including the determination of research methods and tools and the selection of research objects, and describes the data collection process in detail. The research focuses specifically on corporate universities that can be argued to be “excellent”. Based on the existing literature, this study defines an excellent corporate university in the Chinese context as "a company that has won various authoritative awards in the international and domestic competitions of influential learning organizations or corporate universities."

This study obtains great supports from the sponsor of "China Corporate university Ranking List", including providing some first-hand data, interviewing, and filling out questionnaires as follows: With the help of sponsors, the study can closely communicate with many corporate universities that have won authoritative awards or advanced experience. Administrators of those corporate universities were invited to provide some internal data through filling the information collection form, which includes descriptions of the current operations and future directions of the corporate university (see appendix 5 and appendix 6). Corporate universities participating in the survey are required to provide detailed answers to the questions in the questionnaires based on interviews or existing literature and provide information including corporate university management articles of association, experience cases, course presentation PPT, introduction video and audio and other forms of auxiliary supporting materials. As a result, this study obtained more detailed primary and secondary data. This research has received written permission from the head of the corporate university and has also passed an ethical review. Moreover, this study conducts investigations through semi-structured interviews with senior managers, corporate university staff, and head of human resources. The study also distributed questionnaires to corporate university staff and related employees through the campaign to ask them about their views on the development of corporate university enabling organizations.

3.1 Research Sample

According to the latest data, the development of Chinese corporate universities is very rapid, and the total number has exceeded 3,000. There are different types of corporate

universities, such as those established by state-owned enterprises, private enterprises, foreign-funded enterprises and Sino-foreign joint ventures. Corporate universities also come in different sizes, ranging from small (fewer than 10 employees) to large (more than 1,000 employees). Corporate universities may also have different names, such as "Business school" or "School of Management" or "Academy" with the corporate name attached. How to effectively sample in such a large population is a very critical issue. This study draws on the sampling rules of existing case studies, that is, selects typical samples that meet the conditions and have certain differences in the scenario (Zhang Hong et al., 2022).

Because this study chose corporate universities as samplings, the cases corporate universities need to be excellence in their industry, and have advanced practical experiences. Hence, following this principle, this study selected some outstanding corporate universities as the research object. Considering data available and according to the recent authoritative ranking list, including "2019 Chinese Corporate University Rankings" and "2020 Chinese Corporate University Rankings"), this research screened out corporate universities that are at the forefront of the comprehensive ranking (See Section 3.3 and Section 4 for details). Since these corporate universities have certain advanced advantages in terms of development mode and performance, they are models for other corporate universities to learn from, which is in line with the research question of how Chinese corporate universities can empower corporate development. Besides, ensure that the aforementioned corporate universities provide detailed primary and secondary receipts. This rich source material facilitates the progress of the research and ensures effective data analysis.

Based on the total number of corporate universities that responded to the interview in 2019, we selected cases from these respondents based on the following criteria: ① Having received prestigious national and international honors, signifying industry-leading practices. ② Possessing abundant internal operational and management data. Subsequently, the selected cases underwent random sampling, and the sampled results were examined to ensure diversity among the enterprises and industries of each case. In the first round of the qualitative data collection phase, a total of 7 corporate universities were included in the study through this screening process. These universities encompass various ownership types, including 5 from

state-owned enterprises (China Mobile Academy, China General Nuclear University, ZTE University, China Resource Gas College, China Railway Fourth Bureau Group's Corporate University), 1 from a joint venture (Dongfeng Nissan University), and 1 from a private enterprise (Sany Corporation University). The industries represented include machinery manufacturing, infrastructure construction, and mobile communications, among others.

Similarly, we conducted the same screening, sampling, and adjustments for the corporate universities that responded to the 2020 interview. In the second round, a total of 6 corporate universities were included as samples for the study, without any overlap with the universities selected in the first round. These universities represent a diverse ownership structure, including both state-owned enterprises (such as China Construction Bank University and Taikang Business School) and private enterprises (such as New Hope Business School, Geely Corporate University, Red Star Macalline Group's Management School, and Robam University). The industries they belong to cover agriculture, forestry, animal husbandry, fishery, manufacturing, finance, wholesale, and retail, encompassing four major sectors.

In summary, the 13 corporate universities selected as samples exhibit certain differences in industry distribution, the scale and nature of their parent companies, and the development models and practical achievements of the corporate universities themselves. Choosing these 13 representative enterprises for the research expands the coverage of the sample data, reduces sampling bias, and enhances the theoretical saturation and credibility (Wiesche, 2017). For detailed information on the data collection process, please refer to Section 3.3.

3.2 Research Methods

With the increasing degree of marketization in China and the intensification of enterprise competition, talents have become the key factor for enterprises to obtain core competitiveness. The booming development of Chinese corporate universities also reflects the increased emphasis Chinese companies place on talent development. This study focuses on the research problem of "how does corporate university empower enterprises to develop", and the current secondary data on corporate universities is relatively insufficient, especially the latest data on the development model and performance at the corporate university level. However, analyzing

how corporate universities empower companies to develop is inseparable from the in-depth mining of a large amount of primary data.

In order to collect primary data on corporate universities in the Chinese context, this study uses a combination of qualitative and quantitative research methods including bibliometrics, coding analysis, content analysis, and questionnaire surveys, as well as benchmarks widely used in practice. Mixed methods research is a research method that combines quantitative and qualitative research methods to help researchers develop rich insights into a variety of interesting phenomena (Vance et al., 2013). First, this study is based on the existing research foundation, analyzing previous research, and integrating the viewpoints of the existing research. Secondly, the ranking of China's excellent corporate universities was used, from which qualified corporate universities were selected as research samples. Thirdly, to collect the cognition, evaluation, and judgment of corporate university practitioners, corporate executive teams, and industry experts semi-structured interviews were conducted. This happened alongside the collection of a large number of practical data and materials from corporate universities with the help of information collection documents.

After being collated, the data was coded step by step with the help of the content analysis method, and the "story line" of this research is sorted out. Fifth, a questionnaire survey method was used to organize the coding content into the measurement quantity of corporate university empowerment enterprise development. The questionnaires were distributed to corporate university practitioners and related experts in the form of online surveys, which were used to establish the reliability and validity of the theoretical framework of this study. It should be noted that previous studies have also pointed out that the case study method is also applicable to the theme of corporate university research because this study focuses more on the general path of how corporate universities can empower corporate development in the Chinese context, rather than focusing on the uniqueness of individual cases. Practical experience; therefore, this study only draws on some of the methods in the case study, rather than mainly adopting the analytical thinking of the case study.

3.2.1 Coding Method

Qualitative research focuses on evaluating and explaining phenomena, often analyzing data that cannot be quantified, such as interviews, surveys, web pages, audio-visual materials, and journal articles, to solve research problems (Lin Xiaoying, 2015; Zhao Hui et al., 2013). Similarly, in the field of corporate university research, qualitative research is also a widely used method (Parshakov & Shakina, 2018). For example, Meister (1994) interviewed about 100 corporate university presidents and extracted information including how to design corporate universities, and best practices in corporate university learning programs (corporate citizenship, environmental frameworks, employee self-development) Universities become core constructs such as Learning Labs. This research adopts the coding analysis method, through semi-structured interviews with corporate university practitioners, scholars, and experts, as well as the analysis of internal documents provided by the corporate university, to extract a theoretical model that explains how a corporate university empowers enterprise development.

Grounded theory is a research method of inductive phenomena, which aims to gradually abstract the original data through a spiral cycle of induction, comparison, and analysis to form a conceptualization or theory, which has been widely used in many disciplines including management, sociology, psychology, and education. From the perspective of data analysis logic, it can be divided into classical grounded theory grounded theory, proceduralised theory grounded the three major schools of theory and constructivist's approach to grounded theory have different coding processes (Zhou Qing et al., 2020; Jia Xudong and He Liang, 2021). Among them, the representative scholar of the classical grounded theory paradigm is Professor Glaser. The core idea is to emphasize that the formulation of research questions and the formation of theories are a process of natural emergence, and that no theory is presupposed—research questions should be drawn from the research participants. The researcher should always keep an open mind to deal with the issues concerned by the research object, not preconceived but to look at the phenomenon as a subject (Glaser & Strauss, 1967).

The representative scholars of the programmed grounded theory paradigm are Professor Strauss and Professor Crobin. They mainly focus on the causal relationship in the data, and believe that the data analysis process already contains the logic preset by the researcher, which

is the embodiment of the researcher's subjectivity. The representative scholar of the constructive grounded theory paradigm is Professor Charmaz, who combines the views of the two schools of classical grounded theory and procedural grounded theory, and believes that researchers can recognize the laws in objectively existing data, that is, researchers are subjective and a mixture of objectivity. Therefore, the "programmed grounded theory" proposed by Strauss and Corbin is more suitable for the research question of this study. Among the three schools of grounded theory, this research chooses the analysis process from the *programmed grounded theory paradigm* to emphasize the linking of existing experience and theoretical assumptions through causal relationships, thereby forming new conceptual connotations (Glaser & Strauss, 1967).

This study adopts the coding method guided by the selection theory (Li Liang et al., 2020). This method falls between pre-set coding (establishing an initial set of codes before field research) and inductive coding (no pre-setting any codes at all), and proposes a basic explanatory system for selecting theoretical codes, although it does not satisfy certain requirements. A specific situation, but gives a basic scope for the research question within which the researcher can build a code when generalizing. The advantages of this method are: (1) Selecting the reference theory can focus on the research problem and avoid being "submerged in the ocean of data". (2) The choice of reference theory can provide legitimacy for research findings and avoid getting lost in existing research. (3) Reference theory is helpful for researchers to analyze data efficiently.

3.2.2 Content Analysis

Content analysis is a research method for quantitative analysis of text materials, which is widely used in sociology, political science, psychological cognition, and organizational management (Qiu Junping and Zou Fei, 2004). There are three main types of main application methods: conventional, directed, and summative. All three methods interpret meaning from textual data, with the main differences being the coding scheme, the origin of the code, and the threat to credibility (Hsieh & Klenwo, 2012), but all can make the research results more accurate by quantifying the textual information. It is objective and comprehensive and can realize deep mining of text data. Drawing on existing research (Liu Xiaoxuan and Lin Chenghua, 2019; Guo Tengjun and Fang Li, 2018), this study uses content analysis to analyze internal data,

such as articles of management and project cases, provided by corporate universities (see the data collect form in appendix 5 and 6, and case findings in Section 4), to reveal the performance of China's outstanding corporate universities in China's economic environment.

3.2.3 Questionnaire Research Method

Drawing on the existing research paradigm that combines a questionnaire survey and coding method (Zhang Hong et al., 2022), this study integrates the original data and coding results and draws on the measurement scales of relevant literature to compile the measurement items of the enabling behavior of corporate universities. And repeatedly discuss with scholars in related research fields to conduct preliminary revisions to generate initial measurement items. This process should ensure that the items are extensive and cover the theoretical scope of the corporate university empowerment path, and also follow the principle of simplicity in scale development.

3.3 Multiple Data Collection

In this study, data collection for corporate universities is logically divided into the qualitative data collection stage and the quantitative data collection stage. The qualitative data collection includes the first round (2019) and the second round (2020). The material for the second round of data collection was added in consideration of the fact that the material for the first round may not be sufficient. In qualitative data collection stage, this study mainly establishes theoretical models through coding methods. In quantitative data collection stage, this study verifies the rationality of the theoretical model through the questionnaire. During the qualitative data collection stage, a theoretical model is established using coding methods. In contrast, the quantitative data collection stage involves validating the theoretical model's validity through surveys and questionnaires. Modifications, supplements, and prospective discussions of the model and its pathways are conducted based on the collected quantitative data.

3.3.1 Qualitative Data Collection (2019, first-round)

The qualitative data collection of the corporate universities enabling enterprises development research is divided into two consecutive stages. We firstly choose the corporate

universities as samplings, the first stage includes semi-structured interviews, designing data collection forms, and recycling and analyzing the internal documents submitted by 7 corporate universities. Specifically, the researcher distributed a data sheet with open-ended questions to corporate universities to collect rich first-hand data, involving 78 items (see Section 3.3.1.1 and Appendix 5 in details), which were filled in by the person in charge of the corporate university, and at least two people from each corporate university participated in the data sheet Fill out and check work. The content of the feedback from the data sheet describes in detail the management phenomena existing in the practice of corporate universities and provides some basic information for the subsequent qualitative research and analysis.

The second phase added a semi-structured interview on forward-thinking based on the first phase by collecting data from the other 6 corporate universities. The two-stage data collection process presents the characteristics of continuous iteration and optimization, which conforms to the real-time and retrospective principles of the qualitative research data collection stage (Ingstrup et al., 2021), and also conforms to the recognized continuous data iteration principle of existing qualitative research (Jia Xudong and Tan Xinhui, 2010).

3.3.1.1 Study sample selection

The research object of this round of research is corporate universities, and the research time is mainly from June to October 2019. Based on the availability of data and the typicality of the sample, this study sets the following screening conditions for the case corporate university: (1) The case corporate university must be affiliated with an independent legal person enterprise, and the name includes "a certain university, a certain college or a management school", business schools, research institutes, research institutes, development institutes, excluding traditional corporate training centers. (2) Case Enterprise The enterprise where the university is located must be registered in China and at least have the function of serving the talent training of the enterprise. (3) The case corporate university has been officially listed and operated for not less than two years with the calculation time as of December 31 of the research year. Therefore, the time for the first round of interviews and data collection is 2019, and the time for the second round of interviews and data collection is 2020. This study used the first round of qualitative data to form constructs, next, using quantitative data to verify them (see

appendix 5 to 7). Collecting data from different corporate universities in both qualitative and quantitative stages can improve the quality of this study. (4) The case corporate university has achieved some advanced results in operation, and has been honored or selected into a well-known list, which has some advantages. (5) Case corporate universities come from different industries as much as possible and have some differences in operational characteristics.

Therefore, after screening nearly 3,000 corporate universities in China according to the above principles, a total of 120 corporate universities passed the preliminary screening. The researcher made one-to-one contact with the 120 corporate universities that passed the screening through telephone interviews, and emails. A total of 103 companies were willing to participate in the declaration of "China's Best Corporate Universities" - this is the largest and most comprehensive university ranking in China so far. The longest-running and most influential list of corporate universities, which regularly selects corporate universities with a high performance each year. The 103 companies mentioned above agreed to be interviewed and submitted internal operation materials, such as corporate university rules and regulations, organizational structure, training subjects, profiles of trainers and training teams, descriptions of activities and courses, with a response rate of 85.83%.

Existing research believes that the introduction of expert evaluation procedures can allow more experts to participate in the evaluation process as a group, and obtain experts' professional opinions on research objects (i.e., corporate universities), event phenomena (i.e., development performance) and processes (Gerasimova et al., 2017; Matvieieva et al., 2019). In contrast, the evaluation and analysis by the researcher alone has the potential for bias, and a team of experts can often compensate for possible deviations due to individual opinions. Hence, this study organized more than 10 authoritative experts in the field to conduct reviews.

In order to ensure the authority and scientific of the evaluation, the experts participating in the evaluation were people with professional knowledge or practical experience in the field of corporate universities. The organizer of the ranking sent out an invitation to the following people: (1) Academic experts from corporate universities who have published books or research papers. (2) Consultative experts who have long been engaged in corporate university research, consulting, and training in colleges and universities. (3) Principals of corporate universities that

have won awards in various corporate university-related competitions (such as the executive president of TCL University, and the provost of China Resources University). (4) The person in charge of a recognized corporate university evaluation institution in the field (such as the founder of China Learning Organization Network, the rotating vice chairman of the China Corporate university Federation). Specifically, all experts are free of conflicts of interest and come from third-party organizations (such as trade associations and universities) or corporate universities that do not participate in the judging. That is, neither the organizers of the rankings nor the study paid the experts. Expert review tasks for corporate universities include reading corporate university submissions, grading corporate university performance, writing reviews of each corporate university, and assisting with follow-up field and telephone surveys. Based on previous selection by experts, this study further selected the most typical research samples through two steps:

The first step. To ensure the selected corporate university has typical benchmarking, this research systematically organizes the honors issued by domestic and foreign authoritative institutions for corporate universities (see Table 3-1 and Table 3-2), as well as the latest vision, mission, and values of corporate universities (see Appendix 2 and 3 for details). Because the honor of the corporate university represents to a certain extent the recognition of the operation of the corporate university by external stakeholders; while the vision, mission, and values represent the internal expectations from corporate universities. Hence, this study excluding those corporate universities with unclear top-level strategies such as honors and their own vision.

The second step. To ensure that the best corporate university selected as a typical case has greater research value, which means that the research sample needs to have the following characteristics (Martin & Eisenhardt, 2010; Xu Hui et al., 2020): (1) Timeliness The questions requiring this study are of current importance. The information submission form filled in by the case corporate university, as well as the teaching plan, teaching courseware, management system, and other internal operation materials submitted for reference, are all the latest operation materials and experience summaries. (2) The requirement for detailed information reduces information asymmetry. The selection of the best corporate university adopts the

principle of voluntary declaration and filling. Some data submission forms with too few words have the problem of weak effective information and do not have high research and analysis value. The case companies selected for this study all have more than 20,000 Chinese words in their declarations to ensure that the follow-up case analysis is comprehensive and detailed. This study not only relies on the number of words to judge the quality of the information form submitted by enterprises, but also refers to other supporting materials. There are also cases of corporate university submit less than 20,000 Chinese words, but provide other high-quality supporting materials. (3) Diversification of enterprise ownership and industry, that is, there is a certain degree of differentiation between cases, which is convenient for abstracting theories from common laws.

All corporate universities should provide information from five aspects: (1) the mission and functional orientation of the corporate university, (2) the training object of the corporate university, (3) the organizational structure of the corporate university, (4) the management mechanism of the corporate university, and (5) the development process and the Excellent results.

To sum up, during the first round of the qualitative data collection phase, a total of 7 corporate universities passed the screening (as shown in Table 3-1), of which: (1) These excellent corporate universities are at the leading level in terms of overall level, online learning platform construction, learning project design and operation capabilities, and talent development practices. (2) All of them have submitted a wealth of internal operation and management information; (3) The affiliated enterprises take into account the ownership types of state-owned enterprises and central enterprises, private enterprises, and three-funded enterprises, including 5 state-owned or central enterprises (China Mobile University, China General Nuclear University, ZTE Corporation) College, China Resources Gas College, China Railway Fourth Bureau Group Corporate university), 1 foreign-funded enterprise (Dongfeng Nissan University) and 1 private enterprise (Sany Corporate university). (4) The industry covers information transmission, software and information technology services, manufacturing, agriculture, energy, and other industries.

Table 3-1 The basic situation of the first round of case corporate universities (7)

No.	Basic information of corporate university					Basic information of the affiliated company					
	Corporate university name	Year established	Number of full-time staff	Number of part-time staff	Number of internal trainers	Enterprise name	Established time	Location	Enterprise type	Industry type	Enterprise employee count
1	China Mobile Academy	2000	100	0	35	China Mobile Communications Group Co., Ltd.	2000	Beijing	central / state-owned enterprise	Information transmission, software and information technology services	500000
2	CGN the University	2013	230	2545	220	China General Nuclear Power Corporation	1994	Guangdong Province	central / state-owned enterprise	Electricity, heat, gas	40000
3	ZTE College	2003	200	400	7000	ZTE Corporation	1985	Guangdong Province	central / state-owned enterprise	manufacturing	100000
4	Dongfeng Nissan University	2013	15	0	1000	Dongfeng Motor Co., Ltd. Dongfeng Nissan Passenger Vehicle Company	2003	Guangdong Province	Three foreign-funded enterprises	manufacturing	18708
5	SANYO Corporate University	2003	20	-	440	Sany Group Co., Ltd.	1989	Hunan Province	Private Enterprise	manufacturing	10000
6	China Resources Gas College	2012	11	-	-	China Resources Gas (Group) Co., Ltd.	2007	Guangdong Province	central / state-owned enterprise	Electricity, heat, gas	45504
7	China Railway Fourth Bureau Group Corporate university	2016	26	0	21	China Railway Fourth Bureau Group Co., Ltd.	1950	Anhui Province	central / state-owned enterprise	construction industry	23000

Table 3-2 The first round of case corporate university awards

corporate university	years	Awarding unit	Awards
China Mobile Academy	2015	ATD Talent Development Association	Excellence in Practice Award
	2016	China E-Learning Industry Conference	China E-Learning Industry Excellence Implementation Award
	2016	China Talent Development Forum	Outstanding Corporate University Award
	2017	Xinhua Newspaper Media Group "Training" magazine	Outstanding Corporate University Award
	2018	School of Overseas Education, Shanghai Jiaotong University	China's Best Corporate University Outstanding Contribution Award
	2019	Central Enterprise Talent Training Honorary Selection	Advanced Network Academy (Online Platform)
China General Nuclear University	2015	China Enterprise Training and Development Annual Conference	China Excellent Corporate University Elite Award
	2016	China Enterprise and Education Development Conference	China's advanced enterprise education unit
	2017	Harvard Business Review	"CGN Manager Learning Map Project" won the Ram Charan Management Practice Award Excellence Award
	2018	China Enterprise Executive Training and Development Alliance	Advanced Corporate University of the Year
ZTE College	2017	School of Overseas Education, Shanghai Jiaotong University	China's Best Corporate University Trend Leading Award
	2018	Xinhua Newspaper Media Group "Training" magazine	Top 50 National Corporate university
Dongfeng Nissan University	2017	CSTD China Talent Development Community	National Learning Design Competition Silver Award
	2018	Guangdong Talent Development and Management Research Association	Talent Management Innovation of the Year Award
	2019	Xinhua Newspaper Media Group "Training" magazine	Top 50 Chinese Corporate universities
Sany Group corporate university	2018	The 14th China Enterprise Education Top 100	China's Most Valuable Corporate university, China's Top 100 Enterprise Education Advanced Units
	2019	China Academy of Management Sciences	Top 10 National Competitive Private Education Brands in 2019
	2020	Xinhua Newspaper Media Group "Training" magazine	China's Model Corporate university
	2020	2020 Asia Education Forum Annual Meeting	Top 50 National Integration of Industry and Education in 2020
China Resources Gas College	2018	School of Overseas Education, Shanghai Jiaotong University	Best Corporate University in China
	2019	Xinhua Newspaper Media Group "Training" magazine	Top 50 Chinese Corporate universities "Best Learning Project Award", "Top 100 Excellent Micro Courses" in China Micro Course Competition
China Railway Fourth Bureau Group Corporate university	2016	All-China Federation of Trade Unions General Office	National Employee Education Demonstration Site
	2019	Hosted by China Distance Education Magazine and China Corporate university Association	China Enterprise Learning and Talent Development Innovation Award
	2019	Organizing Committee of China's Top 100 Enterprises in Education	China's Most Valuable Corporate University
	2019	Xinhua Newspaper Media Group "Training" magazine	Top 50 Chinese Corporate universities

3.3.1.2 Data collection process

Given the availability of data, currently corporate universities do not publicly disclose their own operational information, so interviews and requests for internal documents from companies are key sources of data. The internal operation data collection of the corporate university is submitted by the responsible team of the corporate university responding to this study. Since 2013, the researcher has continuously tracked the management practices of the best corporate universities in China and has formed a complete set of evaluation indicators (Li Lin et al., 2015). Based on prior evaluation indicators, this study designed an information collection form for the best corporate universities, specifically, including 18 questions about the overall details of corporate universities (full name of the company, time of establishment, nature of the company, scale of the company, affiliation) industry, institutional attributes of corporate universities, number of teachers, service targets, development history), aiming at 60 problems in strategic docking, organizational coordination, business integrity, knowledge integration, resource matching, lean operation, value demonstration. All questions are open-ended questions, and there is no word limit on the filling of the case corporate university, but the information form is required to be filled out by the actual person in charge/team of the corporate university, and the contact information of one filling person and the contact of the top person in charge of the corporate university is provided Ways to conduct online or face-to-face research and communication. The entire information collection form should take no more than 2 months to complete.

The information collection form for internal members of the enterprise - requiring staff to truthfully fill in the various tasks of the corporate university - the shortest return time is 3 days and the longest is 20 days. Therefore, diverse data sources can ensure data complementarity and cross-validation to improve the validity of data analysis conclusions (Zhang Hong et al., 2022). In the end, a total of about 270,000 words of information feedback forms were submitted by the case companies that passed the screening, mainly for the detailed structured questions, including both descriptive text content and pictures, audio, and video materials as supplements (see Table 3-3).

Table 3-3 The first round of case corporate university information submission and interview records

Source	Independent information submission		Semi-structured interview		
	Commit type	Word count	Cumulative number of interviews	Cumulative interview time	Participate in the interview department
China Mobile Academy	documents, pictures	60315 words	3	234 min	3
China General Nuclear University	documents, pictures	25881 words	5	380 min	4
ZTE College	Documentation, Videos	16303 words	4	105 min	3
Dongfeng Nissan University	Documentation	22588 words	5	336 min	2
Sany Corporate University	documents, pictures	26004 words	3	65 min	3
China Resources Gas College	documents, pictures	45891 words	4	479 min	4
China Railway Fourth Bureau Group Corporate university	Documentation, Videos	37272 words	10	417 min	3

Semi-structured interview materials of the case corporate university were investigated and sorted by the researcher successively (as shown in Table 3-3). Each interview involved between three and seven corporate university staff or industry experts, and generally lasted about two to three hours. Considering the professionalism and uniqueness of corporate universities, the intended research objects of the semi-structured interview include both internal and external aspects of the enterprise.

The internal interviewee are senior managers or team leaders of the enterprise, who can actually be responsible for enterprises' strategy and of corporate universities' operation. They are: (1) The corporate university or training professionals (e.g., executives dean, director of college, executive vice president, head of business school); (2) Commissioners of the human resources department of the enterprise and related departments of talent development (e.g. human resources director, assistant to the general manager of HR, and director of talent development and training); (3) Heads of departments related to corporate cultures (e.g, corporate culture supervisors and corporate culture specialists); and (4) Specialists in the extended field of corporate universities (e.g, the manager of the capability development center and the director of platform operations).

Yet, the external interviewee mainly includes expert judges who participated in the selection of the best corporate universities in China, and professional scholars with certain professional knowledge in the field of corporate university research.

This study uses semi-structured interviews with the respondents to collect the information about development of enterprises empowered by different corporate universities and professional perspectives on the development path of corporate universities, such as "what do you think are the core elements driving corporate university empowerment" and "which corporate universities in your corporate university are strategically capable of enabling corporate development" (see appendix 2 for details). It was done in real-time, mainly asking them for their views and judgments on corporate university empowerment activities. Most of the interviewees believe that they are happy to participate in the survey because they can extract and summarize the work of corporate universities based on this, which is also conducive to promoting thinking and benchmarking.

Overall, in the first round of qualitative data, this study follows the principle of "triangulation verification", and collected a total of about 230,000 words of internal documents, as well as 33 hours of records obtained by interviewing 34 people. The data comes from various sources, especially the seven corporate universities that have provided detailed internal documents. Data, curriculum system, organizational structure description. Complete original data is conducive to subsequent triangulation verification.

3.3.2 Qualitative Data Collection (2020, second-round)

This study designed the qualitative data collection at the very beginning including stage one and stage two. The purpose of the first stage is to provide raw material for the coding process, while the purpose of the second stage is to verify that the theoretical framework is useful to explain enabling behaviors from other Chinese corporate universities. Hence, the second round of data collection was a planned part of the research to provide theoretical saturation and had been scheduled for a year after the first round. To avoid autocorrelation, this study collected data from the other six corporate universities and included discussion of the additional question on future direction and why this was introduced in this section.

3.3.2.1 Sample selection

Grounded theory analysis methods emphasize the importance of "continuous comparison" (Suddaby, 2006), which refers to the synchronization of data collection and analysis, that is, while collecting and analyzing, continuously refining and revising the theory, data collection and analysis is always an interactive cyclic process. If newly collected Data can be generalized by existing categories without generating new categories; on the contrary, if newly collected data can generate new categories, then return to the step of revising the theory and incorporate new categories into the analysis (Wang Lu & Gao Peng, 2010). The grounded theory analysis method does not emphasize the assumption of a priori but believes that new data means the possibility of new theories (Glaser & Strauss, 1967).

This study uses sample replacement to improve validity (Garcia et al., 2021). The second round of qualitative data collection needs to focus on revealing the trend analysis of the future development direction of corporate universities based on the latest data available to corporate universities and improve the endowment of corporate universities. A theoretical model of enterprise development. Specifically, the object of the second round of qualitative data collection is still corporate universities. The data collection time is mainly from May to October 2020. The selection criteria of the samples are the same as in the first round. There are a total of 78 corporate universities. In response, there were finally 6 corporate universities as samples in this round, namely New Hope Business School, China Construction Bank University, Geely Corporate university, Red Star Macalline Group Management School, Robam University, and Taikang Business School (see Table 3-4 and Tables 3-5), with no intersection with the first round of case corporate universities. The reason for the decrease of cases in this round is that we deleted some cases that appeared in the first and second round at the same time to ensure that the samples did not overlap. Among those six cases, the ownership type of the enterprises affiliated with the case corporate university includes both state-owned enterprises (CCB University, Taikang Business School) and private enterprises (New Hope Business School, Geely Corporate university, Red Star Macalline Group Management School, Robam University); The industry covers four major industries: agriculture, forestry, animal husbandry and fishery, manufacturing, finance, wholesale and retail.

Table 3-4 The basic situation of the second round of case corporate universities (6)

No.	Basic information of corporate university					The basic situation of enterprises					
	Corporate university name	Established years	Number of full-time staff	Number of part-time staff	Internal trainer number of people	Enterprises Name	Year of establishment	Location	Enterprise type	Industry type	Enterprise employee count
8	New Hope Business School	1995	14	0	426	New Hope Co., Ltd.	1998	Sichuan Province	Private Enterprise	Agriculture, forestry, animal husbandry and fisheries	102558
9	CCB University	2018	472	800	110	China Construction Bank Corporation	1954	Beijing	central/state-owned enterprise	Financial industry	340000
10	Geely Corporate university	2015	40	5	0	Zhejiang Geely Holding Group Co., Ltd.	1986	Zhejiang Province	Private Enterprise	Manufacturing	1 00000
11	Red Star Macalline Group School of Management	2008	11	2014	2014	Red Star Macalline Home Furnishing Group Co., Ltd.	2007	Shanghai	Private Enterprise	Wholesale and retail trade	30000
12	Robam University	2014	10	0	118	Hangzhou Robam Electric Co., Ltd.	1979	Beijing	central/state-owned enterprise	Manufacturing	4800
13	Taikang Business School	August 2016	29	195	1199	Taikang Insurance Group Co., Ltd.	1996 8	Beijing	state-owned enterprise	Financial industry	800 000

Table 3-5 Awards of the second round of case corporate universities

Corporate university	Years	Awarding unit	Awards
New Hope Business School	2012	The 11th China International Talent Exchange Conference	2012 Best Practice Award of China Enterprise Business School
	2016	Xinhua Newspaper Media Group "Training" magazine	2016 China Talent Development Elite Award: Excellent Corporate university, Best Enterprise
	2017 2018	Xinhua Newspaper Media Group "Training" magazine	Best Enterprise for Talent Development in China, Excellent Corporate university
	2019	China Business Foundation, Training Magazine, ISPI (International Performance Improvement Association)	Performance Improvement Best Practice Award
	2020	Cloud Academy, China Europe Business Review	Organization and Learning Development Yuntu Award
CCB University	2019	School of Overseas Education, Shanghai Jiaotong University	China Corporate university Trend Leading Award, Most Influential Corporate university
	2019	"China Distance Education" magazine	Outstanding Organization Award
	2020	Xinhua Newspaper Media Group "Training" magazine	China's first batch of exemplary corporate universities
	2020	China Business Management Research Association, Social Responsibility and Sustainable Development Professional Committee	Top 10 Cases of Corporate Sustainable Influence Brands in 2020
	2020	Financial Professional Committee of China Education Technology Association	Excellent Works First Prize, Excellent Project Organization Award
	2020	China Enterprise Executive Training and Development Alliance	Advanced Corporate university
Geely Corporate university	2017	School of Overseas Education, Shanghai Jiaotong University	China's Most Growing Corporate University
	2018	Xinhua Newspaper Media Group "Training" magazine	Excellent Corporate University
	2019	China Enterprise Culture Research Association	2019 National Cultural Outstanding Achievement First Prize
Red Star Macalline Group School of Management	2012	State Administration of Foreign Experts Affairs, Shenzhen Municipal People's Government	China Enterprise Business School Best Leadership Award
	2013	Xinhua Newspaper Media Group "Training" magazine	The best enterprise for talent development in China
	2014	China Public Welfare Training Alliance	Best Universities for Philanthropy Enterprises in China
	2015	China Chain Corporate university Alliance	Top 10 Corporate Universities
	2016	Xinhua Newspaper Media Group "Training" magazine	Excellent Corporate University of Chinese Enterprises, Best Study Program, China Gold Lecturer
	2019	China Chain Store & Franchise Association	CCFA Outstanding Corporate University Best Innovative Practice Award, CCFA Industry Craftsman Award
Robam University	2018	School of Overseas Education, Shanghai Jiaotong University	Best Corporate University

	2018	Xinhua Newspaper Media Group "Training" magazine	Top 50 Chinese Corporate universities
	2019	"China Distance Education" magazine, China Corporate university Association	2019 China Benchmarking Corporate University Award
Taikang Business School	2018	Xinhua Newspaper Media Group "Training" magazine	Top 50 Chinese Corporate universities
	2019	China Distance Education Magazine, China Corporate university Association	2019 China Benchmarking Corporate University Award

3.3.2.2 Data collection

Similar to the data collection process in the previous round, the participants filled out the information collection form "Application Materials for the Best Corporate Universities in China" to investigate the overall status of corporate universities organized by the corporate system. Considering that when enterprises fill in the information questionnaire by themselves, there will be factors that subjectively choose the advantages and fill in, which only reflects the advantages and highlights of the work, not the shortcomings and difficulties of the work. The information collection form is designed with an open-ended question "The focus and action direction of the corporate university in the next stage". This question appears after "the stage summary of the corporate university's development process", the purpose is to collect the corporate university's future development direction. The information collection form for this round includes the following contents (as shown in Tables 3-6):

Table 3-6 The second round of case corporate university information submission and interview records

Source	Independent information submission		Semi-structured interview		
	Commit type	Word counts	Cumulative number of interviews	Cumulative interview time	Participate in the interview department
New Hope Business School	documentation, Videos	15832 words	5	352 min	2
CCB University	documentation	21586 words	7	578 min	3
Geely Corporate university	documentation	14982 words	3	340 min	2
Red Star Macalline Group School of Management	documents, pictures	31831 words	2	104 min	1
Robam University	documents, pictures	15784 words	6	403 min	3
Taikang Business School	documentation	17684 words	3	116 min	2

(1) Basic information research on companies and corporate universities, including the full name of the company, the time of establishment, the nature of the company, the scale of the

company, the industry it belongs to, the institutional attributes of the corporate university, the number of teachers, service targets, and development history, 18 questions to collect basic information (same question as last round of data collection).

(2) The investigation of the corporate university's operation system, including 17 questions to collect detailed information, including the corporate university's value orientation, project system, curriculum system, teacher system, technical system, support system, management system, and collaboration system.

(3) Research on the forward-looking analysis ideas of corporate universities to empower corporate development. Based on the theoretical model constructed by the previous round of data collection and analysis, two main theoretical aspects are designed. One is to investigate the practical practices of corporate universities in this aspect, and the second is to investigate the forward thinking and related assumptions of corporate universities in this area (see more details in Chapter 4). This is a specific open-ended question. It is a setting of a new information item, which avoids the respondent's strengths and weaknesses in the previous round of data submission, and does not mention or at least mention his own shortcomings and areas for improvement, which can reflect a summary of the inadequacies of previous work in corporate universities. In addition, this open-ended question can reflect the future development ideas and strategic concerns of corporate universities to a certain extent. Therefore, this open-ended question is the focus of this round of grounded analysis, which can further supplement the theoretical model constructed before.

Overall, the second round of the qualitative data collection phase completed the first-hand data collection of 6 corporate universities, including about 120,000 words of internal documents and about 16 hours of interview records provided by 26 people. The purpose of the qualitative data collected in the second stage is to add some analysis that were ignored in the first round. It should be noted that the qualitative data collection process of the first and second rounds is not separated from each other (Jia & Heng, 2021).

3.3.3 Quantitative Data Collection (2021)

The quantitative data collection stage builds upon the preceding two qualitative data collection stages. In other words, the data gathered in the final stage is utilized to validate the conclusions drawn from the initial two stages and to supplement the model. Following methods suggested by Alonso-Diaz and Yuste-Tosina (2015), this study employs multi-source data collection techniques, including questionnaire surveys and interviews. The study gathers quantitative data on enterprise development empowered by corporate universities through self-designed survey questionnaires (the complete questionnaire is available in Appendix 7, and the results are presented in Section 5.3). The data collection process involves the following steps:

First, this study designed the items of the questionnaire according to the constructs generated in the coding analysis process. To improve the reliability, a dozen industry experts were invited to analyze the same text according to the same evaluation criteria. Those experts were mainly independent scholars or government workers, not practitioners of corporate universities, and have no conflict of interest with all corporate universities as cases in this study. This study selected experts were according to the researcher's prior cooperative network to ensure that the experts had mastered the knowledge of the corporate university. After studying and analyzing the data of outstanding corporate universities, describe their most distinctive and important practices to ensure the accuracy of the research constructs to the greatest extent possible.

Second, the questionnaire is distributed to practitioners of corporate universities, corporate leaders and employees with corporate universities, and relevant scholars and experts. The questionnaire is used to verify whether the construct of corporate universities enabling organization proposed by the theoretical model is worth promoting. Therefore, it is the best choice to distribute the questionnaire to corporate university practitioners and experts with relevant knowledge. Among them, except for some targeted questionnaires that were directly delivered to the staff of corporate universities, the rest of the electronic questionnaires were distributed through WeChat public accounts, WeChat, corporate university websites, and other

channels. The questionnaires have indicated whether the answer information is allowed to be used for corporate university research, and a total of 180 questionnaires.

Third, the questionnaire mainly collects the data related to the driving factors and demographic variables of the corporate university empowerment of the respondents and distributes it in the form of an electronic version, which is filled out by the respondents themselves. A total of 180 questionnaires were distributed in this study, and 151 valid questionnaires were recovered, with a response rate of 83.89%. Among them, a total of 137 questionnaires were selected as "Yes" for the question "Do you have a corporate university", which ensured that all the samples were in corporate universities, and 14 questionnaires that did not meet this item were excluded. According to the answering time and the degree of overlap, 18 questionnaires with an answering time of less than 1 minute and 18 questionnaires with consecutive identical answers were excluded, and the final valid questionnaire was 119. In the final analyzed sample, individual statistical characteristics and organizational statistical characteristics are as follows (as shown in Tables 3-7):

Table 3-7 Individual statistical characteristics of the respondents

Variable	Category	Frequency	Percentage
Gender	Male	75	63.0
	Female	44	37.0
Age	20 ~ 30 years old	16	13.4
	30 ~ 40 years old	47	39.5
	40 ~ 50 years old	42	35.3
	over 50 years old	14	11.8
Education level	High school and below	0	0
	college	5	4.2
	Undergraduate	55	46.2
	Postgraduate	55	46.2
	PhD student and above	4	3.4
Working years	less than 1 year	7	5.9
	1-3 years	10	8.4
	3-5 years	14	11.8
	5-10 years	18	15.1

	10+ years	70	58.8
Type of enterprise	central enterprise	54	45.4
	local state-owned enterprises	10	8.4
	Private Enterprise	44	37.0
	Three foreign-funded enterprises	6	5.0
	Other	5	4.2
Type of department	Production department	6	5.0
	Technical department (R&D, testing)	8	6.7
	Functional departments (finance, human resources, public relations, administration, legal affairs)	72	60.5
	Marketing department (marketing, sales)	14	11.8
Team size	5 people or less	3	2.5
	5-10 people	21	18.5
	10-20 people	34	28.6
	20 people	60	50.4
Is a manager/leader	Yes	57	47.9
	No	62	52.1

(1) Gender characteristics: males account for 63 % and females account for 37%.

(2) Age characteristics: 13.4% in the 20-30 age group; 39.5% in the 30-40 age group, the highest proportion; 40-50 years old 35.3%; over 50 years old 1 1.8%.

(3) Characteristics of education level: high school and below accounted for 0%, junior college accounted for 4.2%, undergraduate accounted for 46.2%, master's degree accounted for 46.2 %, doctoral degree and above accounted for 3.4%.

(4) Characteristics of working years: 5.9% of the survey respondents have worked for less than 1 year, 8.4 % have worked for 1 to 3 years, 11.8% have worked for 3 to 5 years, and 11.8% have worked for 5 to 10 years. It accounted for 15.1%, and the proportion of more than 10 years was 58.8% and the highest proportion.

(5) Characteristics of the type of enterprises where they are located: central enterprises account for 45.4 % and the highest proportion, local state-owned enterprises account for 8.4%, foreign-funded enterprises account for 5%, private enterprises account for 37%, and other types of enterprises account for 4.2%.

(6) Type of department: production department accounts for 5%, technical department (e.g., R&D, testing) accounts for 6.7%, functional department (e.g., finance, human resources, public relations, administration, legal affairs) accounts for 60.5% and the proportion of the highest, the marketing department (e.g., marketing, sales) accounted for 11.8%, and other departments accounted for 16%.

(7) Characteristics of team size: a team with 5 or fewer members accounts for 2.5%, a team with 5-10 members accounts for 18.5%, a team with 10-20 members accounts for 28.6%, and a team with 20 or more members accounts for 0.4% and the highest.

(6) Managerial characteristics: Managers accounted for 47.9%, and non-managers accounted for 52.1%, basically the same.

Overall, the quantitative data collection phase attempts to validate the plausibility of the theory constructed by coding through empirical methods, further improving reliability and validity. Therefore, the researcher collected the judgments of professionals in the corporate university field (e.g., corporate university administrators and employees, industry experts) by distributing questionnaires. In this study, 180 self-designed questionnaires were distributed, and 119 valid questionnaires were obtained —this will provide evidence in the robustness test in Chapter 5.

3.4 Chapter Summary

Chapter 3 describes in detail how this research was carried out. First, this study introduces methods, including coding methods, content analysis methods, and questionnaire survey methods. Secondly, this study introduces how the data are collected in the qualitative and quantitative research stages, including what the sample is, how the data collection process is carried out, the response of the research subjects, and the recovery of actual data. It should be noted that this research phase involved three rounds of data collection (qualitative data collection for the first and second rounds, and quantitative receipts for the third round) to ensure that the theory constructed through the subsequent coding analysis was saturated. In the next chapter, this study will introduce 13 corporate universities as case studies and conduct an initial analysis using the content analysis method.

4 CASE DISCOVERY AND TEXT ANALYSIS

The purpose of this chapter is to provide details of the preliminary analysis of the case corporate university data from the data collection stage and to present a three-dimensional image of the corporate university from multiple perspectives. First, this study conducted a preliminary analysis of all 13 corporate universities as cases, which can better analyze the practice of corporate universities. Secondly, the interview materials, archive materials, and third-party data of the case corporate university are summarized, which provides a research basis for the next text analysis and coding analysis.

4.1 Case Findings Based on Best Corporate University Practices

This research collated data focusing on case characteristics and practical results, mainly internal materials and interview materials provided by corporate universities (see *Supplement materials*), including a small amount of third-party media reports. As discussed in section 3.3, this research carries out a preliminary analysis from five aspects: (1) the mission and functional orientation of the corporate university, (2) the training object of the corporate university, (3) the organizational structure of the corporate university, (4) the management mechanism of the corporate university, and (5) the development process and the Excellent results.

4.1.1 China Mobile Academy

First, in terms of the mission and functional orientation of the corporate university, China Mobile Academy serves as a training base for elite talents of the group company, a collaborative hub for technology and business, a learning platform for employee development, a management center for course teachers, and an online portal and ideas for knowledge management. The carrying channel of communication shoulders the mission of cultivating middle and senior management talents, professional and technical talents, and business backbones for the group company, and is a platform for talent elites to learn and exchange ideas and inspire innovative ideas.

Second, in terms of the training objects of corporate universities, China Mobile Academy has always focused on the quality of training. Since its establishment, it has trained management elites, business backbones, and technical experts for China Mobile headquarters and provincial

companies. In addition, China Mobile Academy also relies on the online university and the APP platform of China Mobile Network to carry out online learning for all employees of the group, supporting the company's business training and talent development with low cost and high efficiency. The training objects of China Mobile Academy are mainly divided into senior management talents/reserve cadres, heads of prefecture and city companies, outstanding county managers/team leaders, technical/business backbone and core employees, headquarters employees, and internal trainers. Among them, the internal trainer team is divided into two groups: management personnel trainers and technical business expert trainers, including not only the leadership team of the enterprise but also technical experts from various business lines and business departments. As shown in Figure 4-1, at present, there are 86 group-level technical business experts' internal trainers, nearly 10,000 provincial-level internal trainers, and nearly 10,000 prefecture-level internal trainers; according to professional lines, technical business internal training Teachers are divided into general management, technology, and business. The overall business lecturers are mostly, accounting for more than half, and the number of internal lecturers in each professional line of group-level internal trainers is basically the same.

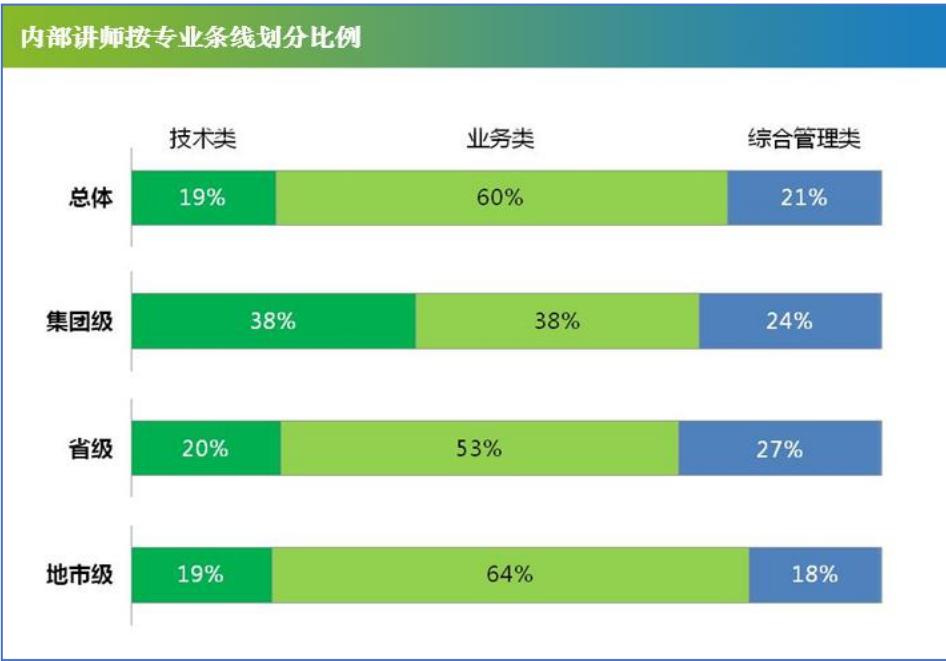


Figure 4-1 Distribution ratio of internal lecturers of China Mobile Academy

Especially, in regard to empowerment, China Mobile University has designed task model for internal trainers, including systematic development path, online and offline learning

resources and standardized ability improvement training, which is conducive to promoting the continuous empowerment and development of group-level internal trainers. As a regular online empowerment course, the *Wood Bird Camp* (in Chinese, “啄客大本营”) has effectively improved the curriculum development ability of internal trainers.

“Original text in Chinese: 中国移动学院基于内训师的TDFC角色任务模型，设计了系统化的发展路径，开发了线上线下学习资源，形成了标准化的线上线下相结合的能力提升项目训练营，赋能集团所有内训师的能力发展(Please see more details in Supplement Material S2.1).”

Third, in terms of the organizational structure of the corporate university, the leadership team of China Mobile Academy is responsible for the overall work of the corporate university, leading the college to efficiently identify key elements of organizational learning, systematically plan and integrate resources from all parties, and promote organizational learning and employee development. As shown in Figure 4-2, from the perspective of the company, the academy connects the subsidiary's training department, group business department, and group human resources department respectively: (1) The academy is responsible for integrating the high-quality training courses and internal trainer resources of each subsidiary company; (2) the college is a partner of the business department of the group, providing professional training services for the business department; (3) the college is the executive department of human resources management of the group, responsible for providing and implementing training programs for the business department As well as managing company-wide training resources. From the perspective of the internal structure of the corporate university, China Mobile Academy has set up four major teaching departments: the Teaching Department, the Online Learning Department, the Party School Education Department, and the Teaching Support Department, service management, operation, and maintenance support departments, and comprehensive and financial functional departments.

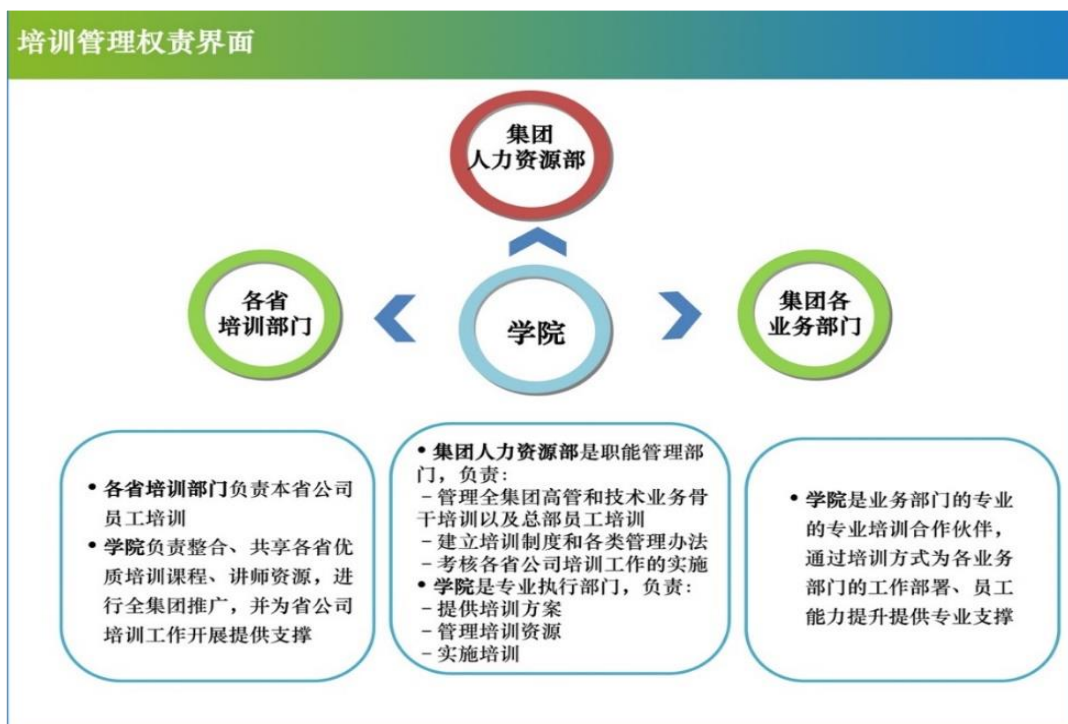


Figure 4-2 China Mobile Academy training management and responsibilities Illustration

Fourth, in terms of the management mechanism of corporate universities, following the ISO10015 training quality management standard, China Mobile College has conscientiously summed up the training operation experience, comprehensively strengthened the teaching quality management, and continuously strengthened the basic management work. It has formed a set of standardized operation management mechanisms, which can Realize fine management of training programs. As shown in Figure 4-3, the training management mechanism includes training project management, student assessment management, training expense management, and training file management; resource management includes curriculum introduction and development management, curriculum application promotion management, internal and external trainer management, and teaching resource management; Teaching and research management includes online learning platform management, R&D project management, learning method management, and cutting-edge theoretical research; educational affairs tool management includes training manual development management, project manual development management, standard tool development management, and system platform use management. Judging from the feedback from internal customers and external customers, this management mechanism has effectively improved its management collaboration efficiency and training

effectiveness. In addition, China Mobile Academy has won the Corporate University Management Practice Award 4 times since 2016, which demonstrates that the corporate university's management mechanism is at the forefront of the industry, with effectiveness and advantages.



Figure 4-3 Illustration of the operation mechanism of China Mobile Academy

(1) Establish a standardized institutional system. China Mobile Academy has complete systems and effective operation. Taking the teaching and training system as an example, China Mobile Academy has formulated the "Teaching and R&D Project Management Manual ", "Course Development Management Manual", "Course Resource Manual", "Lecturer Remuneration Management Manual", "Student Management Regulations", "Online University External Relations". Cooperation Project Management Measures" and "Party School Teaching Management Regulations" system to ensure that all work has rules to follow and unified management.

(2) Establish a whole-process evaluation management system for training (Require - Plan - Implement - Output, RPIO). Effectively evaluating the company's overall training effectiveness and improving the company's overall training effectiveness is an important way

for corporate universities to help companies develop. China Mobile Academy draws on the Analysis-Design-Development-Implementation-Evaluation (ADDIE) training project management process - a method for systematically developing teaching. It mainly includes: what to learn (setting of learning objectives), how to learn (application of learning strategies) and how to judge that learners have achieved learning results (implementation of learning assessment). China Mobile Academy has continuously improved its training implementation idea. As shown in Figure 4-4, China Mobile Academy has built RPIO in practice to realize the whole-process evaluation and management of training quality and effect, and combined with the actual situation of the company, targeted innovation design management from three levels of the company, business department and project tools and methods. Since its application in 2013, RPIO has been used every year to analyze annual training needs research, formulate group training plans and group training projects, and has been recognized and supported by business departments and group leaders.

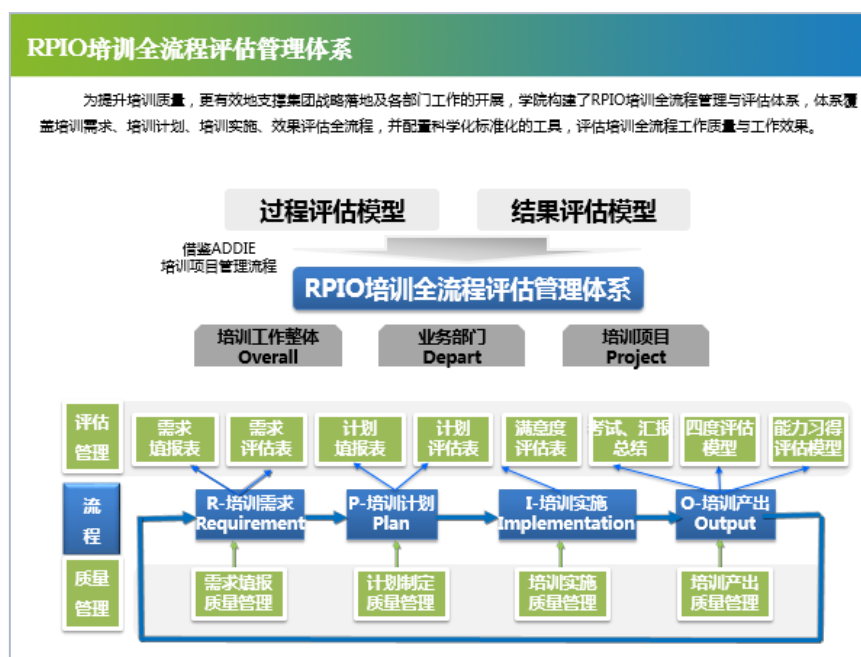


Figure 4-4 China Mobile University RPIO whole process evaluation management model

(3) Design the strategic synergy mechanism of "corporate university-enterprise". According to the corporate university's organizational structure and the actual needs of the implementation of training programs, China Mobile Academy has set up a strategic collaboration mechanism for business departments, equipped with regular interface trainers for

each internal business department and external customers to support business needs and be responsible for annual training programs. This mechanism further regulates the support activities of corporate universities for business departments - China Mobile Academy requires that all needs provided by the business will be responded to within 1 day, The preliminary plan will be given within 3 days, and the overall plan will be given within 7 days. At the beginning of each year, China Mobile Academy will also cooperate with the human resources department of the group company to organize a centralized seminar for the training associates of the business department to precisely focus on the training needs. As shown in Figure 4-5, in the regular seminars, China Mobile Academy has established a working mechanism of "Two Brings", which requires the trainees to submit to the corporate university "a person who has dealt with a certain problem at work" before participating in the training. Best Practices (Cases)" and "A Tough Problem Facing Current Jobs". In response to the best practice cases submitted by students, China Mobile Academy sorts summarize, and produces online courses to form the best-case presentations; and for difficult problems, China Mobile Academy's trainers will lead the Academy to conduct in-depth discussions to form feasible solutions. Program. Whether it is the best case or the solution, it will be published on the online learning platform of China Mobile Academy, not only for the student's learning reference but also for the decision-making reference of the group leader.



Figure 4-5 Examples of regular seminars at China Mobile Academy

Fifth, in terms of the development history and outstanding achievements of corporate universities, China Mobile Academy is a teaching and training institution directly under the China Mobile Group. Table 4-1 shows the milestone events since the establishment and

operation of the college, which shows that since its establishment in 2000, the college has experienced more than 20 years of development experience, and has continued to explore in terms of company talent training and business development. For example, in 2007, China's Internet technology was still in its infancy, but China Mobile Academy discovered the importance and necessity of online learning, and took the lead in establishing China Mobile Online University, building an online learning platform in an all-round way, which facilitated the smooth transformation of corporate universities. The foundation has been laid; in 2015, China Mobile Academy won the industry's highest honor "ATD Excellent Practice Award" and the "2015 China E-learning Industry Application Innovation Award" issued by the China E-learning Industry Conference, and has won online learning for many years since then. The heavyweight honor is the benchmark and model for the continuous development of online learning in corporate universities.

Table 4-1Memorabilia of China Mobile University

Year	Key events
2000	The reorganization system of the Management Cadre College of the Ministry of Posts and Telecommunications is placed under China Mobile Communications Corporation
2000	The college was renamed as China Mobile Group Corporation Training Center
2004	The college went public with China Mobile Group and was officially renamed China Mobile Communication Management College
2007	China Mobile Online University was established to build China Mobile's online learning platform throughout the network
2012	Introduce mobile learning platform to comprehensively promote mobile learning
2013	According to the needs of operation and development, the college officially changed its name to China Mobile College
2014	Won the "2014 Best Learning Project Award for Chinese Enterprises" issued by "Training" magazine; Won the "Best Corporate University Award in China" issued by the Overseas Education College of Shanghai Jiaotong University
2015	Won the industry's highest honor "ATD Excellent Practice Award"; MOOC project won the "2015 China E-learning Industry Application Innovation Award" issued by China E-learning Industry Conference
2016	Won the 2016 "China E-Learning Industry Excellence Implementation Award" selected by the China E-Learning Industry Conference; Won the "2016 China Talent Development Forum" Annual Outstanding Corporate University Award
2017	Won the "E-learning Industry Application Innovation Award" issued by the 5th China E-learning Industry Conference; the online university has been fully migrated to the cloud platform, and the system has been revised and upgraded

2018	Won the "Second (2018) Central Enterprise Alliance Cup Micro Course Competition Second Prize"
2019	Won the "2018 Best Propaganda Award" in the "Honorary Selection of Talent Training of Central Enterprises", "2018 Advanced Network Academy (Online Platform)" in "Honorary Selection of Talent Training of Central Enterprises", and "2018 Excellent Party Building Training Project"

4.1.2 China General Nuclear (CGN) University

China General Nuclear (GCN) University established by China General Nuclear Group, which was one of China's large state-owned enterprises and exhibited different characteristics from other private enterprises. For example, the direct leader is a "party secretary" (a de facto management position in a Chinese communist party enterprise).

First, in terms of the mission and functional orientation of a corporate university, CGN University takes the origin of management change, boosting corporate strategy, incubating talent multiplication, business development partner, spreading corporate culture, and organizing knowledge precipitation as its mission, and is determined to become a benchmark for corporate talent training.

Second, in terms of the training objectives of corporate universities, the training objects of CGN University include not only the junior, middle, and senior managers of the group but also the employees of various functional departments of the group (such as business, finance, human resources, market development), as well as all kinds of professional and technical personnel required for the development of the Group's nuclear power, nuclear fuel, new energy, finance, nuclear technology, and nuclear power service businesses. At present, the whole group has more than 10,000 various courses and more than 4,500 online courses. In regard to empowerment, the CGN University enable internal employees by summarizing and promoting the cases. Relying on this enabling method, corporate university has explored the case extraction process and tools, which can quickly extract highly operational tools from cases.

(Original text in Chinese: 中广核大学举办了两期集团企业案例开发赋能认证培训班, 共48名集团的内训师参加认证, 并出版了《中广核案例建设二期项目管理案例集》, 相关案例已经应用到管理课程中。某区域公司提出“三式学习法”(分享式、体验

式、启发式）学习方式，探索出案例萃取流程和工具，能够快速从案例中萃取操作性很强的工具方法。某事业部发布了《某某部电厂事件案例汇编》，涵盖各电厂实际案例 170 项。通过案例建设，有效地丰富了培训资源。Please see more details in Supplement Material S2.4).

The online learning platform developed by CGN University is the only training management and learning self-service platform of the Group, and it has been applied to the member companies of the Group. The whole group's training business process, teacher management process, and technology authorization process are all managed through the CGN University training and learning platform. CGN Group employees use PCs or mobile phones to conduct self-help learning and sharing on the CGN training and learning platform (as shown in Figure 4-6).



Figure 4-6 Screenshot of CGN mobile learning APP

Figure 4-7 shows the training and learning time of the whole group from 2014 to 2018: In 2018, the group's annual learning time accumulated to 4.61 million hours, an increase of 5.5% year-on-year, and the per capita learning time was 108 hours, an increase of 5% year-on-year. Therefore, the study courses carried out by the corporate university can cover more than 18,000 employees of the whole group.

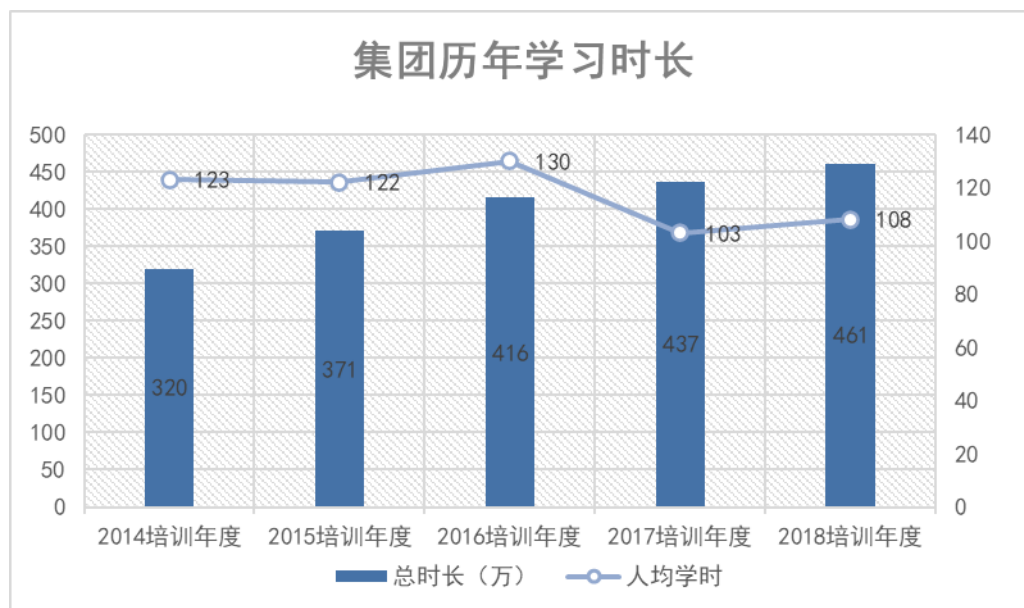


Figure 4-7 Statistics on study time of CGN University

Third, in terms of the organizational structure of the corporate university, due to the nature of the group company as a state-owned enterprise, the president of CGN University is served by the secretary of the party group of the group company, and the director of the school committee is the chairman of the group; the general manager of the group company serves as the first vice president, Deputy Director of the School Board. The organizational structure of CGN University is divided into three levels: (1) CGN University at the group level (including 6 training colleges and training centers), (2) secondary training colleges and directly affiliated training centers (determined by business departments) affiliated to corporate universities, (3) the third-level training center (determined by the business department).

Fourth, in terms of the management mechanism of corporate universities, CGN University is responsible for regularly collecting training needs from various business units and functional departments through face-to-face interviews and distributing questionnaires. Each business unit and functional department trains management cadres for each business unit and functional department conducts general knowledge training, trains certified lecturers, conducts curriculum design, and provides vocational certification training for employees of each business unit and functional department. For example, CGN University is responsible for publishing training management policies and various training management procedures, procedures, and work guidelines at the group level. As a result, member units publish various implementation rules

that fit their departments. With the help of standardized institutional processes, the corporate university has clarified the management and control requirements in the group's training field and the division of responsibilities at all levels, which can ensure the normal operation of the group's huge learning and training system.

Fifth, in terms of the development history and outstanding achievements of corporate universities, Table 4-2 shows the memorabilia of CGN University. Since its establishment in 2005, CGN University has undergone many changes: for example, in 2007, the Corporate University was identified as the group's only training management institution, focusing on providing training services for employees; in 2009, the corporate university was responsible for cultivating The Group's first batch of internal trainers, and formally introduced leadership and employee ability development courses; in 2010, the corporate university started the digital development stage and launched the E-learning platform for online learning; from 2011 to 2015, the corporate university completed its name change. The work has clarified the positioning of the learning and training center undertaken by itself in the group.

Table 4-2 Memorabilia of CGN University

Year	Key events
2005	China General Nuclear Power Group Institute of Nuclear Power was officially established
2007	The group company has clarified that the Nuclear Power Academy is the unified training management organization of the group, and has four training centers under management, operation, engineering, and education and training.
2009	The group's first batch of management training internal trainer certification training was officially launched; The Group Nuclear Power Academy introduced and developed a comprehensive leadership and employee capacity development program
2010	Group training management platform and E-learning system launched
2011	The Wind Power Training Center of the Group Nuclear Power Academy was established
2012	The Leadership Development Center, an entity of the Group's Nuclear Power Academy Management Training Center, was put into operation
2013	The Group's School of Nuclear Power was renamed China General Nuclear University
2015	The Nuclear Power Service Training Center of China General Nuclear University was established
2016	Six colleges (School of Management, School of Nuclear Power Operations, School of Nuclear Power Engineering, School of Nuclear Power Technology, School of Nuclear Fuel, and School of New Energy) under CGN University were unveiled and established

4.1.3 ZTE College

First, in terms of the mission and functional orientation of the corporate university, ZTE College is committed to creating an employee capability development system that is "combat-oriented and connected from top to bottom", lead the growth of employees, become a learning consultant for employee growth, a partner of business units, and an engine for the construction of learning organizations.

Second, in terms of the training objects of corporate universities, the talent training system designed by ZTE University is divided into three levels: new employees, core backbone employees, and management cadres. For example, the corporate university has designed a blended learning covering the three stages of pre-, middle, and post-stage for the training of core backbone employees. As a result, the three-stage blended learning of Central Communication University has improved the enthusiasm and commitment of new employees to learn, and empower the entire process of onboarding and learning for new employees.

(Original text in Chinese: 前训阶段以在职自学为主，强“知”，企业大学主要负责发布在线学习、自测、专家问答、自备案例内容;在培训过程中，为脱产培训，强调“学”，企业大学依靠翻转课堂，经验提取、案例研究、工作坊为新员工提供学习机会；培训结束后就是在职实践，强调“用”，企业大学通过最佳实践分享、反馈、咨询等方式帮助新员工形成规范的职业行为。从实际反馈来看，中央传媒大学新员工培训项目每年培训新员工4000余人（包括校园招聘和社会招聘），并与多所高校建立战略联盟伙伴关系，建立“企业大学精英班”——旨在提前发现和选拔人才，培养具有中兴文化品牌、胜任岗位的新员工；因此，中兴通讯学院三阶段混合式学习提高了新员工学习的积极性和投入度，为新员工的入职和学习全过程赋能。 Please see more details in Supplement Material S2.3).

Third, in terms of the organizational structure of corporate universities, the “Learning and Development Iron Triangle” is constituted of colleges, first-level competence centers in various business fields, and human resources COEs. As a corner of the Iron Triangle and the COE of each competence center, the college undertakes the company's strategy, takes the lead in formulating policies, systems, and processes related to learning and development, empowers

business units with learning and development methods, and manages and implements their learning and development planning and implementation. support.

Fourth, in terms of the management mechanism of corporate universities, as shown in Figure 4-8, ZTE College takes capability improvement as the core, from the basic IaaS support to the capability precipitation and arrangement of the Paas middle platform, and realizes the capability definition and arrangement of SaaS. Encapsulation, to build a learning and development system that is "combat-oriented and connected from top to bottom".



Figure 4-8 Schematic diagram of the learning platform of ZTE College

Fifth, in terms of the development history and outstanding achievements of the corporate university, Table 4-3 shows the milestones of ZTE College: In 2003, ZTE College was established; in 2006, the corporate university was the first overseas management cadre Provide empowerment training programs; in 2009, ZTE College established an industry-university-research cooperation forum and began to provide empowerment training programs for internal and external customers; in 2010, corporate universities provided local talent training programs for overseas projects; in 2015, The corporate university has established a cloud learning platform to link internal and external customers and resources; from 2017 to 2018, ZTE College has won several corporate university honors, which reflects the recognition of the effectiveness of corporate university in empowering enterprise development.

Table 4-3 Memorabilia of ZTE University

Year	Key events
2003	ZTE College was established

2005	The first overseas training center (Russian Training Center) was established
2009	The Industry-University-Research Cooperation Forum was officially established
2014	Zte Institute of Communications Secondary School "Electronic Manufacturing Vocational School" was established
2015	Build a smart cloud learning platform to provide internal and external customers with a learning ecosystem of easy learning, learning and learning
2017	Won the honor of Shanghai Jiaotong University's ranking of the Best Universities for Enterprises in China
2018	Ranked 8th among the Top 50 Enterprises Universities in China by Training magazine

4.1.4 Dongfeng Nissan University

First, in terms of the mission and functional orientation of a corporate university, as shown in Figure 4-9, DFN takes "learning to ignite passion" as its mission, and has played a role in the corporate talent development learning center, corporate management innovation practice center, and corporate culture brand. The role of the four functional positioning of the communication center and the enterprise alliance development and cooperation center is to encourage all employees to learn better by creating a learning atmosphere, creating learning opportunities, creating learning motivation, and adopting various methods to improve employees' enthusiasm for learning.



Figure 4-9 Dongfeng Nissan University's learning vision logo

Second, in terms of the training objectives of corporate universities, DFN University has established a credit system based on employee growth based on the needs of employee growth and development. Corporate universities have formulated "course packages" with different training contents (including four types of "Induction Training Course Package", "Credit Course

Package", "Job Certification Course Package" and "Employee Skill Improvement Course Package", forming a management/ Two sets of training systems for technicians and skilled personnel; therefore, corporate universities take "training hours - earning credits - post-certification" as a triple guarantee to meet the needs of employees to improve their professional skills. The company's talent training. For example, for the training of management/technical personnel, the corporate university has formed a five-step training system of employee integration, employee professionalization, teamwork, management ability, and leadership ability. For the training of skilled personnel (employees with professional operation ability, such as operation workers on the manufacturing site), the corporate university has formed the Dongfeng Nissan "Seven On-site Management" training system based on Nissan's advanced on-site management experience; at the same time, the corporate university has established a The basic skills training system can cover the seven basic skills training systems of "assembly, spraying, stamping, welding, security, quality inspection, and machining" for skilled personnel.

In regard to empowerment, the Dongfeng Nissan University promote the capability of enabling organization through improving employees' abilities and top management members' leadership. Based on those practice, the Dongfeng Nissan University try to answer "how to empower the upstream and downstream partners of the supply chain more broadly", so it established a supply chain school and supply chain research center.

(Original text in Chinese: 其实要能够更好地赋能企业，我觉得首先上来是要出圈，这个圈是怎么形成的呢？东风日产企业大学是怎么一步一步建立起来的？最早是企业需要员工的能力提升，那员工怎么才能让他们提升呢？刚开始是外部招聘，后来发现光靠招聘还不行，还得自己培养，那我们需要培养什么样的能力？于是能力模型就出现了。赋能——原来想的比较多的是内部赋能，基本上又落地到领导力课程，帮高管团队提升提升能力，但课程原来已经有了，是可以延续下去的；但能不能向我们的供应链上下游合作伙伴进行更宽幅的赋能？所以当时建立了一个供应链学院和供应链研究中心。Please see more details in Supplement Marital S1.5).

Third, in terms of the organizational structure of corporate universities, as shown in Figure 4-10, Dongfeng Nissan University undertakes the important functions of general ability and

professional ability training. The two are the "Teaching and Research Office" and the "Career Development Office". The former is guided by corporate strategy and is responsible for researching the strategic direction of corporate training; the latter is oriented to the needs of employees and is responsible for formulating actual vocational training and counseling content, researching vocational assessment tools, and training. Internal trainers; "Three Institutes" are "General Studies Institute" (responsible for operating the credit system for employee learning and development and carrying out general education for employees), "Innovation Institute" (responsible for summarizing processes, projects, knowledge management innovations related to organizational development) Cases) and "Professional College" (responsible for meeting business professional skills training needs, including R&D department, quality department, supply chain department, manufacturing department, and marketing department).

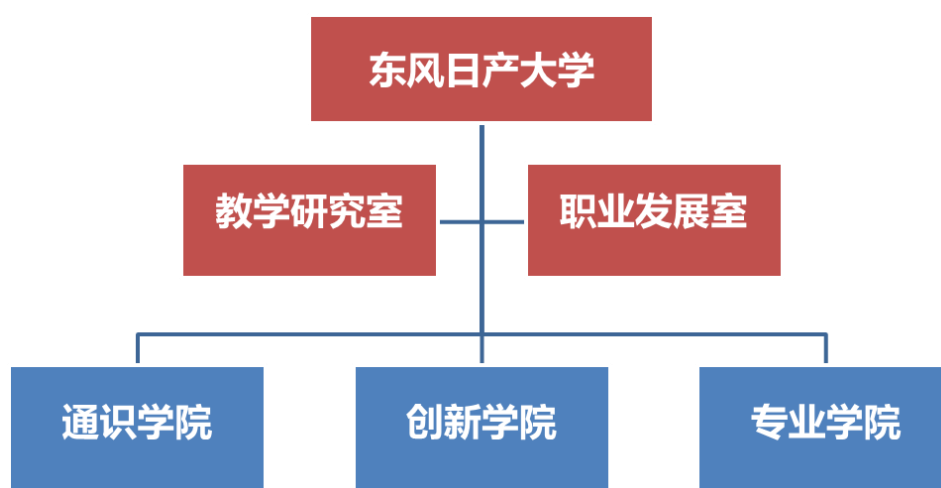


Figure 4-10 Organizational structure of Dongfeng Nissan University

Fourth, in terms of the management mechanism of corporate universities, Dongfeng Nissan University has a unified management and operation mechanism, including "Employee Training Management Measures" for the company's training business, "Training Project Implementation Specifications" for the implementation process of specific training projects, "Training Assets Management Measures" for training resource management, "Management Measures for Internal Lecturers and External Lecturers" for the development of trainer groups, and "Guidelines for the Implementation and Management of Large-scale Training Projects".

Relying on the above-mentioned management and operation mechanism and management plan, DFN provides legal system norms for the training work.

Dongfeng Nissan has determined the training model of the "competition-based learning model", which includes five links: learning, practice, competition, use, and evaluation. As shown in Figure 4-11, in the "learning" link, the corporate university provides face-to-face training courses for employees, and sets up courses and training content in the form of flipped classrooms and online and offline reading clubs; in the "practice" link, employees communicate with internal trainers Participate in speeches, make micro-lectures and practical operations together to realize the practice; next, in the "competition" section, the internal trainer of the corporate university will set up gamification levels for employees, design the game rules of the point system, and encourage employees to participate in the game. In the process of chemical learning; finally, in the "use" link, the corporate university will match the employees into a team, and the internal trainer will lead everyone to carry out team practice and solve the practical problems faced by the business department; in addition, the corporate university does not deliberately set "evaluation" "The sequence of the links, the evaluation runs through the entire project and training process, including the team and individual level evaluations (learning effect evaluation and ability evaluation).

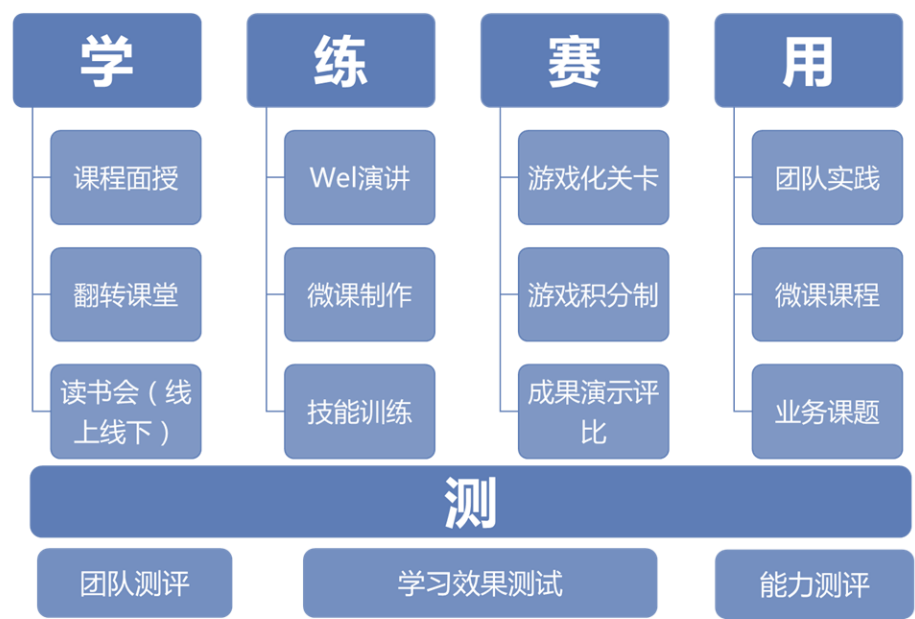


Figure 4-11 Dongfeng Nissan University's competitive learning model

Fifth, in terms of the development history and outstanding achievements of the corporate university, Table 4-4 shows the memorabilia and milestone events of Dongfeng Nissan University: In 2011, in response to the strategic needs of senior managers, how to establish the enterprise's corporate university Schedule; In 2012, Dongfeng Nissan University established an internal trainer training system; in 2013, Dongfeng Nissan University officially operated; in the four years from 2015 to 2019, the corporate university continued to refine the operating mechanism, and carried out several Empowering business training and learning programs.

Table 4-4 Memorabilia of Dongfeng Nissan University

Year	Key events
2011	The Company Executive Forum proposes to build Dongfeng Nissan University and take learning as a company strategy
2012	Dongfeng Nissan University Lecturer Certification Training System Established
2013	Dongfeng Nissan University officially opened
2014	Dongfeng Nissan University Learning Center Breaks Ground
2015	Wind Nissan University credit system officially launched
2016	Dongfeng Nissan University Learning Center was officially completed and put into use in Huadu
2017	Dongfeng Nissan University Happy Night School officially opened for enrollment
2018	Dongfeng Nissan University School of Quality was established
2019	Six professional colleges of Dongfeng Nissan University Manufacturing College and Party Construction College were established

4.1.5 Sany Corporate University

First, in terms of the mission and functional orientation of corporate universities, the sole purpose of corporate universities is to serve industries and enterprises well. According to the development model of the Chinese concept of "one body and two wings", the college takes the development of the service industry as the main body and takes academic education and vocational training as the two wings. With the concept of "creating first-class talents", it cultivates and develops talents that meet the strategic development needs of Sany Group, builds

a learning organization; undertakes external training business from the industry, government, and other social organizations, exports corporate culture, and assumes social responsibilities.

Second, in terms of the training objects of corporate universities, according to the school's development orientation of "one body and two wings", the training objects of Sany Corporate University are divided into two categories : (1) Academic education is oriented towards future employees, to train construction machinery, construction (2) Vocational training is mainly for existing employees within the group, providing differentiated vocational training services for employees of different departments and ranks. At the same time, the corporate university is not limited to the enterprise itself, but also provides training services for corporate partners, industry associations, government departments, and other social organizations in the ecosystem where the enterprise is located.

“One of the things we are starting to do at Trinity is to build talent chains. The employees of the enterprise should continue to develop, the external environment is changing, and people should continue to grow. As vocational schools can also do vocational training, the state requires vocational colleges must do training, they must do. There are a lot of these, if small and medium-sized enterprises do not have corporate university, whether the university can take up the position, I will do it for you, the future is very likely to appear this integration”. (Original text in Chinese: 我们三一大学正在开始做的一件事就是建立人才链，企业的员工要不断进行发展，外界环境在变，人要不断成长。作为职业学校也可以做职业培训，国家要求职业大学必须做培训，他们必须做。这里面有很多，如果中小企业没有企业大学，高校是不是能补位，我来给你做，未来很有可能出现这的融合 (See more details in Supplement Martial S1.1).

For example, as shown in Figure 4-12, Sany Corporate university has developed a "Learning Sany" training program for industry associations and client companies. This program adheres to the project concept of platform sharing, experience transfer, and common development. The diversified training methods of listening, watching, learning, and playing bring students a rich and colorful learning experience. The corporate university refines, shares and promotes corporate culture and outstanding management experience accumulated over a

long period, so that members of industry associations and customers can truly enter the company; in this process, customers as students can perceive and understand Sany Group's benchmark and cutting-edge management concepts and ideas, and promote the communication and interaction between enterprises and enterprises, customers and customers in the ecological chain.

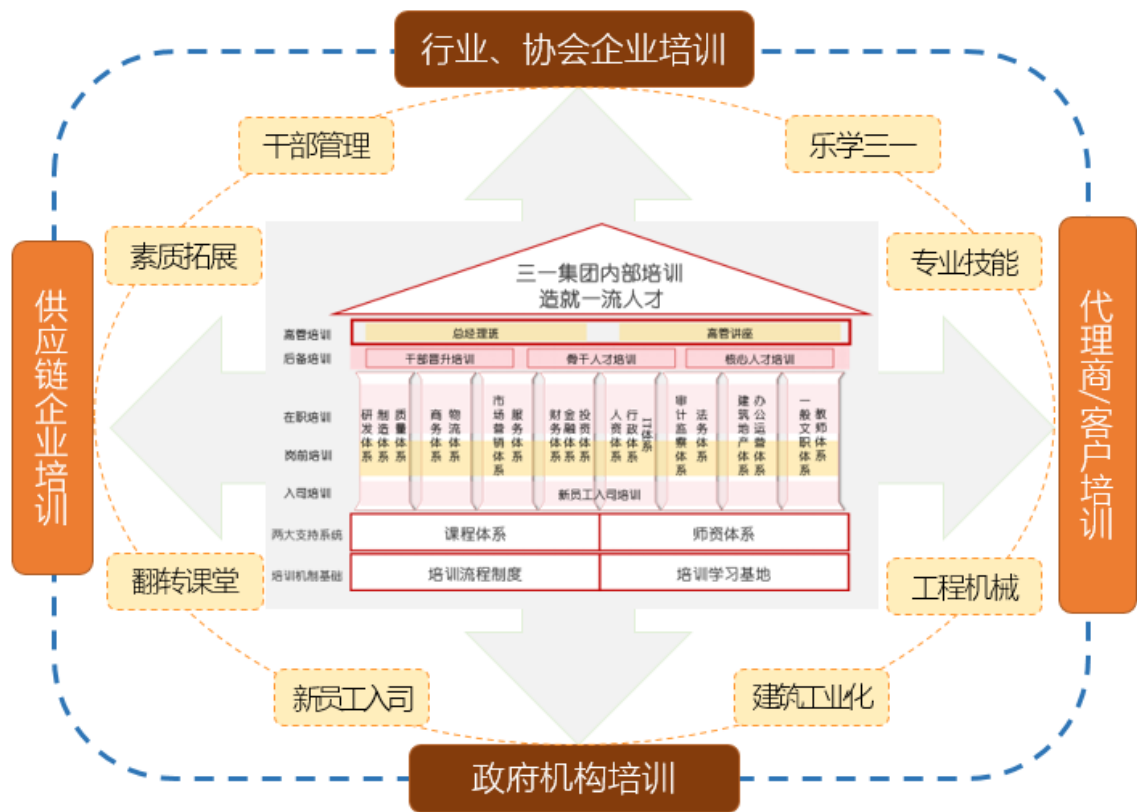


Figure 4-12 Panoramic display of the training objects of Sany Corporate university

Third, in terms of the organizational structure of the corporate university, Figure 4-13 introduces the positioning of the corporate university in the enterprise: the president of the corporate university is the deputy general manager of Sany Group, and the direct superior is the chairman of the Sany Group; From a departmental perspective, accept the management of the Group Human Resources Headquarters.

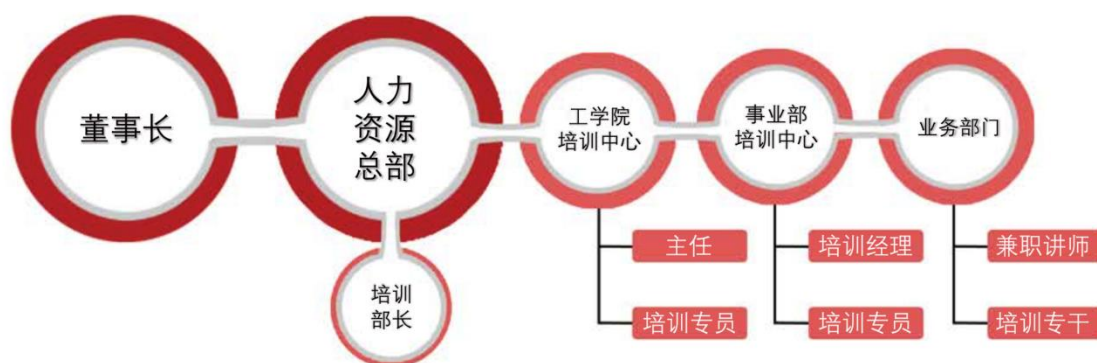


Figure 4-13 Organization chart of Sany Corporate University

Sany Corporate university is divided into two centers, the Engineering College Training Center (diploma education) and the Division Training Center (vocational education). Construction machinery intelligent manufacturing, construction machinery marketing and maintenance, construction machinery productive services, construction industrialization, and information services are five major groups. According to the industrial development trend, the front and back markets of the industrial chain are connected. There are 22 higher vocational majors and 9 secondary vocational majors. specialized. Vocational education has set up a training center, which is responsible for the vocational training business (internal training and external training of the group), and manages and implements a team of more than 20 people. It undertakes the personnel training function of Sany Group internally and undertakes the training business needs of the ecological chain industry. There are one director and one deputy director, and more than 20 training supervisors and training specialists. As a result, the training center of the engineering college is oriented to future employees, and the training center of the business division is oriented to the existing employees of the enterprise and upstream and downstream customers of the industrial chain. The cooperation between the two can better empower the training work of the enterprise.

Fourth, in terms of the management mechanism of corporate universities, Sany Corporate University provides three major enabling systems: learning programs, competence centers, and platform support. The learning empowerment system helps to deepen employees' understanding of corporate culture and enables employees to master basic professional abilities. The capability empowerment system enhances the practical capabilities of employees, HRBPs, and business

experts through Sany Corporate University, and conducts certain quality control over the entire training system. The platform is mainly based on the mobile learning platform, which is fully integrated with the Internet to establish a learning center within the company.

Fifth, in terms of the development history and outstanding achievements of the corporate university, as shown in Table 4-5, the predecessor of Sany Corporate University was established in 2003 and was the training center at that time; in 2010, the corporate university was formally established and In 2016, it first undertook an order for training projects from external customers; in 2017, Sany Corporate university became the only educational college listed on the New Third Board; in 2018, Corporate university invested 11 million yuan to build a digital information platform; 2019 In 2008, Corporate University won the honor of "Best Corporate University" in Shanghai Jiaotong University's list of China's Best Corporate Universities.

Table 4-5 Memorabilia of Sany Corporate University

Year	Key events
2003	Sany Changsha Training Center (predecessor of corporate university) was established
2006	Sany Electromechanical Technical School (predecessor of Corporate university) completed its name change
2010	Sany Corporate University officially established
2016	Sany Corporate University has taken the first step of "marketing training" and undertaken service training projects from external customers for the first time
2017	Sany Corporate University has officially become the only educational college listed on the New Third Board
2018	Sany Corporate University invested 11 million yuan to build the "Smart University" information platform and established 8 computer rooms; In the same year, nearly 400 certified project managers were trained throughout the year
2019	Sany Corporate University was awarded the "Best Corporate University" by Shanghai Jiaotong University in China's Best Corporate University Rankings

4.1.6 China Resources Gas (CRG) College

First, in terms of the mission and functional orientation of the university, at present, China Resources Gas (CRG) Group has established a corporate university in Nanjing, name CRG College. With the advanced experience and multiple abilities, it takes top experts in the gas industry as its core strength and focuses on the development of gas elites. Cultivating and developing, assisting the dissemination of corporate strategic management ideas, summarizing

management practices, inheriting excellent culture, assisting innovation incubation, providing intellectual support for the rapid and sustainable development of CRG, and striving to build a comprehensive gas professional management school.

Second, in terms of the training objectives of corporate universities, based on the resources and management experience accumulated in the past, CRG College has adjusted the past "functional" organizational division model to a "market-oriented" corporate university operating model, providing more than 200 gas companies. Member companies, nearly 50,000 internal customers, and industry partners provide services - which means that the corporate university provides training services not only for employees within the company but also for companies in other industries. The difference is that the internal employees of the enterprise can obtain free training from the corporate university; while the enterprises in the industry need to pay for the training services of the CRG College. Customers provide high-quality customized training and consulting services.

Third, in terms of the organizational structure of the corporate university, CRG College is subordinate to the human resources department of CRG Group. As shown in Figure 4-14, CRG College has undergone three stages of organizational structure adjustment: (1) From 2011 to 2013, this corporate university was established to respond to business needs and cultivate talents in key positions (such as professional managers); From 2012 to 2013 alone, the executive training courses launched by corporate university produced more than 90 financial directors and middle managers, and 13 general managers of enterprises. (2) From 2014 to 2016, CRG College strengthened the "1+5" development model, that is, it grows hand in hand with five regional training centers (Central China, South China, Southwest China, North China, and East China), covering the national business scope. In 2013 alone, the corporate university set up 32 projects with a total of 2,379 class hours and has trained about 7,000 employees accumulatively. (3) Since 2017, CRG College has insisted on building the platform of the corporate university, summed up the Group's experience and extended it to all regional training centers, and strengthened the training of professional managers and business talents.



Figure 4-14 Development history map of China Resources Gas Institute

Fourth, in terms of the management mechanism of corporate universities, CRG College has reviewed the past training system, combined with future development, and proposed to gradually transform the "functional corporate university operation model" that used projects as the main body of labor division and cooperation into one with "back-office engagement". The "market-oriented corporate university operation model" of development and front-end demand". As a result, the organizational structure of the corporate university has been adjusted to three pillars of "talent planning", "knowledge management" and "operational delivery". (1) In terms of talent planning, corporate universities have established a full sequence of talent development standards, curriculum resources, and an examination and evaluation system, and have completed the construction of the knowledge and knowledgeability system for key positions in the front-line operation sequence. Method", solidifying and promoting the unique learning method of gas people. As shown in Figure 4-15, the corporate university focuses on four key groups to formulate a "Blue Flame" talent learning map and match the corresponding course resources. First, the management sequence: mainly focus on the gas "3 C leadership quality model and general management ability quality model". Focusing on the "3C talent" training model, we will continue to pay attention to the leadership and professional ability improvement of key talents at the high, middle and grass-roots level. Second, professional sequence: according to business needs, various training projects and system construction are led by various departments of the headquarters. For example, a grade certification scheme has been formulated for the key groups of cost engineers. Third, the operation sequence: carry out the first-line job skill compliance certification project, and realize the internal certification. The

first-line post-skills compliance certification project includes formulating the content of the standards that should be known and should be understood by front-line personnel. A total of 37 standards should be known and should be developed (involving positions such as safety, transportation, and distribution, customer service), and developed based on these standards. supporting textbooks. Since 2016, we have achieved 100% employment with certificates. Fourth, the sequence of newcomers: Through the organic combination of the "Newcomer Quality Item Model" and the "Newcomer Three-Year Training Plan", the development channels for newcomer selection, training, appointment, and promotion have been fully opened.

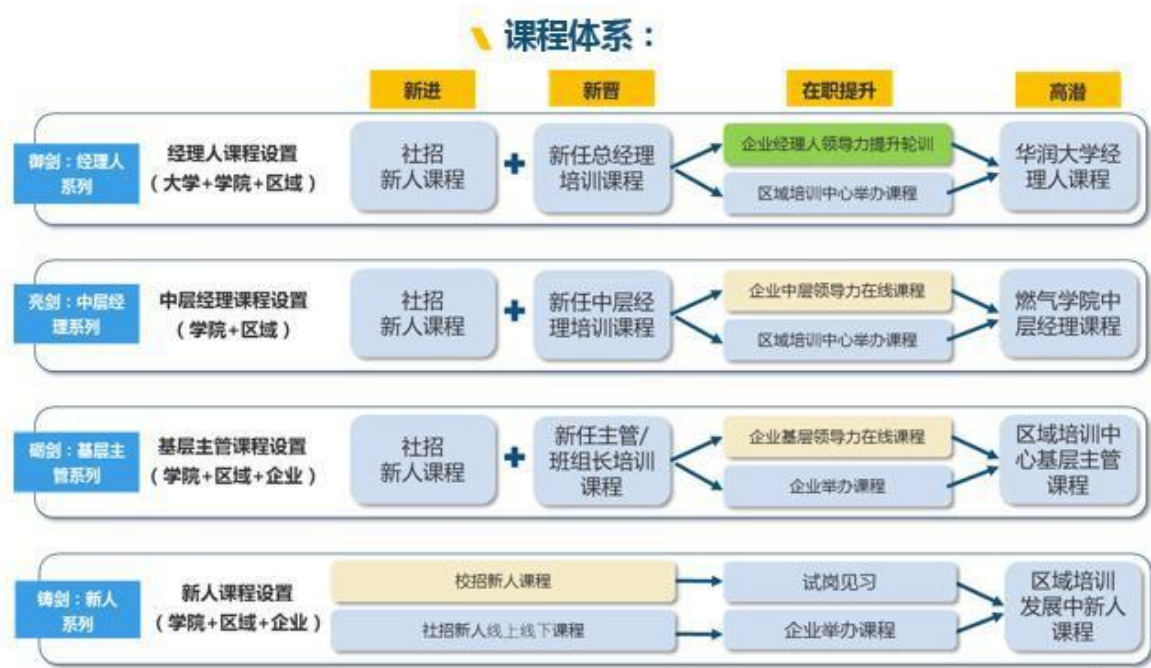


Figure 4-15 An example of the curriculum system of CRG College

In addition, in terms of knowledge management and operational delivery, corporate university has systematically summarized the Group's years of operation and management experience, and developed two internal training materials, "The Way of Gas Business Control" and "Learning Benchmarking Classic Cases". The corporate university has also developed 37 sets of first-line skill module learning toolkits for internal employees (including written learning materials, PPT courseware, videos, question banks, and assessment guidelines) by conducting scenario-based seminars on gas operation and management.

Fifth, in terms of the development history and outstanding achievements of corporate universities, Table 4-6 shows the construction history of CRG College: Since its establishment

in 2013, CRG College has been developing continuously, with increasing training programs and more diversified training forms. In particular, in 2018, the corporate university focused on formulating the "Blue Flame" talent development learning map, which fully reflects the corporate university's empowerment of employees' personalized career development.

Table 4-6 Memorabilia of CRG College

Year	Key events
2012	CRG College (corporate university) was established
2013	CRG College was established, and 32 training programs were set up in the first year
2014	Started the preparatory work for the construction of the online learning platform; a total of 63 training programs were opened throughout the year
2015	189 member companies of the online learning platform have been launched, and the number of students is close to 20,000
2016	CRG College takes the lead in holding a micro-course competition, and gradually guides enterprises to independently develop personalized courses
2017	The headquarters of the CRG College has carried out the preparatory work for the establishment of the entity college from the aspects of software and hardware
2018	CRG College has formulated the "Blue Flame" talent development learning map of China Resources Gas, and other types of learning maps are under continuous development

4.1.7 China Railway Fourth Bureau (CRFB) Group's Corporate University

First, in terms of the mission and functional orientation of the corporate university, the corporate university of the China Railway Fourth Bureau (CRFB) Group is based on cultivating talents, promoting change, managing knowledge, promoting culture, and sharing and communicating, and centered on serving the development strategy of the enterprise, becoming the business of the enterprise. Booster, strategy driver, performance facilitator, and industry coordinator.

Second, in terms of the training objectives of the corporate university, the CRFB Group's Corporate University includes employees at all levels of the group (employee family members), as well as customers, partners, and stakeholders. In regard to empowerment, the corporate university also try to enable internal and external stakeholders. For example, CRFB Group's Corporate University has established the vision of becoming a leader in talent training and organizational development in the field of infrastructure, and has formulated the development goal: one year to build a foundation, three years of practice and five years of maturity.

(Original text in Chinese: 2016 年，企业大学确立了做基础设施领域人才培养和组织发展的领跑者的发展愿景，并制定了“一年筑基、三年实践、五年成熟”的发展

目标。“一年筑基”是指组建一支专业的学习队伍，搭建好全局学习的平台，通过制定相关管制制度以及关键学习项目的实施和运营达到能够较好服务企业的目标。

“三年实践”是指通过高赋能提升培训队伍的专业性，并通过做精学习项目，广聚资源，达到不仅能够服务企业，同时也能够服务客户的目标。“五年成熟”是指通过出人才、出学习产品，铸造企业学习品牌和学习文化，最终达到服务企业生态链的目标。 *Please see more details in Supplement Material S2.2).*

Third, in terms of the organizational structure of corporate universities, a learning committee has been established to adopt the operation mechanism of physical institutions, virtual organizations, and online platforms, while the outputs are products, knowledge, and talents. To realize the university is built in the enterprise, and the enterprise is built in the University, the corporate university of CRFB Group established the Innovation College, Leadership College, and Project Management College, inject impetus into the improvement of the core competitiveness of enterprises.

Fourth, in terms of the management mechanism of corporate universities, the corporate university of CRFB Group implemented the "Four Project", including professionalization of teaching staff, professionalization of trainers, standardization of learning management, and instrumentalization of learning methods. Key elements of the university; focus on the quality of the development of learning products, and formed products including "qualification training and certification of ten key positions in the project", "four types of talents in short supply", "high-level leadership building", and learning technology and tool development.

Fifth, in terms of the development process and outstanding achievements of the corporate university, Table 4-7 shows the construction process of the corporate university of CRFB: After the institutional preparations from 2014 to 2015, the corporate university was officially established in 2016 and was Year to undertake customer training orders. In 2018, enterprises have completed the development and sharing of more than 500 micro-courses by the university.

Table 4-7 Memorabilia of CRFB University

Year	Key events
2014	Create a "silent plan" to formulate a unified thinking on "why, what, and how" corporate universities are built
2015	Promulgated the "Implementation Measures for the Third-level Education and Training System of CRFB Group"; Independently developed the "Training and Certification System for Personnel in Key Project Positions Based on Competency Model"
2016	The Corporate University of CRFB Group was officially established

2017	The corporate university has developed an online learning platform to open a blended learning model; Undertook customer training for the first time, and the construction of a "learning ecosystem" began to test the water;
2018	Corporate university held a micro-course development workshop, creating and sharing more than 500 high-quality video courses
2019	Corporate university provides learning and training for multiple clients or units in the industry one after another

4.1.8 New Hope Business School

First, in terms of the mission and functional orientation of a corporate university, New Hope Business School helps activate the learning energy of employees and industry chain partners, creates a learning environment that stimulates innovative thinking, continuously improves problem-solving abilities, and shapes and spreads professionalism. With business demeanor, improve the skills level of employees and customers, and promote the sustainable development of the company. New Hope Business School is committed to "undertaking the company's strategy, promoting performance improvement, and assisting talent development (from corporate information)", integrating action learning methods, case teaching methods, blended learning, cross-border research methods, and coaching to guide diversified learning form a unique talent training and development model, and embed learning into organizational workflow and practice.

Second, in terms of the training objects of corporate universities, as shown in Figure 4-16, the training objects of New Hope Business School are mainly internal employees of the company, including (1) The "New England Plan" for new employees "(Primary induction training) aims to accelerate the integration of new employees into the company and career growth, and consists of training camps, mentoring, on-the-job training, communication sessions and advanced training for one year. (2) The "Elite Plan" and "Heroic Plan" (intermediate management training) for the middle management team aim to output outstanding managers and output successors for management, professional management, and general management. This project trains about 1,000 outstanding backup employees every year. They have operational awareness and management capabilities and can break through, change and

innovate. They are a powerful impetus in the process of organizational transformation. (3) The "LinkedIn Program" (advanced strategic training) for senior managers mainly introduces external resources, broadens managers' horizons, and enhances strategic leadership.

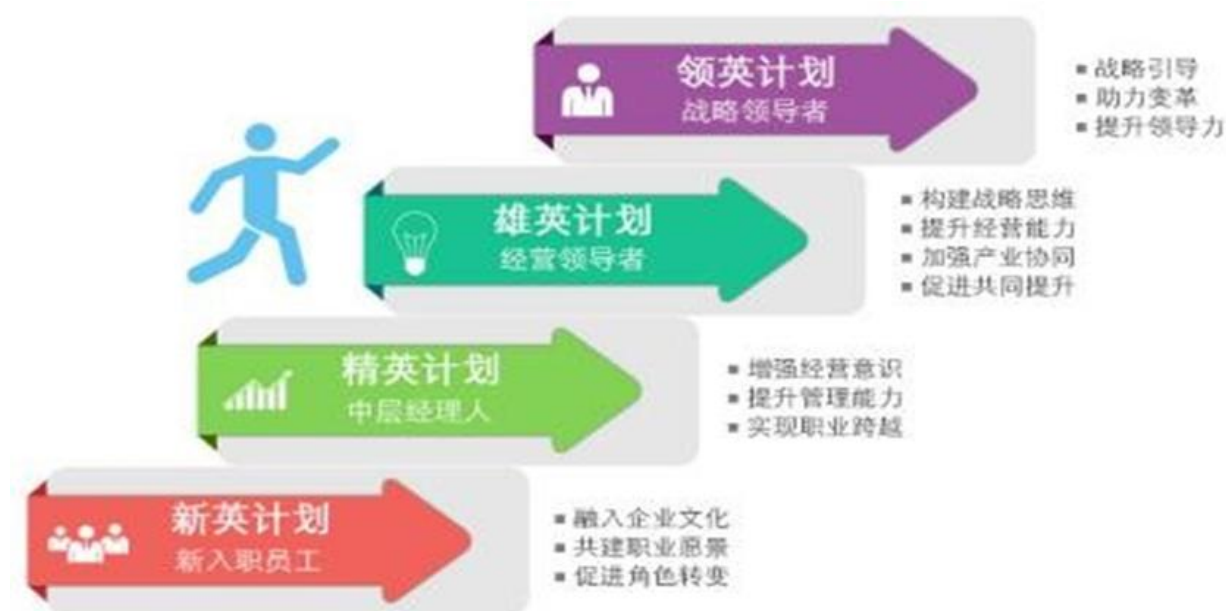


Figure 4-16 Cultivation objects and training programs of New Hope Business School

Third, in terms of the organizational structure of the corporate university, as shown in Figure 4-17, the New Hope University Business School has a strategic core position in the organization and is fully supported by the company's management committee and the president. The corporate university first assumes the function of organizational learning and stimulates the learning motivation and potential of internal employees through a variety of training and learning methods. The corporate university also builds an internal training knowledge base through industry research and case summaries to continuously improve the organization's knowledge management capabilities. In addition, corporate universities provide a platform for employees to share knowledge and experience. Corporate university integrates internal OA, knowledge management database, knowledge sharing platform, and market research system, and absorbs the relevant knowledge of external stakeholders (such as industry chain partners, core users), and integrates knowledge into multiple modules for employees. Learning; corporate universities also use the business school platform to establish cooperative relationships with famous universities at home and abroad to improve the channels for knowledge acquisition.

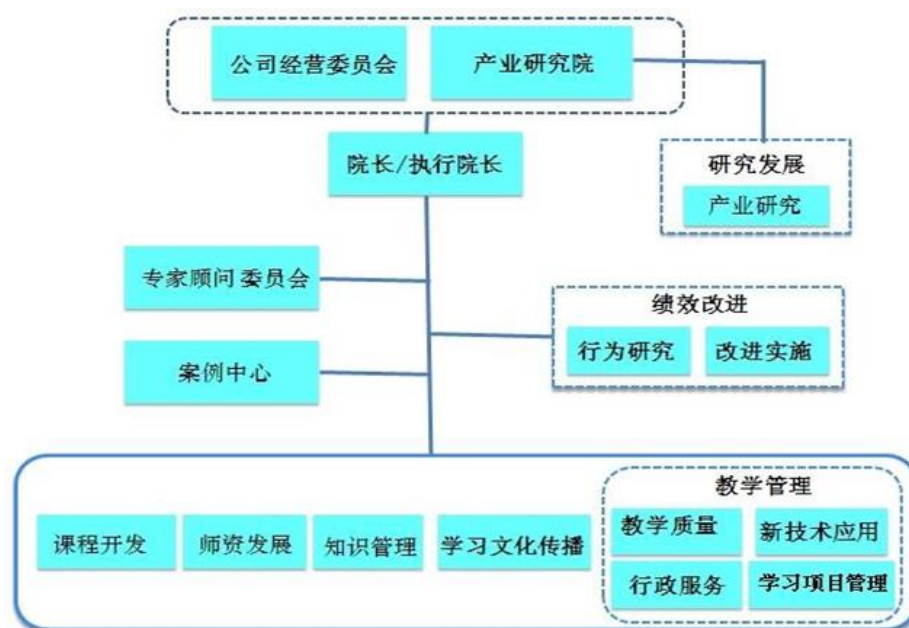


Figure 4-17 The organizational structure of New Hope Business School

Fourth, in terms of the management mechanism of corporate universities, the Business School of New Hope University has a strategic core position in the organization and is fully supported by the company's management committee and the president. In terms of function, it first emphasizes the dissemination of learning culture, that is, to stimulate the learning motivation and potential of each employee, provide a platform for sharing knowledge and experience, and build a knowledge base through industry research and case summary to form and continuously improve the organization's knowledge management capabilities. Second, establish a learning and knowledge management platform, and integrate knowledge through internal OA systems, knowledge management databases, knowledge sharing platforms, and market research systems. At the same time, scientifically and systematically learn and absorb relevant knowledge from the external environment, such as industrial chain partners and core users, to enrich the knowledge base. Use the business school platform to establish cooperative relations with famous universities at home and abroad, and improve the channels for knowledge acquisition. Third, lead the communication and knowledge sharing among employees, such as through online platforms and online and offline social platforms, to vigorously promote the exchange of knowledge and the collision of ideas among employees in various departments.

The business school also encourages employees to be willing to innovate, take risks, and make bold changes by promoting the construction of an internal innovation culture.

Fifth, in terms of the development process and outstanding achievements of corporate university, as shown in Table 4-8, since corporate university has not disclosed all the key events, New Hope Sixth Business School still shows the forward-looking layout of cloud classroom and online learning platform. Currently, New Hope University has two major practice directions in its enabling business, inducing process performance improvement and professional ability training and certification.

(Original text in Chinese: 新希望大学赋能业务方面有两个重点实践方向：（1）流程绩效改进，侧重对业务流程进行分析，跨部门、快业务的组织研讨会改进流程，促进协同增效。新希望大学曾经为金融担保等部门系统梳理过业务流程。（2）专业能力培养和认证方面，为重点专业线路和关键岗位，萃取最近实践、构建能力和任务模型，评鉴人才能力和绩效关联度。提供专业能力培养和认证的解决方案。 Please see more details in Supplement Material S2.2).

Table 4-8 Memorabilia of New Hope Business School

Year	Key events
2013	Awarded as "Excellent Enterprise" of China Online Education for 15 years
2014	UCloud leads online education into the cloud
2018	Won the award of "2018 China's Best Corporate university"
2020	Built an online learning platform and quickly iterated the content of the platform

4.1.9 China Construction Bank (CCB) University

First, in terms of the mission and functional orientation of corporate universities, China Construction Bank (CCB) University is not only a vocational education platform for internal employees but also an important platform for providing vocational education and training services to society and realizing the integration of industry and education. Open cooperation, co-construction, and sharing are the most distinctive features of CCB University that differentiate it from traditional corporate universities.

Second, in terms of the training objectives of corporate universities, CCB University has developed special teaching and training tools, focusing on cultivating new financial talents, cultivating employees by classification and stratification, improving employees' professional ability and comprehensive quality, and helping improve individual and organizational performance. Following the three stages of productization, systemization, and colocalization, the construction of an employee growth system with CCB characteristics has been comprehensively promoted. For example, guided by product thinking, it launched a training program for all new employees of the group, provided a project plan integrating courses, operations, evaluation, and branding for institutions under its jurisdiction, and guided the whole bank to carry out standardized training for 14,000 new employees annually. Another example is that the corporate university has developed a capability improvement project for key employees (i.e., outlet account managers), and built a post-examination area on the online platform to facilitate employees to use the fragmented time to learn business knowledge.

Third, in terms of the organizational structure of the corporate university, the functional departments of the Party Committee of CCB University include the Organization Department of Party Committee and the Office of the Disciplinary Committee. In terms of operation and execution, it includes two major sections: the school headquarters and the campus. The headquarter of CCB University consists of four major departments: the General Management Department (Office of the School Committee), the Basic Teaching and Research Department, the Professional Training Department, and the Support and Development Department, which are respectively managed by four executive vice presidents. The campuses are divided into three types: regional campuses, professional campuses, and branch campuses, including 10 regional campuses, 10 professional campuses, and 37 branch campuses.

Figure 4-18, CCB University is composed of four major departments: the General Management Department, the Basic Teaching and Research Department, the Professional Training Department, and the Support and Development Department: (1) The General Management Department and the Office of the School Committee are co-located and responsible for administrative management, daily operation, and resource guarantee. Including

the General Office, the Planning and Scientific Research Management Office, the Teaching Operation Office, the Staff Growth Office, the Science and Technology Development Office, the International Development Office, and the Administration and Finance Office; (2) The Basic Teaching and Research Department is responsible for the general, basic and strategic teaching Research. It consists of the *Ideological and Political Teaching and Research Department*, the *Strategy Teaching and Research Department*, the *Humanities Teaching and Research Department*, and the *Leadership Teaching and Research Department*. (3) The *Professional Training Department* is responsible for leading the training and research of the business sector, covering and closely connecting all business departments of CCB. It consists of eight training institutes: *Inclusive and Retail Training Institute*, *Housing Finance Training Institute*, *Fintech Training Institute*, *Customer Relations Training Institute*, *Asset Management and Investment Banking Training Institute*, *International Finance Training Institute*, *Risk Management Training Institute*, and *Finance and Auditing Training Institute*. (4) The *Support and Development Department* provides support for the professional development advantages of CCB and consists of two sections: support and development. Supporting sections include labs (offices) including big data labs, financial technology labs, and product innovation labs; digital education learning centers (university network platforms); case study teaching centers (knowledge management centers); curriculum design management centers and careers Qualification Development Center. The expansion section includes *Foreign Cooperation Center*, *Financial Media Center*, *Brand Operation Center*, and *Alliance Secretariat Office*.

and external customers through the CCB University online learning platform. Currently, the CCB University still follows the leadership's empowerment strategy planning and empowers the organization from the concept, action, structure, capacity and other levels.

(Original text in Chinese: 董事长强调：“当你用一个新金融逻辑和思维去思考这个问题时就会发现，没有什么事情是跟我们没有关系的，一切痛点皆是机遇”。建行大学从理念上首先给组织进行破冰和重塑，为组织从理念、行动、架构、能力等层面都进行赋能。 Please see more details in Supplement Material S2.4).

Table 4-9 Memorabilia of CCB University

Year	Key events
2018	CCB University was officially established; in the same year, the CCB University network platform was officially launched
2019	Won the "2019 National Excellent Financial Media Works"
2020	Won the "2020 China Internet Education Technology Innovation Enterprise Award" and "2020 China Internet Education Quality Platform Product Award"

4.1.10 Geely Corporate University

First, in terms of the mission and functional positioning of the corporate university, Geely University adheres to the purpose of empowering business and achieving every Geely fellow traveler and takes talent delivery, cultural empowerment, and organizational development as its functional positioning. It will further cooperate with upstream and downstream industry chains, industry institutions, and partners of major universities around the world to create an ecological chain of talent training throughout the career cycle and help Geely's global business continue to develop.

Second, in terms of the cultivation objects of corporate universities, Geely Corporate University advocates that everyone is a teacher and everyone is a student. Customized, scenario-based, and personalized training programs, provides employees with learning space at different stages of career development. with resources. As for the career development stages of different employees, corporate universities have provided differentiated training resources. At present, a leadership training system covering the senior, middle and grass-roots levels of the enterprise have been established, such as the flight plan and the dolphin plan for management

cadres, and the geese for reserve talents. Program and GM1000 postgraduate training program, leader program, and skill master program aimed at improving the ability of skilled talents, as well as special forces and special training camp programs for professional talents, each different training program maximizes the potential of employees and shapes the enterprise learning atmosphere.

Third, in terms of the organizational structure of the corporate university, the organizational structure of Geely Corporate University can be divided into first-level colleges, professional colleges, regional campuses, graduate schools, and operation management departments (as shown in Figure 4-19).

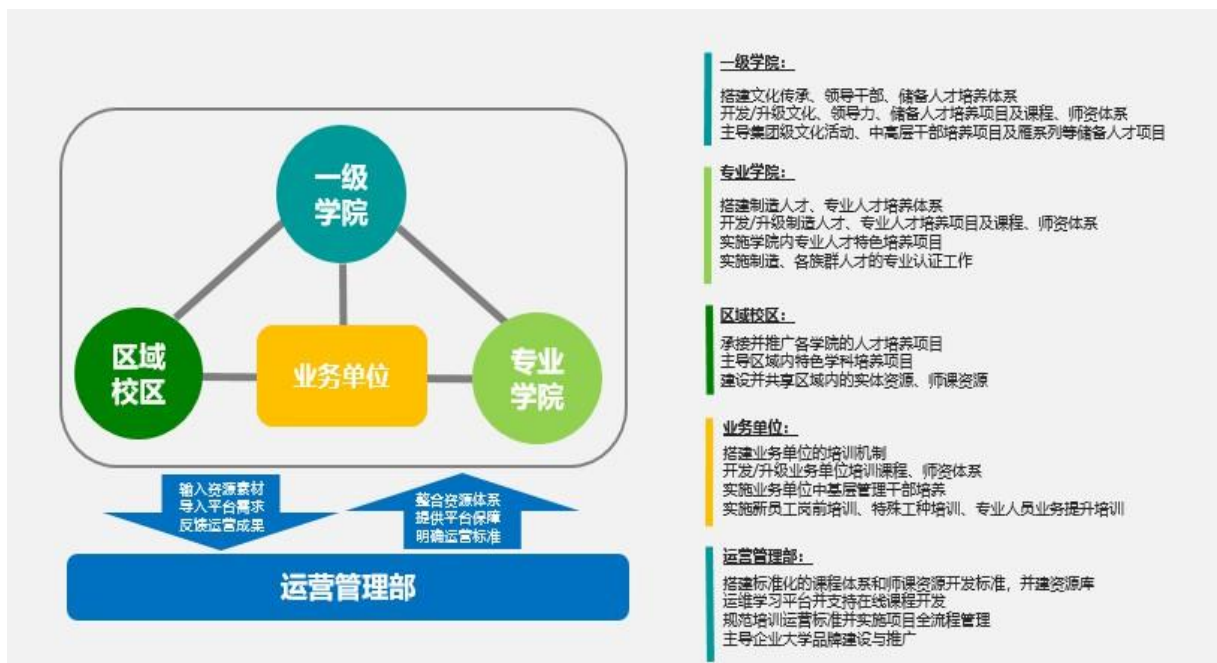


Figure 4-19 The organizational structure of Geely Corporate University

Fourth, in terms of the management mechanism of corporate universities, the operations of Geely University are mainly divided into four categories. First, the training manager/supervisor of the first-level college: mainly responsible for cultural construction, leadership training, reserve talent training, general ability improvement module system construction, and training mechanism construction, and responsible for the development, design, and promotion of talent training for the whole group project. Second, the training HRBP of professional colleges: mainly responsible for the establishment of the professional talent training system and mechanism for the corresponding ethnic groups of the college, long-term

assignment to business units, in-depth exploration of customer needs, focusing on business pain points and professional talent echelon construction, providing upstream and downstream industries. Partner training services. Third, the campus operation team: is mainly responsible for the physical construction of the campus and the operation of the physical educational affairs, according to the individual needs of the docking area, gathering the internal and external resources of the area, and designing and developing characteristic talent training projects for the service business units. Fourth, the corporate university operation team: is mainly responsible for the overall operation, system planning, and resource construction, with an internal professional team responsible for the development and upgrading of course resources, lecturer resources, supplier resources, and online learning platform resources. On the above, the operation team also went to all business lines, colleges, campuses, and subsidiaries of the group to diagnose and empower the training system.

Fifth, in terms of the development history and outstanding achievements of corporate universities, Figure 4-20 shows the development history of first-level colleges from 2017 to 2019, most of which were established jointly by corporate universities and business departments. For example, Automobile Manufacturing Research Institute (a first-level college of corporate university, jointly established by corporate university and regional manufacturing base), quality management college (co-constructed by corporate university and group quality management center) and power training institute (co-built by corporate university and group power system center) were established in 2017. As a result, each college or campus has set up senior managers, training development managers, training supervisors, cultural construction supervisors, training BPs, development instructors, and training specialists; as of 2020, there is more than 40 full-time staff, and more than 4,000 part-time lecturers and instructors. Currently, based on those fundamentals, the Geely Corporate University has built a group-level knowledge management platform, which is used to set up learning resource systems and standards, guide and empower the construction of learning resources for each sub-group. At the same time, Geely University has also built a management platform for internal trainers at the group level, formulated standards for the selection, training, employment and retention of internal trainers,

guided and empowered the construction of internal trainer teams in each sub-group, and promoted the construction of a learning organization.

(Original text in Chinese: 吉利大学建设了集团层面的知识管理平台——用来搭建学习资源体系及标准、指导并赋能各子集团学习资源建设工作。同时，吉利大学也搭建了集团层面的内训师管理平台，制定内训师的选、育、用、留标准，指导并赋能各子集团内训师队伍建设工作，推进学习型组织建设。Please see more details in Supplement Material S2.3).



Figure 4-20 Geely University development process

4.1.11 Red Star Macalline (RSM) Group's Management School

First, in terms of the mission and functional orientation of the corporate university, the Red Star Macalline (RSM) Group Management School revolves around the strategic guiding principle of "enhancing the awareness of the protagonist, standardizing the operation process, and accelerating the cultivation of talents" by the senior executives of the RSM Group. The corporate university followed the goal of "the Whampoa Military Academy in the home furnishing industry". The RSM Management School was born out of the RSM Home Furnishing Talent Training Program. Previously, RSM Group had accumulated rich school-running experience and faculty through joint school-running and internal training. To better amplify and radiate the advantage energy generated by these accumulations to the entire home furnishing

industry, the corporate university came into being. Through this carrier, the RSM Management School integrates the upstream and downstream resources of the industry, from the aspects of talent training, industry research, trend guidance, and cultural dissemination, to build the first real Whampoa Military Academy in the home furnishing industry.

Second, in terms of the training objects of corporate universities, the RSM Management School focuses on talent training and basic operation training in the home furnishing industry within the group. In addition, the corporate university also opens training services to external customers, providing education upgrading training, non-academic upgrading training, and job skills training programs for manufacturers, distributors, and store managers in the entire industry chain, so as to comprehensively improve the quality of employees in the industry. Improve the overall image of the industry.

Third, in terms of the organizational structure of the corporate university, the RSM Management School has kept pace with the times and changed its management model, adopting a project-based management model, and has three functional departments under its jurisdiction: the Training Department, the Management Department, and the Lujia Promotion Department. As shown in Figure 4-21, the training department is responsible for industry research, formulating training plans, designing curriculum systems, training system construction, and the implementation of pilot projects for some projects, and is responsible for the implementation and implementation of specific training projects; general management department: responsible for the implementation of corporate culture, the construction of training knowledge system and the docking and management of each branch; the Lujia Promotion Department is responsible for the relevant training of upstream and downstream enterprises in the supply chain. In addition, RSM Management School has established branch branches and subordinate units of regional branches, and its business is reported to the corporate university.

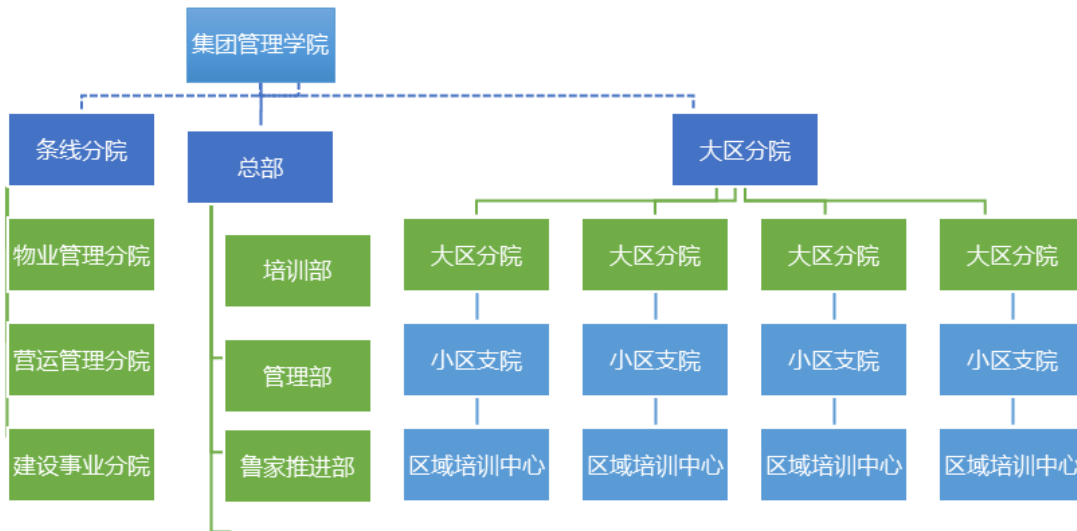


Figure 4-21 Organization chart of Red Star Macalline School of Management

Fourth, in terms of the management mechanism of the corporate university, the cross-departmental management mechanism of the RSM Management School is reflected in: (1) The corporate university is juxtaposed with various functional departments and reports directly to the top management. A flat and flexible structure is more suitable for Effective coordination with different functional departments; (2) Corporate University adheres to the sharing of professional resources, meets the needs of the operation, investment, enterprise, finance, and construction business departments, provides professional training resources and presents matching courses to employees; (3) Corporate University has to Form a standard annual training budget management mechanism, that is, the corporate university can formulate financial plans and execute budgets as an independent operation center.

Fifth, in terms of the development history and outstanding achievements of corporate universities, as shown in Table 4-10, the RSM Management School was established earlier (2008). In 2014, the corporate university officially launched an online learning platform; in 2015, the corporate university set up a training system that combines competition, training, and learning, and optimized the promotion process for internal trainers; in 2018-2019, the corporate university Focus on building digital training programs on online platforms. Currently, in regard to empowerment, RSM Management School is continuing to further extend Luban culture and brand culture to the upstream and downstream of the industry through the combination of

production, education and research, and carry out cross-value chain enabling activities through the Leadership College of Luban University and university alliance.

(Original text in Chinese: 未来红星美凯龙管理学院将继续依托红星体系，紧密围绕红星文化，通过产学研相结合等方式，进一步将鲁班文化、红星企业文化、品牌文化向产业上下游延伸、拓展，通过鲁班大学领袖学院和高校联盟开展跨价值链赋能活动。 Please see more details in Supplement Material S2.5).

Table 4-10 Memorabilia of RSM Management School

Year	Key events
2008	Corporate University officially established
2013	RSM Management School officially established Luban Home Furnishing College
2014	Corporate University continues to launch EMBA education in the home furnishing industry in cooperation with well-known colleges and universities to deeply cultivate industry talents; In the same year, Corporate University established an E-learning platform
2015	Starting from this year, the RSM Management School has optimized the promotion process of trainers through the form of lecturer competitions and the principle of "leading training through competition and promoting learning through competition".
2018	Corporate University launched the "Red Star Online Education Strategy Project" to empower employees with online training
2019	Corporate University launched the course "30,000 Vehicles to Build New", covering all employees of the group, with 50,000 study times, and organized online exams for all employees, with a final pass rate of 100%

4.1.12 Robam University

First, in terms of the mission and functional orientation of the corporate university, Robam University, which established by the Robam Group (an Chinese listed enterprises that makes electrical appliances), cultivates outstanding employess, cultivates our leading talents, and is committed to contributing to the sustainable development of small and medium-sized entities in China, becoming a consultant for employee growth, a business development partner, and a heritage of organizational culture. The corporate university aims to improve employees' abilities and promote career development; help improve business capabilities, focus on comprehensive talent management; inherit corporate culture and assist in organizational reform; promote ecological collaboration and cultivate industry talents to keep moving forward, and become an

enabler for the company's organizational capacity building, to continuously create value for the sustainable development of small and medium-sized enterprises in China.

Second, in terms of the training objectives of corporate universities, Robam University has multiple programs for different positions. As shown in Figure 4-22, corporate universities have set up different training empowerment projects, including Talent Development Projects (the Lemon Project, the Blue Whale Project, and the Sunflower Project) and the Excellent Leader Program for senior managers.

(Original text in Chinese: (1) 人才培养项目包括“柠檬计划”或“蓝鲸计划”、“向日葵计划”、“常春藤计划”和“彩虹计划”。其中，“柠檬计划”是由企业高校和人力资源部门开展的校园招聘员工培训项目，重点针对高潜力后备人才（同年毕业的本科、硕士、博士）进行入职培训和战略认可培训。与之相对应的是“蓝鲸计划”，社会招聘员工赋能项目，是企业为规范新员工入职培训管理，快速胜任新工作，满足公司发展需要而开展的线上线下项目。一个联合授权计划。通过“蓝鲸计划”的培训过程，新员工可以更多地了解公司的历史、发展和企业文化，认同和认同公司的事业和企业价值观，进而强化自己的职业选择，理解和接受公司的共同语言和企业文化。行为准则。向日葵计划是一项旨在培养和提高专业人才素质的专业能力赋能培训计划。主要培训对象是入职 2 - 3 年的员工。企业大学将为这些员工配备培训讲师，为他们搭建平台，引导他们向专业渠道发展，从而培养更多具有较高业务素质的业务骨干和青年专家型人才。此外，“常青藤计划”是一个中层领导培训计划，是公司中层管理干部相互学习，继续传承企业文化的平台。“彩虹计划”是一个授权项目，旨在为公司提供一个后备领导团队。企业大学通过组织价值观和领导力的培养，选拔出一批能力强、素质高、业绩优秀的骨干员工，最终形成未来。业务领导团队。“彩虹团队”由 8-10 人组成，采用领导力发展项目、企业经营战略行动学习、标杆考察等方式进行多元化提升和培训。(2) 优秀领袖项目更多针对高级管理人员，注重拓展高级管理团队的全球视野和大视野，系统提升其经营理念、管理能力、管理素养、领导艺术和社会责任，全面发展其企业家精神，充分调动其二次创业的激情和热情。Please see more details in Supplement Material S2.3).

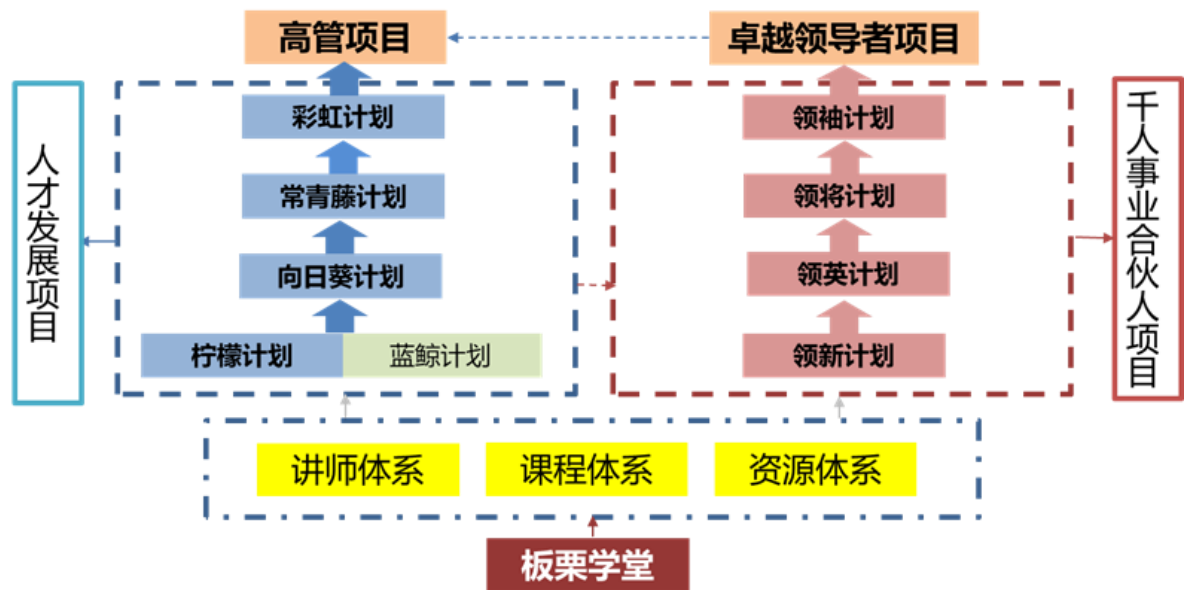


Figure 4-22 Robam University training system composition

Third, in terms of the organizational structure of the corporate university, at the beginning of its establishment, the corporate university and human resources were two independent departments, but they cooperated. At present, secondary colleges of Marketing College, Lean Manufacturing College, and Technology Innovation College have been established, with executives from various business fields serving as the dean of the secondary colleges, focusing on the implementation of the training programs of the secondary colleges. At the same time, all the executives are special senior lecturers of Robam University, serving as the vice forum, talent development project consultant, and core senior lecturer. Senior executives serve as talent appraisers, participate in quality model construction and talent inventory projects, and provide guidance and guidance.

Fourth, in terms of the management mechanism of corporate universities, Robam University has formulated the "Management Standards for Internal Trainers" to motivate and manage internal lecturers at different levels. The current internal trainers are divided into four levels: reserve lecturers, junior lecturers, intermediate lecturers, and senior lecturers. The system includes professional skills for internal trainers in quality, procurement, service, marketing, and production departments, as well as internal trainers teaching management skills. and in-house trainers from the business unit. Corporate University fully empowers employees

and teams and encourages employees to become both the extractors of knowledge and experience and the disseminators of knowledge.

Fifth, in terms of the development history and outstanding results of corporate universities, as shown in Table 4-11, after the establishment of Robam University in 2014, it attached importance to strategic partners (including institutions of higher learning and client companies), and in 2016 The online learning platform "Chestnut School" was launched to fully empower employees' online learning needs.

Table 4-11 Memorabilia of Robam University

Year	Key events
2014	Robam University was officially established; in the same year, Robam University and Cheung Kong Graduate School of Business signed a strategic agreement on talent training
2015	Corporate university has established a "Thousand Talents Training Program" internally and a "High-end Brand Training Program" externally
2016	Robam University mobile learning platform "Chestnut School" officially launched
2017	Robam University Quality Institute was established and several strategic projects were launched; Robam University's first book "Hyperlink" was officially published

4.1.13 Taikang Business School

First, in terms of the mission and functional orientation of a corporate university, Taikang Business School is the home of the the Taikang people' s spirit, a Taikang spirit inheritance ceremony, a Taikang cultural experience base, and a Taikang strategic communication center; Taikang Business School is the cradle of Taikang talent training, focusing on Create international leaders, professional cadres, professional sales teams, and market-oriented professional teams; Taikang Business School is a new starting point for Taikang people's career, and it is the integration of production, education and research and the cultivation of big health industry. It is a high-end platform for Taikang people, focusing on strategic advantages to cultivate core talents, and continuously linking with external first-class resources. It is Taikang's corporate university.

Second, in terms of the training objectives of corporate universities, Taikang Business School has established a complete training system for Taikang core talent learning programs, including new employees, new cadres, young and middle-aged classes, leadership programs, executive training, professional empowerment, and business empowerment. Ability, covering managing self, managing others, managing team, managing business, managing organization and culture core leadership programs, and systems. As shown in Figure 4-23, each year, based on the development needs of the company's strategy, business, and organizational staff, we design and implement corresponding projects, continue to upgrade the curriculum system, teacher system, and business cases, and keep up with strategic requirements, business rhythm, and organizational transformation. Investigate needs, analyze and diagnose, evaluate and review, and at the same time, make different dimension evaluations for each subsidiary, different functional sequence, and position, to better review the project, track the effect, and realize the process tracking, evaluation, and iteration of the whole project.

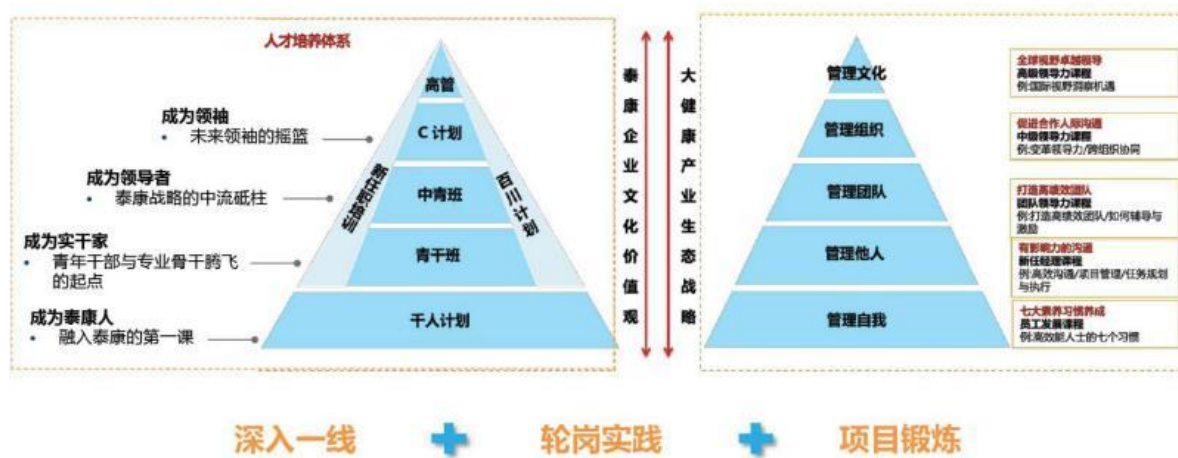


Figure 4-23 Taikang Business School's cultivation objects and projects

Third, in terms of the organizational structure of the corporate university, Taikang Business School has a teaching and training department, a special research and development department, an operation management department, and a general management department. For example, the teaching and training department includes teaching management, training management, training operation, and platform support positions. Its main functions focus on core talent training, training system construction, teachers, curriculum system construction, learning platform

construction and operation, teaching design, course development, cost management, and archives. For example, the Teaching and Training Department leads the Taikang-Harvard Digital Leadership Bootcamp Program, which continues to empower the organization for one year. More than 200 core cadres participated in the study, and the total study time exceeded 2200 hours, with an average study time of nearly 10 hours. During this period, some cadres got the change of new posts and were able to adapt to development quickly.

(Original text in Chinese: 泰康商学院的“泰康-哈佛数字化领导力训练营项目”持续赋能一年，超过 200 位核心干部参与学习，总时长超 2200 小时，人均学习近 10 小时，期间部分干部得到了新的岗位角色的转变并能快速适应发展。 Please see more details in Supplement Material S2.3).

Fourth, in terms of the management mechanism of corporate universities, Taikang Business School mainly includes two core businesses: teaching empowerment and physical services. Teaching empowerment includes the training of core talents of the whole group, the development of in-demand courses, authorized lecturer certification, and empowerment programs. Design, training project operation, learning platform construction, resource management, with dedicated and part-time personnel for different business needs, and the overall operation is based on a project system. Good delivery. In addition, in terms of physical services, there is a fixed organizational structure, and various services are provided in a standardized manner according to the fine division of labor in the team so that internal customers can obtain a better learning experience.

Fifth, in terms of the development history and outstanding achievements of corporate universities, as shown in Table 4-12, Taikang Business School took shape at the beginning of 2008, after 8 years of construction, the entity college officially landed in 2016; The corporate university has launched the Taikang Leadership Program and the Group's New Cadre Empowerment Program to improve the Group's complete core talent training system. More recently, digital training programs have fully empowered enterprises in their digital transformation process.

Table 4-12 Memorabilia of Taikang Business School

Year	Key events
2008	Taikang Internet College (predecessor of Corporate university) was established
2010	The core talent training brand project "Taikang Thousand Talents Plan" was launched
2013	The core talent training brand project "Taikang Youth Class" began to be implemented
2015	Mobile learning platform (Taikang Micro Academy) launched
2016	Taikang Business School was officially completed
2018	The corporate university started the Taikang leadership program and the group's new cadre empowerment program and improved the group's complete core talent training system
2019	Corporate University Opens Taikang Digital Leadership Training Camp to Help Digital Transformation

4.1.14 Summary of Cross-cases analysis

Based on the aforementioned case descriptions and analytical content, this study identified the fundamental characteristics and key events of the 13 case corporate universities. The commonalities and differences have been summarized in Table 4-13.

Table 4-13 The service object, organizational structure, and function positioning table of the case corporate university

Name	Functional positioning	User oriented	Organizational structure	Management mode
1. China Mobile Academy	China Mobile Academy serves in the areas of party school education, management training, online learning, and business support. It focuses on training party members and cadre, strategic promotion, and talent development. The academy aims to enhance its teaching platform, support business development comprehensively, and contribute to the company's strategic implementation and sustainable growth	Senior management talents/reserve cadres, heads of prefecture and city companies, outstanding county managers/team leaders, technical/business backbone and core employees, headquarters employees, and internal trainers	China Mobile Academy has set up four major teaching departments: Teaching Department, Online Learning Department, Party School Education Department, and Teaching Support Department, service management, operation, and maintenance support departments, and comprehensive and financial functional departments	Its operation mechanism includes training management, resource management, teaching and research management, educational administration management
2. China General Nuclear University	CGN University takes the origin of management reform, corporate strategy boost, talent doubling incubation, business development partner, corporate culture dissemination, and organizational knowledge accumulation as its mission, and is determined to become a benchmark for corporate talent training.	Junior, middle, and senior managers, strategic executives, and talents in various functional lines of the group (such as business, finance, human resources, and market development); the group's nuclear power, nuclear fuel, new energy, finance, nuclear technology, and nuclear power services All kinds of professional and technical personnel required for business development	The party secretary and chairman of the group company personally serve as the president and director of the school committee of China General Nuclear University (Party School), and the general manager of the group company serves as the first vice president and deputy director of the school committee.	Collect training requirements from business units and functions on a regular basis. Carry out general knowledge training, train certified lecturers, carry out course design and provide vocational certification training for employees of all business units and functional departments
3. ZTE College	A learning consultant for employee growth, a partner for business units, and an engine for building a learning organization.	New employees, core backbone employees, management cadres	"Learning and Development Iron Triangle" is composed of colleges, first-level competence centers in various business areas & BUHR, and human resources COE	The learning and development system of "facing actual combat, connecting the top and bottom"
4. Dongfeng Nissan University	Dongfeng Nissan University undertakes the four functional positioning of enterprise talent development learning center, enterprise management innovation	Management and technical personnel, skilled personnel	Dongfeng Nissan University has a teaching and research center, a career development center, a general education, an innovation college, and a professional college	The unified management operation mechanism and management program provide legitimacy for carrying out training work

	practice center, enterprise culture brand communication center, and enterprise alliance development cooperation center			
5. Sany Corporate University	The primary goal of a corporate university is to serve industries and businesses effectively. Following the "one body and two wings" development model, the university focuses on the service industry's growth while also emphasizing academic education and vocational training.	This corporate university provides external training services to industries, government, and social organizations. It shares Sany's management expertise and cultural values while fulfilling its social responsibilities.	Academic education and vocational education, among which the internal training business is guided by the Group Human Resources Headquarters	Internally cultivate and develop talents that meet the needs of the group's strategic development, and build a learning organization; To undertake training business of industry, government and other social organizations, export management and culture, and assume social responsibility
6. China Resources Gas College	The cradle of talent training in key positions, the center for summarizing and disseminating management practices, the platform for managers to inherit cultural experience, and the window for external exchanges of innovative practices	Member companies, nearly 50,000 internal customers, and industry partners	Gas College has fully communicated and cooperated with the Organization Development Team of the Human Resources Department in talent training.	Market-oriented enterprise university operation model
7. China Railway Fourth Bureau Corporate University	Cultivating talents, promoting change, managing knowledge, promoting culture, sharing communication	Including employees at all levels of the whole group (employee family members) and customers, partners, and stakeholders	The Learning Committee of the Bureau adopts the operation mechanism of "physical institution + virtual organization + online platform" to output "products, knowledge, talents", and establishes the Innovation College, Leadership College, Project Management College, and Jianxin College	Professionalization of teaching staff, professionalization of trainers, standardization of learning management and instrumentalization of learning methods
8. New Hope Business School	New Hope Business School focuses on implementing the company's strategy, enhancing performance, and nurturing talent. It utilizes various teaching methods, research techniques, and coaching to create a unique training model. This model integrates learning into the organization's workflow and practices effectively.	The leading team, comprehensive management talents, new business talents, professional leadership talents	The Business School of New Hope University is a first-level functional organization in the enterprise. It is subordinate to the headquarters of the joint-stock company and reports to the president. At the same time, it cooperates closely with the human resources function in combination with the human resources talent development strategy. At the same	Build learning and knowledge management platforms

			time, the business school has an independent operation function.	
9. China Construction Bank University	Openness, cooperation, joint contribution and shared benefits	Branch account managers, new employees, branch managers, fintech talents, wealth management personnel	The functional departments of the Party Committee of CCB University include the Organization Department of the Party Committee and the Office of the Disciplinary Committee. At the operational execution level, it includes two major sections: the school headquarters and the campus.	Create an online and offline life-long learning ecological platform for employees
10. Geely Corporate university	Geely University adheres to the tenet of empowering business and achieving every Geely fellow traveler and takes talent delivery, cultural empowerment, and organizational development as its functional orientation	Management cadres, professionals	The organizational structure of Geely Corporate university can be divided into first-level colleges, professional colleges, regional campuses, graduate schools, and operation management departments	Corresponding to the four major institutions to implement differentiated operating mechanisms
11. Red Star Macalline School of Management	Home furnishing industry's outstanding talent training	It is aimed at personnel training and basic operation training in the home furnishing industry. At the same time, it provides education upgrading training, non-academic upgrading training, and job skills training projects for manufacturers, distributors, and store managers in the whole industry chain.	The Group Management College adopts a project-based management model and has three functional departments: Training Department, Management Department, and Lujia Promotion Department.	Adopt a cross-departmental management mechanism: the enterprise university and various functional departments side by side to form a standard annual training budget management mechanism, as an independent operation center to develop financial plans and implement budgets
12. Robam University	Cultivate excellent employees, cultivate leading talents, and contribute to the sustainable development of small and medium-sized enterprises in China	Headquarters and branch general managers, middle management cadres, professionals, new employees	Human resources and human resources are two independent departments, but they cooperate. At present, secondary colleges of Marketing College, Lean Manufacturing College, and Technology Innovation College have been established. Executives from various business fields serve as deans of secondary colleges, focusing on the implementation of	Corporate universities encourage employees to be both extractors of knowledge and disseminators of knowledge

			training programs in secondary colleges.	
13. Taikang Business School	Building a high-end platform for the integration of production, university and research and cultivating the large health industry, focusing on strategic advantages to cultivate core talents, and continuously linking to external first-class resources, is Taikang people's own enterprise university	Including new employees (School Recruitment - Thousand Talents Program, Social Recruitment - Hundred Rivers Program), new cadres, youth classes, leadership programs, executive training, professional empowerment, business empowerment	Taikang Business School has a teaching and training department, a special research and development department, an operation management department, and a general management department.	Teaching empowerment and physical services

Through the analysis of the 13 case corporate universities, this study identified commonalities among exemplary corporate universities as follows: First, the case corporate universities exhibit notable commonalities in their mission and functional orientation. All corporate universities regard knowledge management as one of their most crucial functions. This involves systematically managing the corporate knowledge base, summarizing and disseminating experiences, and driving the development of learning organizations and knowledge-sharing platforms. Ultimately, this process empowers the development of the companies.

Second, the case corporate universities play a significant supportive role in talent development. Through diverse training and learning activities, these corporate universities have established multi-level talent pipelines within the companies. They particularly emphasize the cultivation of core talents, constantly enhancing the skills of their employees. This talent development extends beyond the internal workforce; it also encompasses the nurturing of core talents within business teams. This approach provides continuous momentum for the company's transformation in the VUCA era. Moreover, the scope of these corporate universities' nurturing programs is not limited to internal staff; it extends to the entire ecosystem, including suppliers and customers. This expansion implies that the impact of corporate universities goes beyond internal company boundaries, reaching out to the entire ecosystem and driving talent enhancement throughout the entire industry chain.

Third, there are significant differences in the management mechanisms and operational models among the case corporate universities, with most having an independent but collaborative management system. These corporate universities operate independently from the human resources departments, granting them greater flexibility and innovative space. Their independence allows them to develop training plans and courses more flexibly, better meeting the needs of both employees and the company. Despite their independent operations, these corporate universities maintain close collaboration with the human resources departments. This collaboration ensures the alignment of training and development activities with the company's overall strategic objectives. Through this close cooperation with the human resources

departments, corporate universities gain a better understanding of the company's strategic requirements, enabling them to provide training that aligns more closely with the company's development direction.

Finally, the case corporate universities demonstrate "different paths leading to the same goal" in empowering enterprise development, where they share a common overall objective but employ varying specific mechanisms. This study will focus on detailing the specific paths through which corporate universities empower enterprise development in the next chapter.

A Case-Based Text Analysis of Corporate University

The information entered into the analysis software not only includes basic information (such as enterprise scale, and corporate university scale), but also includes the strategic classification information of docking, organizational coordination, knowledge integration, resource matching, value demonstration, lean operation, and business completeness provides a research basis for the next text analysis and coding analysis.

Drawing on the processing methods of existing studies (Peng Min and Zhu Dequan, 2020; Qu Gan, 2018), this study uses NVivo software to visually analyze the collected primary and secondary data. NVivo 12 is a research tool for analyzing various forms of unstructured data. It has strong analysis functions for different formats such as documents, PDFs, pictures, audio, and video data sources. The discovery of repeating themes provides tools (Stock & Boyer, 2009). NVivo 12 is not designed for a particular method, it can meet the needs of qualitative research no matter what method is used. As shown in Figure 4-24, this research uses NVivo to analyze the data, and realize the visualization of the data by establishing nodes and coding (Wang Zhong et al., 2012).

The analysis of the original data of 380,000 words in this study is carried out as follows: First, the content of the questionnaire is extracted and cleaned to obtain basic data, the software is used to quantitatively analyze the word frequency, and the high-to-low order of high-frequency words is obtained, and NVivo 12 is drawn. A coded map and word frequency table (Meshram & O'Cass, 2013) to determine the focus of corporate university work. Secondly, relying on the three-level coding program, different levels of coding and cluster analysis were carried out to establish the relationship between each node (Zhou Qing et al., 2020), and the key elements of corporate university empowerment to the organization were sorted out; The visualization of the university's ability to improve organizational capabilities can clarify how organizational capabilities can be improved through learning activities, and provide constructive guidance and suggestions for enterprises to better improve their competitiveness through learning activities.

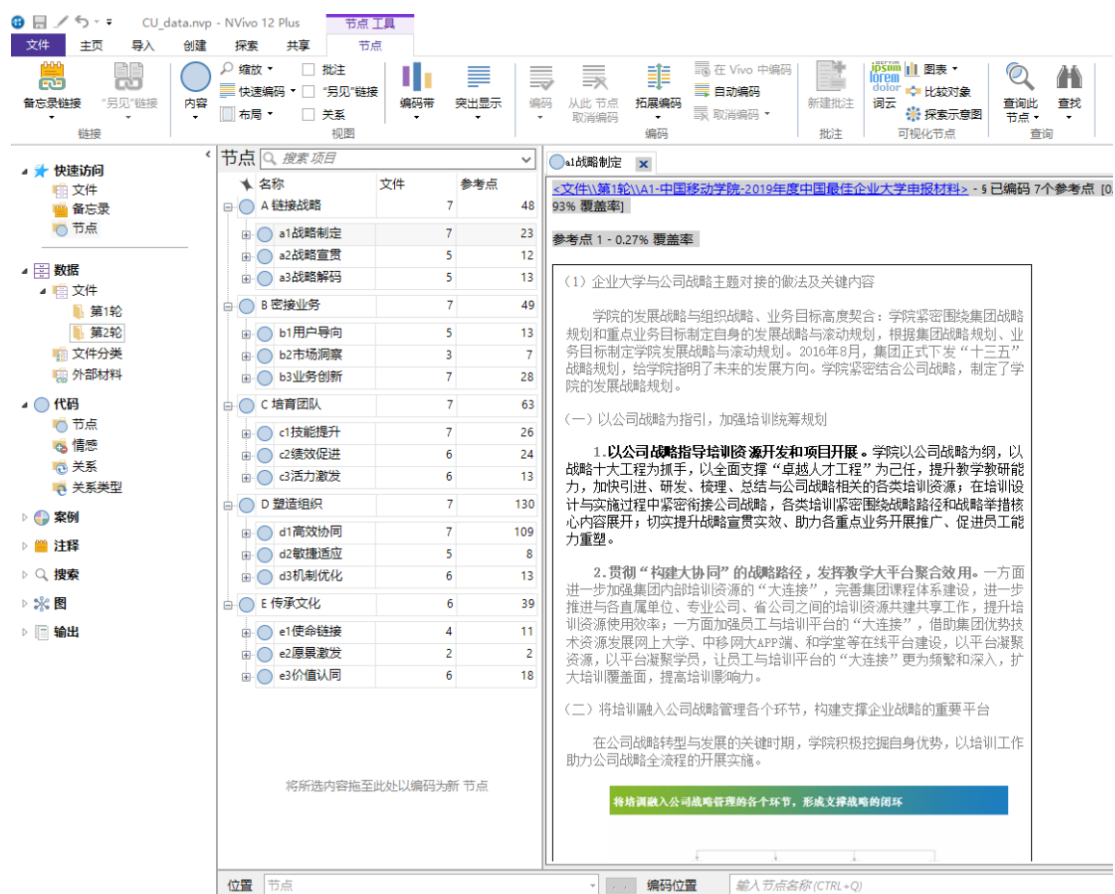


Figure 4-24 NVivo 12 software operation interface

4.1.15 Word Frequency Analysis

First, word frequency analysis is performed on qualitative data. Words are the most basic units of meaning in texts. Counting the frequency of words used in texts can reflect the trends and characteristics related to a topic (Ran Hua, 2018). Word frequency refers to the number of times a given word appears in the document and is used to evaluate the importance of a word in a document (or corpus, information set) used to express a concept. If a word appears more in the materials and interviews provided by the corporate university, it means that the corporate university pays more attention to the business/resource/capability. The word frequency analysis function of the NVivo software can list the most frequently occurring words or concepts in the raw data and can assist in identifying possible topics, analyzing the most common words used by the leaders of the best corporate universities when introducing the operation of corporate universities. To show the importance of word frequency in the total amount of text, it is

necessary to assign different proportional coefficients of word frequency, called weighting, which indicates the importance of an index item in the index system.

The specific steps are as follows: (1) Data import. With the help of the "Explore" function of the NVivo software, set the minimum word length to 2 Chinese characters, and select "Exact Match Query". Table 4-14 shows the frequency analysis of 50 high-word words with a minimum word length of 2 words. (2) Set stop words. Irrelevant words such as "aspect, situation, establishment" are set as stop words in this study, which do not help clarify the research topic (e.g., "aspect" can be used with any word to form a clear meaningless by itself) to reduce interference with the results. (3) Analyze the content and weight of high-frequency words. Previous studies have shown that the key subjects of the research can be effectively analyzed through the analysis of the proportion of word frequency (Ran Hua, 2018).

Table 4-14 Top 10 Word Frequency Analysis of Case Corporate university

No.	Word	Frequency	Percentage	No.	Word	Frequency	Percentage
1	learning	4523	2.26%	1	manager	238	0.12%
2	project	2723	1.36%	2	group company	214	0.11%
3	talent	2045	1.02%	3	training course	150	0.08%
4	nourish	1856	0.93%	4	digitizing	141	0.07%
5	develop	1798	0.90%	5	informatization	125	0.06%
6	business	1669	0.83%	6	APP	120	0.06%
7	course	1596	0.80%	7	the Internet	112	0.06%
8	Staff	1548	0.77%	8	targeted	99	0.05%
9	ability	1332	0.66%	9	general manager	92	0.05%
10	organize	1278	0.64%	10	influence	82	0.04%

In Table 4-14, the results of this study on the qualitative data of the best corporate universities show that the word *learning* has the highest frequency, with a total of 4,523 occurrences (accounting for 2.26%), which verifies the research subject corporate university. The core work of the university is all about learning; secondly, the word frequency of *project* is 2723 times (accounting for 1.36%), indicating that the best corporate university relies on various learning projects to carry out specific work. In addition, Table 4-14 shows that the top 10 in the word frequency analysis table are *talent* (word frequency 2 045, accounting for 1.02%), *cultivation* (word frequency 1,856, accounting for 0.93%), *development* (word frequency 1 798,

accounting for 0.90%), *business* (word frequency 1 669, accounting for 0.83%), *course* (word frequency 1 596, accounting for 0.80 %), *employee* (word frequency 1 548, accounting for 0.77%), *capability* (word frequency 1,332, accounting for 0.66%), *organization* (word frequency 1 278, accounting for 0.64%).

Because a phrase in the Chinese context refers to a structure composed of two or more Chinese characters. The more Chinese characters, the more specific the information reflected by the phrase. As shown in Table 4-14, this study classifies the high-frequency words obtained from the exact match query. Compared with the high-frequency words with a length of 2 Chinese characters, the high-frequency words with a length of more than 3 Chinese characters specifically include *manager* (word frequency is 238, accounting for 0.12%), *group company* (word frequency is 214, accounting for 0.11%), *training course* (word frequency is 150, accounting for 0.08%), *digitalization* (word frequency is 141, accounting for 0.07 %), *informatization* (word frequency is 125, accounting for 0.06%), *APP* (word frequency is 120, accounting for 0.06 %), *internet* (word frequency is 112, accounting for 0.06%), *targeted* (word frequency is 99, accounting for 0.05%), *general manager* (word frequency is 92, accounting for 0.05%) and *influence* (word frequency is 82, accounting for 0.04%). The above will be used as the keywords to form nodes in the subsequent third-level or second-level coding and become the core elements of the best corporate university empowerment framework for key analysis.

4.1.16 Visual Analysis of Word Cloud Graph

According to previous research, a word cloud graph is a visual presentation of a word frequency summary, which is more conducive to grasping key features (Zhang Gongrang et al., 2021). This can more intuitively and comprehensively reflect the key subjects involved in the questionnaires filled out by the best corporate universities, and show the concerns of the leaders of corporate universities. This research uses the exploration function of NVivo12, selects word frequency statistics, and selects the imported questionnaires filled out by the 7 best corporate universities. Since the information of two-character phrase responses in the Chinese context is too general, this study sets the visual word length to be greater than or equal to 3. In addition, to ensure that the research focuses more on the research problem of the corporate university

empowerment path, this research sets the visualization rule as " exact match query “, and sets words similar to " 2018 " that are irrelevant to this research as stop words.

As shown in Figures 4-25, the practical activities of corporate universities show certain common factors, which correspond to the current research status of the theoretical circle and the practical status of the business circle: First, the words " managers, group companies, training courses " Although it occupies a central position on the word cloud map and has a large font size, it is synonymous or inclusive with *leader*, *company*, and *project* in the word frequency analysis results with a minimum length of two words. Secondly, *digitalization*, *informatization*, *APP* and *internet* appear with larger font sizes, all of which appear more than 100 times, with a weighted percentage of more than 0.05 %. Among them, digitization, *informatization*, *APP*, *internet*, *intelligence* and *industry chain* are all important words to express the characteristics and technical means of the times, reflecting that the practical activities of corporate universities have been closely related to the characteristics of the times. Combined, it exhibits organizational-level characteristics that are consistent with the broader context of digital transformation.



Figure 4-25 Word cloud map of corporate university

Note: The larger words shown in the picture are “administrators”, “group company”, “digital”, “information technology”, “training program”, “Internet”, “commission”, “professional”, “targeted”, “all-round”, “standardized”.

4.2 Chapter Summary

In this chapter, this study introduces the corporate university as a case, to better understand the sample situation. These 13 corporate universities are typical representatives of China's outstanding corporate universities, and their companies are also companies with excellent performance. Therefore, the samples in this study have been used to analyze the path and mechanism of corporate universities to empower companies to develop. In addition, this study conducted a brief content analysis of all cases and grasped the key points with the help of the visual word frequency table and word cloud map, which provided cognitive ideas for the coding analysis process in the next chapter. This research will use the three-level coding procedure in Chapter 5 to conduct an in-depth study of the path mechanism of corporate universities empowering enterprises.

5 THE ANALYTICAL PROCESS OF CORPORATE UNIVERSITIES EMABLE ENTERPRISE DEVELOPMENT

This chapter mainly explains the coding process for a sample of corporate universities, discusses which key elements of corporate universities are more valuable for cultivating organizational capabilities, and enrich literature on corporate university. This study will help enterprises to understand the effective practices of excellent enterprises in the industry and excellent corporate universities in terms of talent training and organizational development, conduct benchmarking through the research process, and accelerate enterprises to improve talent quality, organizational ability and competitive advantage through training. This study involves analyzing 7 distinguished corporate universities in China initially chosen from the questionnaire. In the second stage, an additional 6 corporate universities were included for trend analysis. NVivo 12 software was utilized for coding, employing the three-level coding method rooted in theory. This approach aimed to summarize and analyze the pathways and mechanisms through which outstanding corporate universities empower corporate development.

This study focuses on how corporate universities can empower companies to develop. Many research perspectives have been proposed for corporate universities (e.g., organizational learning theory, organizational capital theory, and knowledge-based views). Therefore, this study uses reference theory to focus on issues and analysis. Data, starting from the collection of facts, deduces the best theoretical explanation, and realizes the gradual theorization of corporate university empowerment research.

Drawing on the three-level coding procedure of the procedural grounded theory, this research mainly focuses on manual coding, with the help of NVivo 12 qualitative analysis software for auxiliary coding, and gradually conducts *open coding*, *axial coding* and *selective coding* for the data collected from multiple channels. First, this study imported the initial materials provided by each corporate university into the software, and each enterprise was individually set as a case, using NVivo 12 The software encodes the data, and reads the material line by line for open coding. Next, it integrates similar content and extracts the secondary coding after completing the initial coding of all enterprises in the first stage; Third, this study

concludes five categories and establish a path model for corporate university to empower enterprise development. Finally, according to the path model, further refine the driving factor system of corporate university, complete the story line of core categories, and construct a path mechanism model for corporate universities empowering corporates development.

5.1 The Three-level Coding Process of Corporate Universities Enabling Enterprises' Development

5.1.1 Open Coding

Open coding is the first coding process of the procedural grounded paradigm, a process of decomposing, examining, comparing, conceptualizing, and categorizing data (Glaser & Strauss, 1967). This process emphasizes breaking up the data, coding it line by line (or paragraph, word), and then recombining it in a new way (Wu Xianming and Su Zhiwen, 2014; Jia Xudong and Tan Xinhui, 2010). The open coding process of this study is as follows.

The first step is "labeling". Before coding, this research needs to exclude paragraphs and chapters that are not related to this research (e.g., detailed introduction of the enterprise where the corporate university is located), and collect the interview materials, secondary data, import audio and video data into NVivo software and create different case folders. Code is a label, a meaningful unit of analysis. The process of coding is to "label" the original text (Li Liang et al., 2020): This study tries to use the original language that is close to the respondents to mark the original data, or use simplified and the initial refined words to name the nodes. Figure 5-1 shows the coding process of a case corporate university. It can be found that the coding at this level is relatively disordered. This study summarizes the key elements reflected in its main practice as nodes.

(China Railway Fourth Bureau University) builds a platform for overall learning, and achieves the goal of better serving the enterprise through the formulation of relevant control systems and the implementation and operation of key learning projects. Improve the professionalism of the training team through high empowerment, and improve the Learning projects and gathering resources to achieve the goal of not only serving the enterprise, but also serving customers. [Code b11, thinking about customer value]

This research extracts the above data content, that is, corporate universities can provide multi-channel and multi-form course resources with customers as the center, therefore, the above content is labeled as "thinking about customer value (Code b11)".

Corporate universities continue to explore new training models for modern enterprises, and use online universities to realize the functions of pre-training organization, mid-term monitoring, and post-tracking, and promote the integration of online universities, physical teaching and mobile terminals:

Corporate university promotes change learning, comprehensively enhances the organization's active adaptation and continuous innovation capabilities [code d11, cultivating organizational capabilities], reaches organizational consensus, and promotes enterprise reform and development.

In this study, the above-mentioned raw data were labeled as "cultivating organizational capacity (Code d11)". It is worth noting that the same content may correspond to multiple tags at the same time, which is allowed in the grounded analysis method:

In order to meet the group's multi-project, multi-base, cross-regional, and large-scale talent training needs, and to solve the learning problem of employees in remote areas, CGN University has built a mobile learning platform for employees, which breaks through the limitations of time and space and enables group learning to officially enter mobile The Internet Age. [Code d123, Mechanisms for Unblocking Information; Code d134, Digital Organization]

Since this piece of material can be identified as "Information Unblocking Mechanism (Code d123)", it has also been labeled as "Digital Organization (Code d134)". And so on, this study codes each case corporate university one by one.

Furthermore, to reduce the effect of selective perception of investigators' individual biases, two research colleagues independently coded and their coding was compared to the researcher's. They are senior practitioners with 10 years of experience in corporate universities, doctoral students majoring in business management, trained in coding technology, and have qualitative coding experience. They can consider issues from the perspectives of researchers, corporate universities themselves, and third-party scholars, so as to avoid too much convergence of ideas, and then they can form a variety of ideas to form cross-validation to improve the reliability and validity of coding. Finally, the coding was checked by three senior experts who were not involved in the coding, who were researchers in the field of corporate universities independent of this study, and were able to make independent corrections to the coding process without interference from the researcher. To sum up, after independent coding and mutual verification by two researchers, as well as inspection and supplementation by three own experts, this study established a total of 324 free nodes in the "labeling" stage.

The second step is "conceptualization". In this study, the free nodes belonging to the same phenomenon are classified under one node, and a complete definition is made. The basis of qualitative analysis is the initial conceptualization of the original data. Based on the free nodes obtained by "labeling", combined with the practical content of corporate universities, this study further refined and summarized the free nodes, and obtained 76 conceptualized nodes.

The third step is "categorization". In this study, conceptual nodes related to the same phenomenon were merged again, and in the process, the study obtained a total of 23 categories. Two examples are used here to show how conceptual nodes can be combined into categories:

[China Mobile University] have carried out special training for new technology and new business talents , combined with the company's new technology and new business promotion needs [Code b211, solve practical problems] , apply online live broadcast, learning area, series A series of theme trainings are carried out by combining online

and offline micro-courses and face-to-face training, such as technical strategy special training, Internet of Things\big data\cloud computing series live broadcast, VoLTE marketing special study. All kinds of teaching and training programs of corporate universities are closely related The company's strategy and business development need to be carried out and integrated into the business development and professional team building, so it has been actively involved and closely cooperated by the business department [Code b212, helping the business discuss key tasks], which has laid a solid foundation for the success of the project.

The business unit regards us [Dongfeng Nissan university] as a popular platform for the company's new technology and new business [Code b213, to solve practical problems], and a partner trusted and relied on by the business unit. [Code b213, to gain the trust of the business unit]

From this, this research conceptualized three nodes belonging to the same category of concepts in the material, namely "solving practical problems (Code b211)", "helping the business discuss key tasks (Code b212)" and "gaining the trust of business units (Code b213)" is aggregated into one category - "Helping businesses identify markets (Code b21)".

According to the original data of China Mobile Academy:

The Goose Team Project of Corporate University for the employees of the company's headquarters conducts project rotation training every year, focusing on theoretical improvement, vocational skills, execution, communication skills, thinking skills, psychological quality, innovation ability, and team building. Focus on improving the teamwork ability of the headquarters staff, create a departmental 'geese' culture, improve organizational cohesion, and promote the common improvement of individual and team performance. [Code c211, high team performance]

Further evidence from Sany Corporate university was used to identify the same code:

Through training to help employees learn and grow independently, shorten their time to become talents, build scale leadership, and improve team combat effectiveness from the level of skills. Through the introduction of Huawei and communication

benchmarking enterprises, technical design training, on the basis of knowledge management, can effectively enhance the creativity of the enterprise, thereby enhancing the performance and competitiveness of the enterprise. [Code c211, high team performance]

Similarly, using data from China General Nuclear University further coding was done:

The corporate university research group attaches great importance to improving the employee learning and development system, cultivating employees' study habits, and improving their ability to be competent for the job; corporate universities discover the advantages of employees through the career guidance system and better match their advantages to work [Code c 212], improving performance and taking into account people-oriented orientation] corporate university helps employees understand the value and meaning of work, discover internal driving forces, and devote more attention to work.

Therefore, this study aggregates "high team performance (Code c211)" and "improving performance while taking into account people-oriented orientation (Code c212)" to obtain the category "improving team performance (Code c21)".

In the open coding stage, this study extracted 76 conceptual nodes from 324 free nodes describing the development of enterprises empowered by corporate universities, and further abstracted 23 categories. Table 5-1 presents some of the source data evidence for the categories.

Table 5-1 Selective coding and typical evidence content

Open coding category	Category connotation	Example citation
Participate in high-level seminars (a11)	Can be invited to participate in high-level seminars on company management topics	Provide information, methods and tools for topics closely related to the Group's strategic development through the high-level EDP training program. On this basis, the participating executives discuss and output opinions and suggestions for the Group's strategic reference (China General Nuclear University)
Assist high-level decision-making (a12)	Plan research and provide high-quality reports or solutions in line with the company's high-level concerns	Through the feedback of students, the college forms a report on the problems found during the implementation of the strategy and submits it to the leaders of the group company and relevant business departments as an important basis for subsequent strategy formulation, publicity and implementation and refinement of measures (China Mobile Academy)
Strategic Execution Capability (a21)	Possess the ability to design and operate solutions that drive the company's strategy through execution	The college takes the company's strategy as the key link, takes the ten strategic projects as the starting point, and takes the comprehensive support of the "Excellent Talent Project" as its own responsibility, improves teaching and research capabilities, and accelerates the introduction, research and development, sorting out, and summarizing various training resources related to the company's strategy; In the process of training design and implementation, the company's strategy is closely linked, and various trainings are carried out closely around the core content of strategic paths and strategic measures; effectively improve the effectiveness of strategic publicity and implementation, assist in the promotion of key businesses, and promote the reshaping of employees' capabilities (China Mobile Academy)
Strategic Landing Capability (a31)	Keep up with the development of the industry, plan the action theme according to the company's strategy and convert it into a landing project	Through the development of organizational learning blueprint and implementation of strategic decoding, the implementation and implementation of corporate strategies are promoted by publicizing or embedding learning strategy decoding courses in various key and key groups of learning projects (China Railway Fourth Bureau Corporate university)
Think about user value (b11)	Improve employees' thinking about problem solving from the perspective of customer value	Carry out rapid iterative course development projects for the new technologies and new businesses that the company focuses on to promote the rapid popularization of new technology and new business knowledge. Carry out targeted learning projects to effectively improve the new business capabilities of the company's key talents and ensure the effective development of the company's business transformation

Customer Management Capability (b12)	Improve staff skills in customer experience management and service	The business class went to Huawei to benchmark and learn the concepts and models of business-driven digital transformation, the manufacturing class went to Shanghai Volkswagen and Bosch to benchmark to learn intelligent manufacturing, the business class went to Shanghai Baosteel to benchmark to learn procurement informatization construction, and the marketing class went to BYD to visit and exchange Parts sales and after-sales service (Dongfeng Nissan University)
Help business identify markets (b21)	Help business units identify and gain insight into market and customer changes	Special training for new technology and new business talents: Combined with the company's new technology and new business promotion needs, apply online live broadcast, learning area, series of micro-courses, and face-to-face training to carry out a series of theme training, such as technical strategy special training, Internet of Things\Big data\ Cloud computing live broadcast, VoLTE marketing special study (China Mobile Academy)
Help business iterate (b31)	Help business units innovate and iterate on businesses, products and services	Improve business operation capabilities. The expansion of the enterprise scale has resulted in uneven quality of the managers of the regional centers and material supply centers and insufficient professional quality. Therefore, by establishing a complete set of scientific learning system and training programs, they can quickly master the various requirements based on the enterprise strategy. Knowledge and skills, improve job competency, to ensure production supply and service quality (China Railway Fourth Bureau Corporate university)
Nurturing high-quality employees (c11)	Help the company develop employees who can actively undertake and perform high-quality work	Through training, we help employees to learn and grow independently, shorten their time to become a talent, and at the same time build large-scale leadership, and improve team combat effectiveness from the skill level. Through the introduction of technical design training from Huawei and communications benchmarking companies, based on knowledge management, it can effectively enhance corporate creativity, thereby enhancing corporate performance and competitiveness (Trinity University)
Cultivating Employee Competitive Advantage (c12)	Helping companies develop employees who better meet customer needs than their competitors	Starting from the needs of employee growth and development, DFN has established a credit system based on employee growth. Promote talent training with triple guarantees of hours, credits and certification, and design four course packages: induction package, credit package, certification package, and improvement package. Two complete training systems for management technicians and technicians have been formed (Dongfeng Nissan University)
Improve team performance (c21)	Help companies develop team leadership that balances	Goose team project for the employees of the company headquarters: carry out project rotation training every year, focusing on theoretical improvement, professional skills, execution, communication skills, thinking

	performance and people orientation	skills, psychological quality, innovation ability, and team building, focusing on improving the teamwork ability of headquarters employees, creating Departmental "geese" culture, improve organizational cohesion, and promote the common improvement of individual and team performance (China Mobile Academy)
Stimulate team creativity (c31)	Formed effective programs and practices in nurturing and stimulating team vitality	Through training, we help employees to learn and grow independently, shorten their time to become a talent, and at the same time build large-scale leadership, and improve team combat effectiveness from the skill level. Through the introduction of technical design training from Huawei and communications benchmarking companies, based on knowledge management, it can effectively enhance the creativity of the enterprise, thereby enhancing the performance and competitiveness of the enterprise. To jointly promote the strategic development of the group with training ability, learning ability, combat ability and creativity (Trinity University)
Develop organizational skills (d11)	Clearly define company organizational capabilities and be able to diagnose and nurture them	In terms of organizational capability improvement: By building a cloud-based employee capability improvement system that is "combat-oriented and connected from top to bottom", we will build a capability middle platform, quickly obtain business needs, and complete capability supply, so as to achieve normalization of front-line empowerment, globalization of resource construction, and capability Customized configuration, real-time capability scanning, co-oriented to customers and front-line, dynamic response of capability on demand (Zhongxing University)
Efficient collaboration mechanism (d12)	Help companies promote smooth information communication and cross-team collaboration	Training creates value, and learning achieves the future. In order to continuously expand the vision of management cadres at all levels and continuously improve their leadership and management capabilities, CGNPC strives to build a systematic management talent training system with its own characteristics, and has implemented the Egret Plan, Changwan Leadership Forum, and Executive EDP training. , A series of high-quality training programs in the CGN Lecture Hall, forming a pattern of "two wings flying together" for management and technical personnel training (CGN University)
Optimize organizational structure (d13)	Help companies optimize organizational structure and resource allocation to improve overall performance	The company establishes an online learning platform to provide employees with personalized empowerment and self-growth methods without time and geographical restrictions, realizes the construction and management of the digital system of corporate courses and teachers, and establishes a new learning environment for online learning resources and employee online learning communities. , Through the management of online learning, realize the application of O2O teaching mode, digital class management and operation, staff online examination, ability evaluation and powerful online learning management system,

		reduce the offline training cost of enterprise employees, and create a corporate learning-oriented cultural organization (Trinity University)
Agile to the customer (d21)	Drives the organization and scheduling of tasks throughout the company around customer needs	Significant increases in organizational agility and ability to respond to national policies. For example, according to the national PPP investment policy, timely design and implement investment and financing project training, and timely design and implement overseas personnel learning solutions for the Belt and Road Initiative (China Railway Fourth Bureau Corporate university)
Responding to customers agilely (d22)	Drive rapid response to market and customer changes throughout the company	The college has become a booster for the implementation of the ten strategic projects: focusing on the ten strategic projects, focusing on strategic priorities, and cooperating with the group's business departments to agilely develop new technologies and new business series for the Internet of Things, converged communications, network operation and maintenance, 5G, cloud computing, and big data Courses, collect best practice cases, and open strategic ten engineering study areas through online universities and China Mobile University APP (China Mobile Academy)
Organizational Authorization (d31)	Promote the company's reasonable authorization to make employees the protagonists in the work	The "Group Training Management System" clearly stipulates that each unit of the group should take "authorization to work, full staff training, lifelong education" and "training-assessment-authorization-employment" as the training management concept and training management principle; Establish an authorization training system, that is, employees must undergo appropriate training before performing various tasks, and only after passing the assessment can they be authorized to engage in relevant positions. Different positions require different authorizations (China General Nuclear University)
Reduced administrative redundancy (d32)	Promote the company to optimize the management process to make the work more concise and efficient	Through the reform of the post salary system, the number of core personnel in each post and department of the group has been reduced by 20%, and the total salary has been increased by less than 20% compared with that before the optimization and adjustment. The quality and quantity of incoming and retained talents meet the requirements of organizational development (Sany the University)
Create incentives (d33)	Help the company to establish an incentive and restraint mechanism with clear guidance and standards	By organizing the study, practice, review and transfer of the advanced training program of learning technology, a learning and research and development team of more than 30 people has been established, which has become the core technical force for the future development of corporate universities; through more than 20 training certifications for trainers, it has established A team of more than 1,200 internal trainers, introduced a team of 200 external trainers, held teacher forums, and honored outstanding internal trainers for three consecutive years; Personnel training manager team (China Resources Gas College)

Call of Duty (e31)	Help employees understand the meaning of work from the company's mission	In the special seminar held by the SASAC branch of the Central Party School, China Mobile Party School, as an outstanding representative of the SASAC branch system, shared the experience entitled "Strengthening political leadership and keeping in mind the mission and responsibilities". "Party construction assistant" introduced the teaching management practice of Mobile Party School in insisting on face-to-face and strict school governance and assisting the group's party construction work (China Mobile Academy)
Vision Inspiration (e32)	Help employees inspire work commitment from company goals and vision	China Railway Fourth Bureau is creating a happy enterprise. With the cooperation of Tsinghua University, the corporate university has established a happy enterprise evaluation system with Tsinghua University, and spreads the meaning of a happy enterprise and a happy life to all employees through the moral lecture hall (China Railway Fourth Bureau Corporate university)
Value Inspiration (e33)	Helping companies gain employee buy-in to core values	One of the positioning of DFN University is the corporate culture brand communication center: improve the construction of corporate culture system, inherit the excellent genes of "integration" and "transcendence", enhance employees' sense of belonging and identity to the company, and communicate corporate culture to the media and the public. , enhance the overall image of the enterprise. Dongfeng Nissan also integrates the company's behavioral values into training, for example, the training of managers constantly emphasizes the principle of customer-oriented process sorting (Dongfeng Nissan University)

5.1.2 Spindle Coding

Spindle coding emphasizes causal conditions, sorting out the context of phenomena to obtain results. Comparing and analyzing the 23 categories finally obtained by open coding, merging similar expression nodes, 15 main categories of "strategy formulation", "strategy publicity and implementation", "strategic decoding", "user orientation" and "market insight" emerged. The main category is defined based on the viewpoints of existing literature (as shown in Table 5-2).

Table 5-2 Spindle coding process

Main category (15)	Connotation of the main category
Strategy formulation (a1)	Corporate universities can be invited to participate in high-level seminars on company management topics, and plan research and provide high-quality reports or solutions in line with the company's high-level concerns
Strategic publicity (a2)	Corporate University has the solution design and operational capabilities that drive companies to implement strategy through execution
Strategy decoding (a3)	The corporate university closely follows the development of the industry, plans the action theme according to the company's strategy and transforms it into a landing project
User orientation (b1)	Corporate university can promote the organization and arrangement of tasks around customer needs, and promote rapid response to market and customer changes.
Market insights (b2)	Corporate University can help business units identify and gain insight into market and customer changes
Business innovation (b3)	Corporate University helps business units innovate and iterate on businesses, products and services
Skill up (c1)	A corporate university can help companies develop employees who are motivated and perform high-quality work, and who can meet customer needs better than their rivals
Performance promotion (c2)	Corporate University can help companies develop team leadership that balances performance and people-oriented orientation
Vitality stimulation (c3)	Corporate University has formed effective programs and practices in nurturing and stimulating team vitality
Efficient synergy (d1)	Corporate University clearly defines the company's organizational capabilities and can diagnose and cultivate, which can help companies promote smooth information communication and cross-team collaboration, optimize organizational structure, and improve resource allocation efficiency
Agile adaptation (d2)	Corporate University can drive the company to organize and arrange tasks around customer needs, and respond quickly to market and customer changes
Mechanism optimization (d3)	Corporate university can promote the reasonable authorization of the company to make employees the protagonists in the work, optimize the management process to make the work more concise and efficient, and help the company to establish an incentive and restraint mechanism with clear orientation and standards
Mission connects (e1)	Corporate University can help employees understand the meaning of work from the perspective of the company's mission

Vision inspiration (e2)	Corporate University can help employees inspire work engagement from company goals and visions
Value identity (e3)	Corporate universities can help companies gain employee buy-in to core values

First, the main category *Strategy Formulation* (Code a1) is condensed from two subcategories—Participating in high-level discussions (Code a11) and assisting high-level decision-making (Code a12)—meaning that corporate universities can be invited to participate in high-level corporate management topics, seminars, and plan research and provide high-quality reports or programs that are closely related to the company's high-level concerns.

The main category *Strategy Advocacy* (Code a2) comes from a single subcategory (Code a21, strategy implementation), which means that corporate universities have the ability to design and operate solutions to promote the company's strategy throughout the implementation.

The main category *Strategy Decoding* (Code a3) also comes from a single subcategory (Code a31, strategy landing), which means that corporate universities closely follow industrial development, plan action themes according to the company's strategy, and convert them into landing projects.

Second, since "strategy formulation→strategic publicity and implementation→strategic decoding" is a classic process and link for enterprises to formulate strategies, this study grouped three main categories into one category, and named this enabling mechanism for corporate universities to provide strategic services for enterprises for the "linking strategy". The main category *User Orientation* (Code b1) is condensed from two subcategories (Code b11, thinking about user value; Code b12, customer management ability), which means that corporate universities can promote the organization and arrangement of customers' needs from top to bottom the mission is to drive the company up and down to respond quickly to market and customer changes. The main category *Market Insights* (Code b2) comes from a single subcategory (Code b21, helping businesses identify the market), which means that corporate universities can help business units identify and gain insight into markets and customer changes. The main category *Business Innovation* (b3), derived from a single subcategory (code b31, Helping Business Iteration), refers to the corporate university's ability to help business units innovate and iterate businesses, products and services.

Third, the main categories of "user orientation", "market insight" and "business innovation" all belong to the close contact between the corporate university and the business

department of the enterprise. Serve. Therefore, this study names the enabling mechanism of corporate universities to provide services to business departments as "close-knit business".

The main category *Skill Improvement* (Code c1)" is condensed from two sub-categories (Code c11, cultivating high-quality employees; Code c12, cultivating employees' competitive advantage), which means that corporate universities can help companies cultivate talents who can actively undertake and high-quality employees who are better at meeting customer needs than their rivals. The main category *Performance Promotion* (Code c2) comes from a single subcategory (Code c21, improving team performance), which means that corporate universities can help companies develop team leadership that can balance performance and people-oriented orientation. The main category *Energy Stimulation* (Code c3) comes from a single subcategory (Code c31, stimulating team creativity), which refers to the effective programs and practices that corporate universities have formed in cultivating and stimulating team vitality.

Fourth, the empowerment mechanism of corporate university is closely related to the team. If the individual skills of employees are improved, it may lead to the improvement of team performance, which in turn stimulates the vitality of the team. Therefore, this study named the enabling mechanism of corporate universities to stimulate team vitality through skills and performance training as "cultivating teams". The main category *Efficient Collaboration* (d1) is composed of three subcategories (Code d11, cultivating organizational capabilities; Code d12, efficient collaboration mechanism; Code d13, optimize organizational structure), which refers to the clearly defined corporate university. It can help the company to promote smooth information communication and cross-team collaboration, optimize the organizational structure, and improve the efficiency of resource allocation. The main category *Agile Adaptation* (d2) is aggregated by two subcategories (Code d21, agile to adapt to customers; Code d22, agile to respond to customers), which means that corporate universities can promote the organization and arrangement of the company around customer needs tasks, and respond quickly to market and customer changes. The main category *Mechanism Optimization* (Code d3) consists of three subcategories (Code d31 , organization authorization; Code d32 , reduce management redundancy; Code d33 , establish incentive mechanism), which means that corporate universities can promote reasonable authorization of companies Let employees become the protagonists in the work, optimize the management process to make the work more concise and

efficient, and help the company to establish an incentive and restraint mechanism with clear guidance and standards.

Fifth, "efficient collaboration", "agile adaptation" and "mechanism optimization" reflect that corporate universities can promote organizational development by cultivating organizational capabilities, optimizing organizational structures, rapidly responding to customer needs, promoting corporate empowerment, and establishing mechanisms. The corporate university is named "Shaping the Organization" through the enabling mechanism that optimizes the organizational structure. The main categories *Mission Connection* (Code e1), *Vision Inspiration* (Code e2) and *Value Identity* (Code e 3) all emerged from a single subcategory, which refers to the corporate university that can help employees understand work from the perspective of the company's mission. Meaning and from the company's goal and vision to stimulate work engagement, help the company to gain employees' recognition of the core values. Corporate university can build a bridge of identity between employees and company mission, company vision and company values. This study named the mechanism of corporate university empowering enterprise development by building corporate culture as "inheritance culture".

To sum up, the five dimensions of corporate university empowerment enterprise development identified in the spindle coding stage are shown in Figure 5-2 (Figure 5-3 is in English) , showing the analysis process of the spindle coding stage: this study lists 15 categories as There are five categories, corresponding to the five dimensions of corporate university empowerment of enterprise development , and named as "linking strategy", "closely connecting business", "cultivating team", "shaping organization" and "inheriting culture".

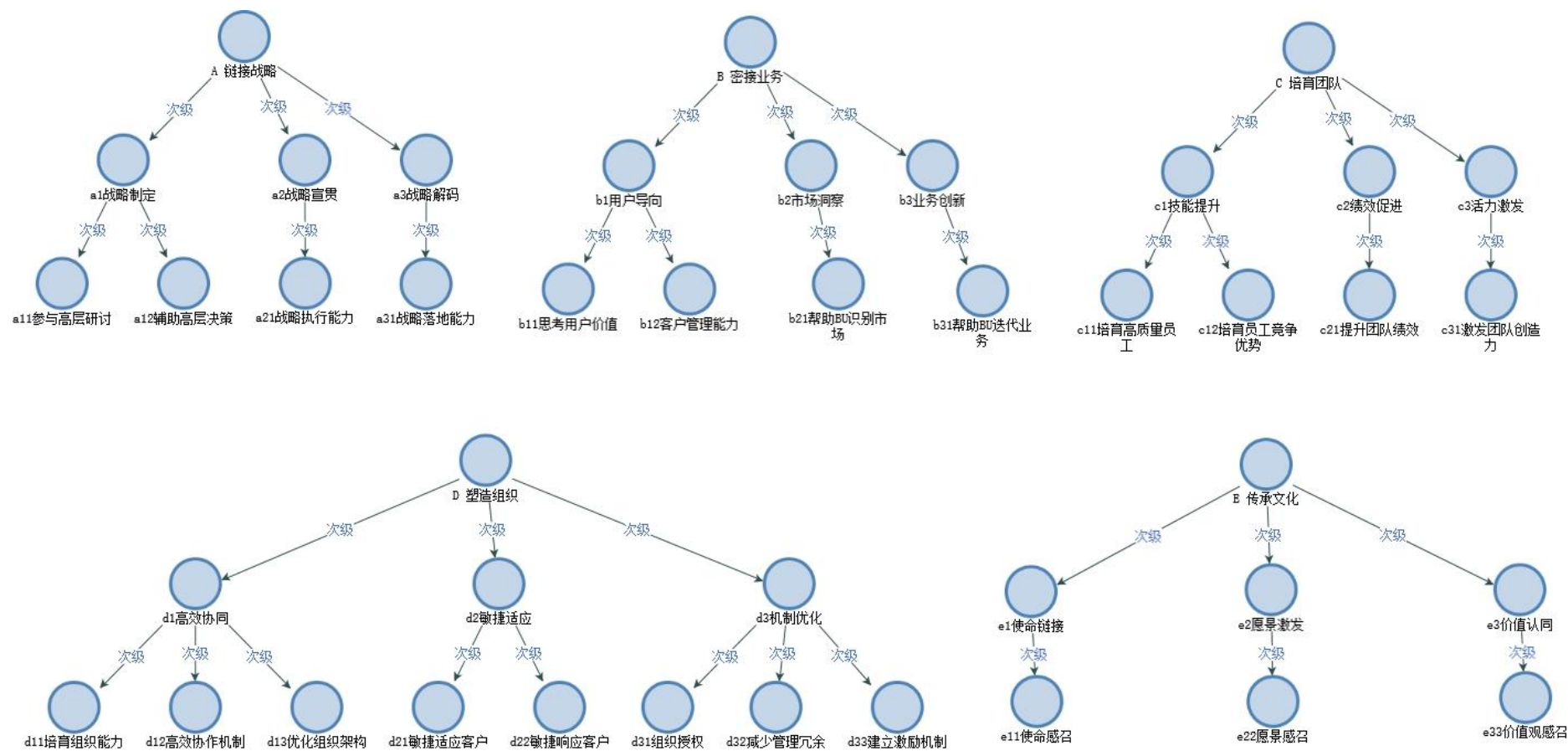


Figure 5-2 Spindle coding process

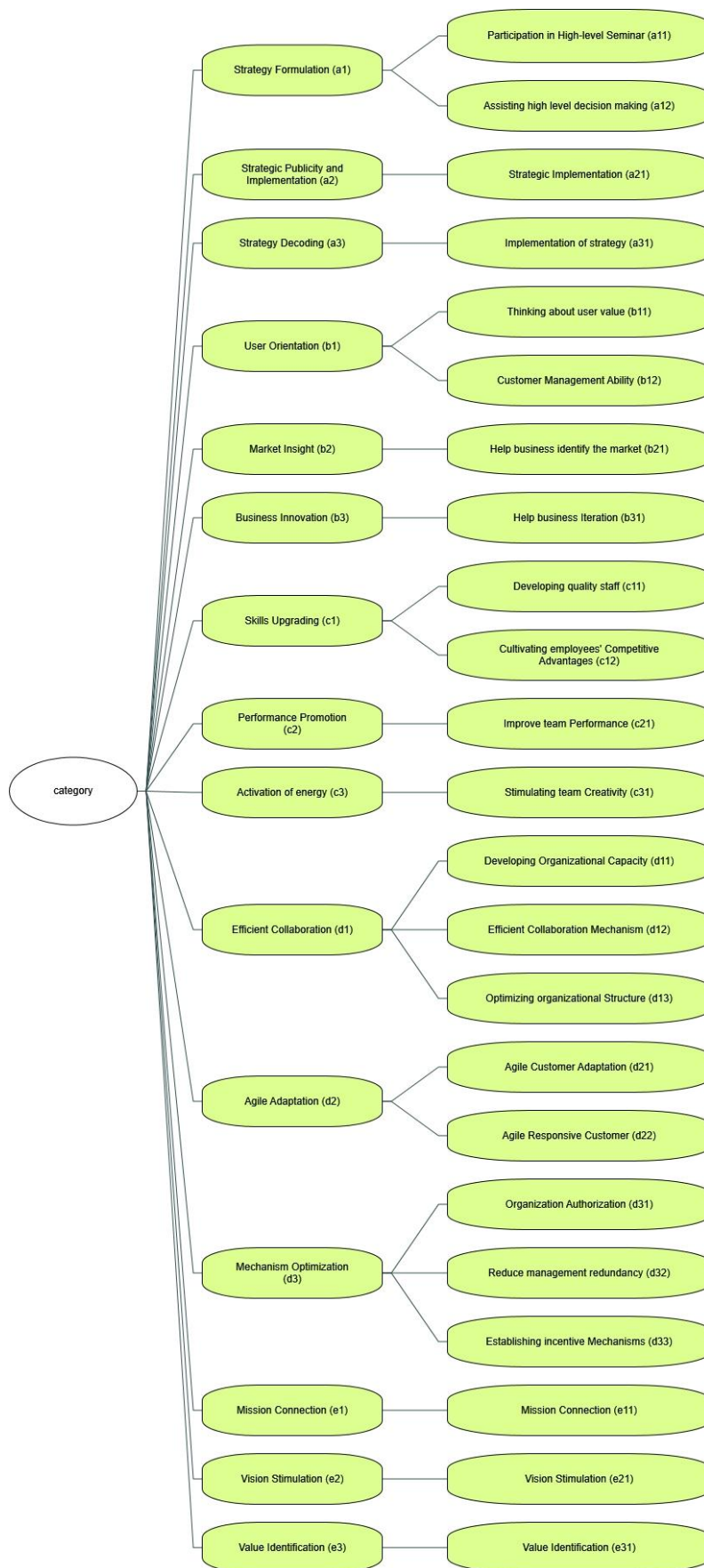


Figure 5-3 Spindle coding process (in English)

5.1.3 Selective Coding

Selective coding summarizes and refines existing categories, and finally obtains a core category that can summarize all categories, establishes the relationship between core categories, main categories and other categories, and describes phenomena and driving factors through a clear "story line". And then developed into a complete theory (Zhou Qing et al., 2020; Jia Xudong and Tan Xinhui, 2010). Therefore, the causal mechanism identified in the selective coding part is "the path of corporate university empowerment of enterprise development", that is, the corporate university based on the Chinese context through enabling strategy, enabling business, enabling enterprise, enabling team and enabling culture Five paths, improve the ability of individuals in the organization, and ultimately achieve the improvement of organizational ability and performance, thus building a path model for corporate university to empower enterprise development. Figure 5-5 shows the path model of corporate university empowering enterprise development.

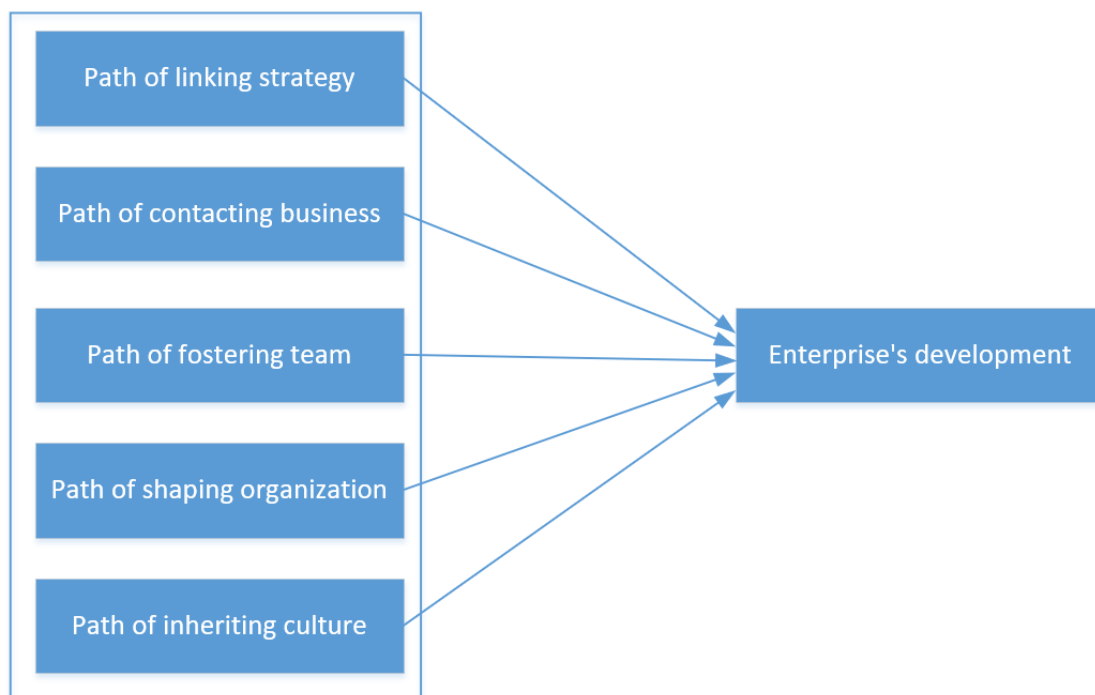


Figure 5-4 A path model ("storyline") for how corporate universities enable enterprise development

5.2 Explanation of the Path Model of Corporate Universities Enabling Enterprises Development

Combining existing research and case enterprise practice, this research establishes a path model and analyzes how corporate universities can empower enterprise development through five paths, including linking strategies, close business connections, nurturing teams, shaping organizations, and inheriting culture.

5.2.1 Corporate Universities Enabling Path of Linking Strategy

Path of Linking Strategy refers to the way that corporate universities connect themselves and the company's long-term strategies through strategy formulation, strategy publicity and strategy decoding, and use the top-down knowledge management system to empower the sustainable development and development of enterprises. We consider corporate universities as the primary driving force behind knowledge creation, especially in the whole process of knowledge management (Chen et al., 2021). Corporate universities actively engage in knowledge management activities, driving innovation and business development. This pathway emphasizes how corporate universities transform internal knowledge into concrete business strategies, enhancing the competitiveness of enterprises.

Business model innovation. Through dialogue with the existing literature, this study found that the corporate university's empowerment of enterprise development through the "strategic" path has been recognized by many scholars. Rademakers (1988) proposed that corporate universities should be integrated with corporate strategic goals. Heinz (2001) also proposed that, as an important part of an enterprise, corporate universities will consider resource actions based on the company's long-term strategic goals and help enterprises drive the appropriate resource mix. Furthermore, Antonelli et al. (2013) argue that corporate universities play an important role between corporate strategy and internal knowledge. Some scholars also analyzed the challenges brought by the digital economy to corporate universities, and believed that the development of corporate universities of information and communication companies must be linked with the development of innovation strategies, and further support business model innovation (Rhéaume & Gardoni, 2017).

The analysis results of this study are in line with the existing research point of view, Elkeles and Phillips (2017) The biggest feature of corporate universities is the strategic nature of helping organizations realize value, and it also supports the strategic statement of corporate universities proposed by Able and Li (2012), namely the educational goals of a corporate university are aligned with the strategic needs of the organization. In addition, the analysis results of this study are also extended on the existing theoretical basis. For example, the key behavior of participating in the original content of the high-level discussion reference point is described as "being invited to participate in high-level discussions on the topic of company operation and management", "during the company's strategic transformation and transition period, the academy organized strategic experts from various provincial companies to conduct strategic discussions and research" (China Mobile Academy), "organized executive lectures including board members, and business management training attended by executives of various business units. and regular book clubs" (Dongfeng Nissan University).

The opinions and reports formed by participating in high-level discussions can often play a supporting role in decision-making. For example, China Mobile Academy "at the same time, the feedback and suggestions from the students are formed into reports on important strategic themes and businesses, and submitted to the leaders of the group company and strategy Department, as an important reference for strategy formulation", another example is the practice of China General Nuclear University "for the top management cadres of the group, a 2-day Leadership Forum' is held every year, and the participants include group members and department heads. and the top leaders of all second-tier member companies. The leadership forum usually combines the outstanding issues faced by the group in that year, such as execution, cadre management, strategic direction adjustment, and key issues of corporate culture." Dongfeng Nissan University "The annual executive forum is the Nissan's important business activities and many major strategic decisions originate from group dialogues in executive forums. The V-up promotion team has always been responsible for the design of executive group dialogues", so the above two points can be combined into "strategy formulation", It is defined as a corporate university that can support decision-making by participating in the formulation of corporate strategies and realize the empowering function of linking strategies. It is consistent with the view of Elkeles and Phillips (2007).

In addition, the word frequency visualization analysis of a single enabling path is performed in sequence according to the five first-level nodes (as shown in Figure 5-6), and the search range setting is set to each construct in turn. The grouping is set to "specific", the larger the font, the higher the frequency of the phrase, and the more important the weight is. The larger words in the "Path of Linking Strategy" are *strategy*, *company*, *enterprise*, *idea*, *action*, and *learning*, followed by *decoding*, *implementation*, *theory*, *support* and *strategy*. Combined with the three sub-paths (*Strategy formulation*, *Strategy implementation*, *Strategy decoding*) in the link strategy path, confirms that the case corporate university is empowered from the three major parts of strategy formulation, publicity and implementation and decoding, which involves participating in high-level discussions, assisting high-level decision-making, and improving strategic decision-making and implementation capabilities.



Figure 5-5 Word cloud map of "Path of Linking Strategy"

Note. The larger words are: *strategy*, *company*, *enterprise*, *action*, *thought*, *learning*, *decoding*, *implementation*, *leadership*.

5.2.2 Corporate Universities Enabling Path of Contacting Business

Path of Contacting Business means that corporate universities empower business departments through internal knowledge management systems, help business departments establish user-oriented thinking, market insight capabilities, and ultimately achieve business innovation. In this path, corporate universities are regarded as the platform for knowledge transfer and training, dedicated to strengthening knowledge management activities within the

internal faculty. Therefore, the "closely-knit business" pathway emphasizes that corporate universities are allowed to adopt the more “business-like” approach (Al Mahameed et al., 2023), to enhance employees' teamwork, collaboration, and innovation capabilities through the internal dissemination of knowledge and training.

The result of the analysis suggests that corporate universities could provide "business scenario" which is used to help identify, understand, and document business requirements (Viergutz & Knust, 2014). This study identified a series of curriculum development, talent development project design and implementation, as well as supporting training methods and learning tools in the case corporate university. Among them, corporate universities are most typical in enabling businesses as digital learning platforms, providing fast-response, high-quality learning channels and knowledge management platforms for leaders, business departments and employees.

A number of case companies and universities have proposed to gradually transform into digital learning platforms. Through online course sharing and building a mobile and PC-side collaborative learning platform, they have established a digital learning system in the fields of strategic consulting, business management, and learning and training. For example, CCB University's curriculum construction centers around the mission of "serving the society, serving strategy, and serving employees", and conducts online, offline and digital courses through independent development, cooperative development, and course introduction. In the improvement of digital management capabilities and the training of digital talents, CCB University has developed and launched the "CCB-Harvard Digital Talent Online Training Camp". The international team of CCB University and the team of enterprise learning experts from Harvard Business School will focus on the ability of enterprises to adapt to the development of the digital economy era and cultivate digital competitive advantages, focus on the development needs of China Construction Bank's digital business strategy and the bottleneck of digital talent training, and introduce domestic and foreign theories. With the cutting-edge research wisdom and business cases of practice, closely linked to business scenarios, learning content can be customized and designed on demand. CCB University has implemented a new digital learning model through live and recorded lectures, online case

catalysis, and group discussions, and built a cross-border digital learning and cross-line discussion and exchange platform for trainees.

At the same time, through agile iterative learning materials and continuous optimization of operational actions to improve employees' digital learning experience and learning effectiveness, it truly realizes that the online content of the course is a high-quality product, which is worthy of employees repeated self-learning and sharing with the team, so as to maximize the precipitation and transformation of learning results. value. Therefore, the above approach is an important form of proving that corporate universities can fully empower corporate development in business scenarios, become a promoter of business performance, and ultimately cultivate excellent talents for the company.

Therefore, this study considers that corporate university, as an internal platform that undertakes training and learning functions, promotes two-way information communication between business departments and enterprises. This echoes the views of the existing literature: Taghizadeh et al. (2021) analyzed the relationship between learning and knowledge management, that is, how learning promotes knowledge management, and how knowledge management promotes organizational learning, and found that training and many processes in enterprises are Highly integrated, including management, research and development, marketing and communication, sales, external network.

In terms of supporting business, some scholars have also proposed that corporate universities can support business model innovation: the top management of enterprises must consider the implementation of corporate universities from a long-term perspective - which means that corporate universities must be carefully designed, not only to Become the core of innovation strategy and further support the business model innovation of enterprises (Rhéaume & Gardoni, 2017). With the development of corporate university, the knowledge management function of corporate university is also continuously enhanced. There are also studies that the knowledge management functions of corporate universities include knowledge transfer, knowledge creation and knowledge services for intra-enterprise entrepreneurship. Corporate universities empower business sectors by transferring, creating and providing knowledge, expanding the knowledge network and knowledge scope of enterprises (Chen et al., 2019) .

In addition, as shown in Figure 5-7, *learning*, *business*, *training* and *work* in the Path of Contacting Business with larger fonts. *Courses*, *cooperation* and *innovation* fully reflect the supporting role of corporate universities as an internal learning platform for business, such as fully reflecting user-oriented thinking, user value and customer management capabilities, and fully conducting market insights and business innovation, including helping business identification market and iterative business to achieve innovation, also have manufacturing, digitalization and improvement to correspond.



Figure 5-6 Word cloud map of "Path of Contacting Business"

Note. The larger words are: business, study, work, training, thought, manufacture, group, knowledge, system.

5.2.3 Corporate University Enabling Path of Cultivating Teams

Path of Cultivating Team refers to that the corporate university empowers each team within the enterprise through systematic training, learning courses and internal consulting activities, and promotes them to achieve skill improvement, performance promotion and vitality stimulation. Thus, this path emphasizes the need for corporate universities to align themselves with team cultivation (Al Mahameed et al., 2023), starting with the smallest business unit, "team," to develop the skills and capabilities of employees, thereby enhancing team performance. Although the existing literature has put forward the idea that corporate universities can empower enterprises to develop from a team perspective, it has not been able to clearly explain the internal mechanism (Wufeng, 2016).

Specific to the practice of corporate universities, the corporate university of China Railway Fourth Bureau has developed different training tools for different employees in the process of building a learning team. The "4D Leadership" learning project was developed to improve the overall quality of grassroots managers; in response to the problem of improving the skills of internal trainers, the corporate university carried out training courses for training managers and internal trainers to strengthen the construction of learning and training management teams; To meet the strategic improvement needs of the management team, the corporate university provides the executive team with on-site experience (e.g., experience 5G technology, BIM+3D printing, brain science technology, future space) to enhance their perception of new technologies, and specific Action learning programs (e.g., group discussions, case teaching, experience sharing, benchmarking learning, online learning) to gather team wisdom.

During the interview (see appendix 2), when asked how the HR department leaders view the relationship between the corporate university and the department, the interviewee said: "The human resources department of the company has a complete corporate culture management system, so the corporate university should spread some corporate culture concepts in a way similar to pre-job training". When asked how the R&D department views the relationship between the corporate university and the department, he said: "The business, product innovation function of the company is definitely still in our department, the corporate university may focus on management innovation, or some knowledge dissemination through training."

In the actual practice of the corporate university of China Railway Fourth Bureau, this study believes that corporate university can achieve empowerment of enterprise development by empowering team development, and by improving the skills, performance and vitality of the internal team of the enterprise. This supplements the research content of the existing literature, and provides a new perspective of internal mechanism analysis for the study of corporate university empowerment enterprises. In addition, as shown in Figure 5-8, the fonts with larger fonts include *talent*, *change*, *employee*, *learning* and *training*, which are analyzed in combination with the sub-paths of the team-cultivation path. In terms of skill improvement, the emerging *knowledge*, *career path* and *course* reflect that corporate universities improve

employees' abilities by carrying out diversified training courses and fully empower employees' career paths; in terms of performance promotion, *team* and *cooperation transformation* and *internationalization* reflect the contribution of corporate universities to the improvement of team performance; in terms of vitality, although there is no obvious expression of words that stimulate team creativity, there are employees with improved skills and teams with improved performance, team creativity naturally improves.



Figure 5-7 Word cloud map of the "Path of Fostering Team"

Note. The larger words in the picture are: change, talent, learning, training, manufacturing, curriculum, capability, action, organization.

5.2.4 Corporate University Enabling Path of Shaping Organization

Path of Shaping Organization refers to that the corporate university empowers the entire organization to achieve long-term development by building an efficient and coordinated organizational structure, agile and adaptive institutional systems, and continuously optimized operating mechanisms. This enabling path constructed in this study includes three dimensions: *efficient collaboration*, *agile adaptation* and *mechanism optimization*. On the one hand, the dimensions of *efficient synergy* and *mechanism optimization* echo with existing literature views, and organizational design enables firms to acquire external knowledge in the process of exploiting opportunities (Foss et al., 2013). There are also studies that believe that the existence of corporate universities is beneficial for organizations to attract and retain talents, and can help employees pay attention to individual career development. Corporate universities (i.e., Organizations that offer them development opportunities which will maintain their

employability, and ensure they do not become deskilled through their work routines, will be more attractive to them than those that do not), will be favored by employees - because employees can perceive that they are taken seriously in the enterprise (Blass, 2005). For example, the practice of Sany Corporate University, as illustrated by their own material, shows that professional corporate universities can provide better services for organizational development and employee growth:

Sany Corporate university puts forward the development concept of being based on the enterprise, facing the society, and deeply integrating into the industrial chain, focusing on cultivating innovative and applied talents for sustainable development. The enterprise provides employees with career development channels of the nineteen major systems in two major sequences (i.e., management and professional), and establishes career management processes for employees from the organizational and individual levels. Provide employees with induction training, pre-job training, on-the-job training, and leadership training according to their career development paths; the centralized training at the organizational level adopts a new training model of "video + seminar + sharing + action", and professional lecturers are invited to serve as the Coaches organize learning.

Sany Corporate university formulates the company's annual (monthly/quarterly) training plan according to the company's strategic development, and supervises the implementation of the human resources department of each subsidiary. The human resources department can give feedback to the corporate university according to its own business situation, and adjust the training plan according to the actual situation. In addition, corporate universities can provide support for business and functional departments with training resources (e.g., training feasibility consulting services and provision of teachers).

On the other hand, the dimension of *agile adaptation* is constructed by this study in combination with the actual situation of the case corporate university, which is highlighted by the fact that the corporate university can respond in a timely manner to the urgent needs of the enterprise according to the business adjustment, structure adjustment and project construction needs, and provide effective learning solution. In addition, *agile adaptation* also breaks through

the traditional boundaries of corporate university empowerment enterprises, and extends the behavior of empowering organizations to outside the organization, such as promoting excellent experience and cases to partners and enterprises in the same industry chain:

By establishing an agile organizational mechanism and system operation mechanism, the China Railway Fourth Bureau Corporate university closely cooperates with each link of the learning value chain with business departments and subsidiaries to quickly solve practical problems, including the establishment of an organizational learning service platform (online platform, online platform offline platforms and mobile platforms), update and integrate learning (courses, lecturers, case resources), and establish a competency evaluation system. As a result, thanks to the services provided by the corporate university, the organizational agility of enterprises and their ability to respond to national policies have been significantly enhanced. For example, in response to the public-private partnership (PPP) model advocated by the state , corporate universities provide relevant employees with timely PPP project training; another example, in response to the “One Belt, One Road” national strategy, corporate universities provide overseas employees with Provides timely learning solutions.

Based on the advantages of the enterprise ecological chain, Dongfeng Nissan University conducts research on enterprise practice cases, summarizes the rules, and promotes it to suppliers and dealers. Corporate university integrates and promotes the excellent cases and practical experience of Dongfeng Nissan and the industry to strategic alliance partners.

In addition, reviewing existing research, Alonso-Gonzalez et al. (2017) proposed an integration strategy across organizational boundaries for corporate universities in large companies. Since the company has already reached the highest level of efficiency in its value chain, it is necessary to integrate corporate universities Boundaries extend into the value network of internal and external processes, including business-related SMEs, suppliers, governments, and university external partners. The cases in this study are all corporate universities of large multinational enterprises, state-owned enterprises and private enterprises, with strong own strength and better resources and infrastructure. In the era of VUCA, the

dimension of *agile adaptation* is one of the important components for the development of excellent corporate universities to empower enterprises.

Figure 5-9, the fonts with larger fonts are *learning*, *work*, *training*, *department*, *organization* and *resources*, which combine the three sub-paths of efficient collaborative organizational development, agile customer response capability and organizational mechanism optimization. Typical phrases such as *collaboration*, *informatization*, *process*, *mechanism* and *innovation* emerged, indicating that corporate universities have fully empowered them in focusing on organizational process improvement, cultivating organizational capabilities, and responding to customers.



Figure 5-8 Word cloud map of "Path of Shaping the Organization"

Note. The larger words in the picture are: study, work, training, organization, movement, department, promotion, development.

5.2.5 Corporate University Enabling Path of Inheriting Culture

The "Path of inheriting culture" means that the corporate university can inspire individual employees to fully recognize the corporate vision, mission and values through mission links and visions. Corporate universities can also help shape an organization's culture (Blass, 2005). Corporate universities are seen as bearers of corporate culture, aiding employees in better comprehending and internalizing the core values and cultural traits of the company (Abel & Li, 2012; Singh et al., 2019). This path emphasizes how corporate universities nurture employees' cultural identity through knowledge management activities, enhancing employee loyalty and job satisfaction towards the company. For example, corporate universities in the energy sector

Note. The larger words in the picture are: enterprise, employees, learning, culture, belonging, happiness, promotion, ideas, values

Finally, a new round of open coding, selective coding and theoretical coding was carried out for the second round of case companies in accordance with the coding procedure, and it was found that there were no new concepts and categories, and the categories were in line with the original logical relationship. This study will empirically verify the constructed theoretical framework in the next section.

5.3 Robustness Test of Corporate Universities Enabling Path Model

This research uses the coding analysis method to construct the corporate university empowerment path model, including five dimensions of "linking strategy", "closely connecting business", "cultivating team", "shaping organization" and "inheriting culture". In any case, due to the small sample size, the case study has the limitation that the generalizability of the conclusions is limited. While mixed methods research is a research method that combines quantitative research and qualitative research methods, it can help researchers develop rich insights into various interesting phenomena (Vance et al., 2013). Some Chinese scholars have proposed to test the theory constructed by the qualitative analysis process through empirical methods to improve the universality of the conclusions (Jia Xudong and Heng Liang, 2021).

Therefore, this study adopts a combination of qualitative and quantitative methods. After constructing the theory of corporate university empowerment in the first stage of qualitative research, based on the definition of each path in the main axis coding stage of this study, the corporate university empowerment theory is developed. The scale of the pathway (see Annex 6 for the full content of the scale). This research is distributed online to corporate executives with corporate universities, corporate university practitioners, and HR teams through questionnaire surveys, and uses quantitative methods to verify the model constructed by coding analysis. Drawing on existing research practices (Hollebeek et al., 2014), exploratory factor analysis was used to define and conceptualize the path of corporate university empowerment in qualitative research, and the results of factor analysis were observed to judge factor structure, convergent validity and reflection theory. degree, and then provide a robustness test for the conclusion.

5.3.1 Exploratory Factor Analysis (EFA)

In order to test the validity of the measurement data and ensure the validity of each latent variable, according to the recommendations of Anderson and Gerbing (1988), this study used the variables involved in SPSS 25.0 to conduct exploratory factor analysis (EFA) to obtain each Measure the convergent validity of the items on the corresponding factors, as well as the discriminant validity among the potential factors. Among them, all samples participated in this exploratory factor analysis (sample = 119). This study takes χ^2/df as an important indicator for judging the degree of fit, and analyzes the degree of fit of the model with reference to the statistics and empirical reference values of each indicator, combined with the correlation coefficient method.

First, this study uses the KMO sample measure and Bartlett's sphere test to analyze the initial scale items (A1 ~ E3) to judge whether factor analysis is suitable, and then conduct exploratory factor analysis to revise the scale. Using SPSS 25.0 , using principal component analysis method to carry out exploratory factor analysis on relevant items, through orthogonal rotation and following the statistical principle of extreme value estimation, the relevant factors are extracted according to the characteristic root greater than 1, and the basic selection according to the factor load should be greater than 0.5 Standard, and consider the selection principle that an item cannot simultaneously have a loading on both factors exceeding 0.4 for item selection. The test results showed that the KMO value was 0.92, and the Bartlett test value was significant ($p < 0.001$), which met the conditions of exploratory factor analysis. According to the principal component analysis method, a total of 4 factors with eigenvalues higher than 1 were extracted (the gravel diagram is shown in Figure 5-11); the cumulative sum of squares of the four factors after rotation is 73.35 % (as shown in Table 5-3).

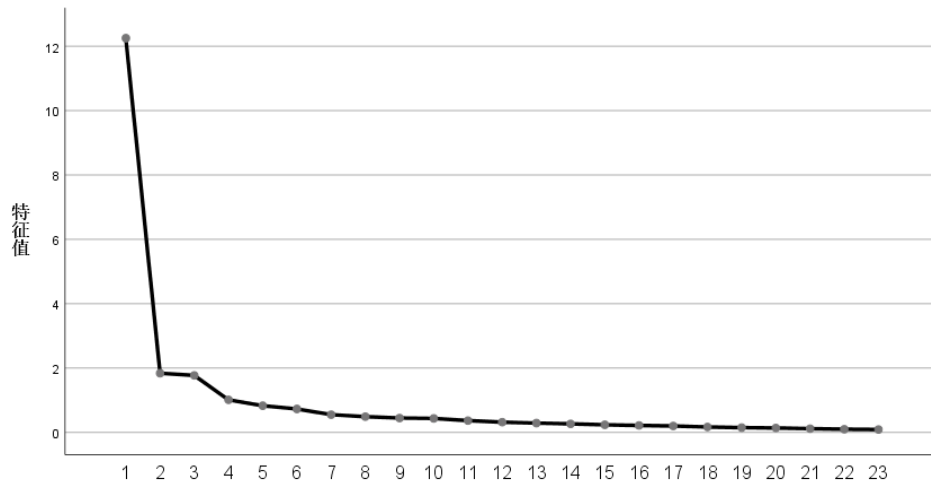


Figure 5-10 Gravel diagram (full sample)

Table 5-3 Exploratory factor analysis factor loading table (full sample)

Element	Initial eigenvalues			Extract the load sum of squares			Rotational load sum of squares		
	Total	Percent variance	Accu-mulation	Total	Percent variance	Accu-mulation	Total	Percent variance	Accu-mulation
1	12.25	53.26	53.26	12.25	53.26	53.26	5.552	24.14	24.140
2	1.838	7.993	61.253	1.838	7.993	61.253	4.172	18.139	42.279
3	1.77	7.696	68.949	1.77	7.696	68.949	3.802	16.532	58.811
4	1.012	4.4	73.349	1.012	4.4	73.349	3.344	14.539	73.349
5	0.829	3.605	76.954						
6	0.728	3.167	80.122						
7	0.551	2.397	82.519						
8	0.49	2.13	84.649						
9	0.446	1.941	86.59						
10	0.434	1.889	88.479						
11	0.365	1.588	90.066						
12	0.317	1.379	91.446						
13	0.289	1.258	92.704						
14	0.266	1.157	93.86						
15	0.237	1.028	94.889						
16	0.215	0.937	95.826						
17	0.199	0.867	96.693						
18	0.169	0.735	97.428						
19	0.148	0.644	98.072						
20	0.138	0.6	98.671						
21	0.116	0.505	99.177						
22	0.099	0.432	99.608						
23	0.09	0.392	100						

Second, use SPSS 25.0 was used for exploratory factor analysis, and the correlation factors were extracted according to the characteristic root greater than 1. The results of the component matrix and the rotated component matrix are shown in Table 5-4 respectively.

Table 5-4 Exploratory factor analysis after rotation factor (full sample)

Items	Elements				Rotated ingredients				
	1	2	3	4	1	2	3	4	1
C2	.816	-.196	.193	-.067	D8	.819	.191	.225	.069
C3	.807	-.145	.196	-.161	D3	.817	.215	.250	.168
B4	.798	.211	.049	-.207	D4	.779	.091	.298	.147
D3	.783	.036	-.435	-.001	D7	.766	.327	.094	.233
D1	.782	.124	.025	.240	D6	.741	.285	.089	.199
B3	.774	.148	.129	-.338	D5	.727	.193	.255	.240
C4	.772	-.189	-.028	-.154	D2	.613	.331	.256	.224
D7	.766	-.150	-.378	-.074	C4	.499	.439	.138	.443
D5	.753	.058	-.339	-.058	E1	.290	.847	.134	.161
C1	.751	-.231	.310	.158	E2	.243	.833	.136	.194
D2	.745	-.040	-.211	.020	E3	.219	.810	.107	.145
A3	.742	.420	.125	.208	C1	.192	.692	.341	.325
A4	.723	.418	.167	.189	C2	.343	.572	.258	.485
E1	.719	-.494	.151	.248	A2	.259	.124	.775	.223
D8	.718	.030	-.493	.056	A1	.069	.104	.767	.140
D6	.713	-.126	-.387	-.064	A3	.331	.189	.765	.235
D4	.713	.157	-.439	-.010	A4	.288	.188	.759	.260
E2	.699	-.481	.199	.221	D1	.432	.398	.558	.172
B1	.694	.149	.357	-.299	B2	.128	.257	.148	.832
A2	.666	.473	.159	.202	B1	.158	.229	.392	.700
E3	.639	-.488	.181	.239	B3	.391	.179	.342	.671
B2	.637	-.036	.372	-.501	C3	.343	.495	.249	.559
A1	.502	.477	.267	.272	B4	.459	.173	.432	.547

Notes: (1) Extraction method: PCA method (4 principal components were extracted); (2) Rotation method: Caesar normalized maximum variance method (The rotation has converged after 6 iterations.)

According to the fitting results, this exploratory factor analysis can prove that the four dimensions of "linking strategy", "close business", "organization" and "inheriting culture" fit well. Specifically: (1) "Linking strategy" (A)" dimension has a high degree of fitting, and A1, A2, A3, and A4 are classified as one factor, which is consistent with the theoretical model constructed in the qualitative research stage. In addition, the item in the "Shaping the organization" dimension appears in this factor ("D1 identifies the company's organizational

capabilities and diagnoses and nurtures them"). (2) "Close business (B)" dimension has a high degree of fitting, B1, B2, B3, and B4 are classified as one factor, which is consistent with the theoretical model constructed in the qualitative research stage. In addition, the item in the dimension of "nurturing the team" appears in this factor ("D3 helps companies cultivate team leadership that takes into account both performance and people orientation"). (3) The dimension of "Shaping Organization (D)" has a high degree of fit, and D2, D3, D4, D5, D6, D7, and D8 are classified as one factor, which is consistent with the theory constructed in the qualitative research stage. The model is consistent. In addition, the item in the dimension of "cultivating the team" appeared, that is, "C3 cultivated and stimulated the vitality of the team to form an effective program practice". (4) The dimension of "inherited culture (E)" has a high degree of fitting, and E1, E2, and E3 are classified as one factor, which is consistent with the theoretical model constructed in the qualitative research stage. In this factor, the items in the dimension of "cultivating the team" appear, namely "C1 helps the company cultivate employees who actively undertake high-quality work" and "C2 helps the company cultivate employees who better meet customer needs than its competitors". (5) The item of "cultivating team (C)" dimension is mixed with "closely connected business", "organization" and "inheriting culture" to a certain extent, and there is a certain deviation from the model constructed in the qualitative research stage. Combined with the interviews with the management team of corporate universities, some corporate universities' learning, training, and corporate culture publicity projects have the phenomenon of "organization → business → team" at the execution level, that is, the organization of "team" and "business" in reality. The boundaries are not clear, leading to a misunderstanding of some respondents.

Since the practical results of corporate universities will affect respondents' judgments on the empowerment path of corporate universities, the more satisfied respondents are with corporate universities, the more clearly, they may have a clearer understanding of the empowerment paths of corporate universities. As a result, the questionnaire sets up the item "Satisfaction with Corporate University" (increasing from 1 to 10). First, screen the whole sample, and retain the group with a higher degree of satisfaction (≥ 6 points) with the company's corporate university; secondly, check whether it is a case company in the qualitative research, and retain the sample from the case company. To sum up, the "excellent corporate

university group" (respondents from 13 corporate universities) were selected as subsamples for exploratory factor analysis (sample = 104, accounting for 87.39%).

First, this study also used the KMO sample measure and Bartlett's sphere test to analyze the initial scale items (A1 ~ E3) to determine whether factor analysis was suitable, and then conducted exploratory factor analysis using principal component analysis and orthogonal rotation. Precipitation characteristic root greater than 1 precipitation correlation factor. The test results showed that the KMO value was 0.91, and the significance level of the Bartlett test value was 0.00 ($p < 0.001$), which met the conditions of exploratory factor analysis. According to the principal component analysis method, a total of five factors with eigenvalues higher than 1 were extracted (the gravel diagram is shown in Figure 5-12 and Table 5-5 shown). Compared with the whole sample, the number of factors in this exploratory factor analysis conforms to the theoretically constructed five-dimensional model, and the explanatory power is enhanced ($74.79 \% > 73.35 \%$).

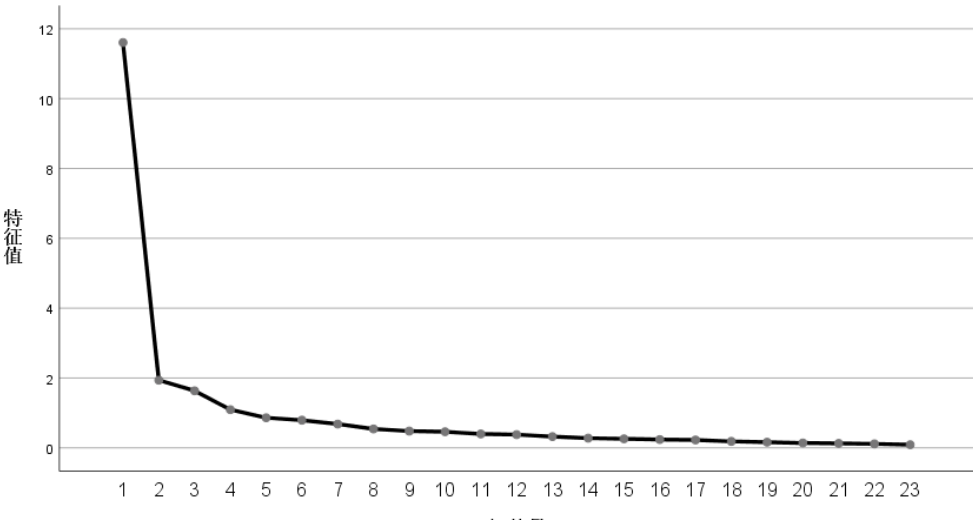


Figure 5-11 Gravel Chart (subsampling)

Table 5-5 Exploratory factor analysis (subsampling)

Factor	Initial eigenvalues			Extract the load sum of squares			Rotational load sum of squares		
	Total	Percent variance	Accu-mulation	Total	Percent variance	Accu-mulation	Total	Percent variance	Accu-mulation
1	11.61	50.462	50.462	11.61	50.462	50.462	4.635	20.152	20.152
2	1.937	8.423	58.885	1.937	8.423	58.885	3.896	16.941	37.092
3	1.634	7.104	65.989	1.634	7.104	65.989	3.668	15.947	53.04
4	1.093	4.753	70.742	1.093	4.753	70.742	3.06	13.306	66.345
5	0.861	3.745	74.487	0.861	3.745	74.487	1.873	8.142	74.487
6	0.793	3.446	77.933						

7	0.681	2.962	80.895						
8	0.541	2.351	83.246						
9	0.48	2.086	85.332						
10	0.461	2.002	87.334						
11	0.397	1.726	89.06						
12	0.379	1.646	90.706						
13	0.322	1.401	92.106						
14	0.278	1.207	93.313						
15	0.258	1.121	94.435						
16	0.239	1.037	95.472						
17	0.226	0.983	96.455						
18	0.184	0.802	97.257						
19	0.161	0.702	97.959						
20	0.139	0.606	98.565						
21	0.127	0.552	99.117						
22	0.114	0.494	99.611						
23	0.09	0.389	100						

Notes: (1) sample = 108; (2) Extraction method: principal component analysis.

As shown in Tables 5-6, from the perspective of the rotated factors, the results of this confirmatory factor are basically consistent with the theoretical model, and are closer to the dimensions involved in the theoretical model than the results of the full sample, and the coverage of the items is relatively high. Specifically, (1) The first factor is consistent with the dimension of "Shaping Organization (D)", including D2 , D3 , D4 , D5 , D6 , D7 and D8 items, the degree of fit High; (2) The second factor is mainly consistent with the "linking strategy (A)" dimension, including items A1, A2, A3 and A4 , and item D1 , with a high degree of fit; (3) The third factor is mainly consistent with the dimension of "cultivating team (C)", including C1, C2, C3, C4 items, and B3 and B4 items, the degree of fit is acceptable; (4) The fourth factor is mainly consistent with the dimension of "inherited culture (E)", including E1, E2, and E3 items, with a high degree of fit; (5) the fifth factor is mainly maintained with the dimension of "close contact business (B)" Consistent, including B1 and B2 items, the degree of fit is acceptable.

Table 5-6 Exploratory factor analysis after rotation factors (subsample)

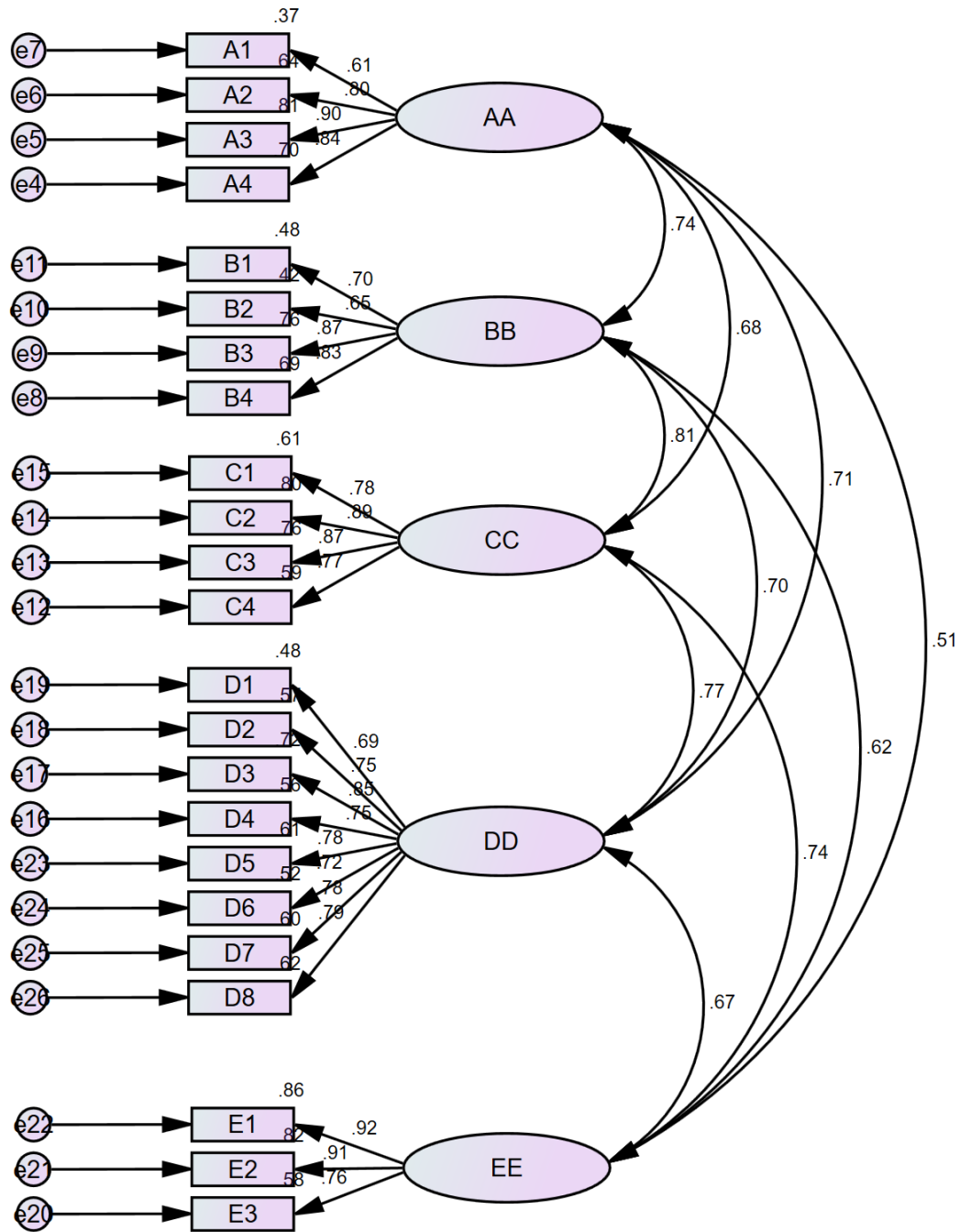
Items	1	2	3	4	5
D8	.787	.219	.170	.214	.000
D6	.787	.093	.038	.260	.265
D7	.769	.060	.173	.328	.165

D3	.751	.223	.419	.090	-.019
D4	.697	.311	.257	.044	.020
D5	.579	.343	.282	.311	.085
D2	.552	.271	.517	.110	.017
A2	.273	.810	.026	.136	.237
A4	.261	.749	.255	.152	.091
A1	.061	.734	.113	.054	.106
A3	.329	.726	.342	.135	.096
D1	.269	.590	.506	.284	-.113
C3	.299	.183	.676	.254	.322
B4	.298	.356	.657	.114	.204
C2	.341	.186	.653	.310	.305
C1	.087	.401	.558	.532	.117
C4	.455	.076	.550	.262	.317
B3	.223	.371	.525	.171	.439
E3	.267	.116	.020	.836	.208
E2	.277	.160	.258	.804	.127
E1	.254	.165	.390	.773	.093
B2	.136	.154	.260	.200	.848
B1	.005	.529	.277	.200	.581

Notes: (1) Number of samples 104; (2) Extraction method: principal component analysis method; (3) Rotation method: Kaiser normalized maximum variance method (rotation has converged after 8 iterations).

5.3.2 Confirmatory Factor Analysis (CFA)

After the exploratory factor analysis was passed, this study used SPSS 25.0 and AMOS 25.0 to verify the reliability and validity of the measurement model (as shown in Figure 5-13). Table 5-7 shows that the Cronbach 's alpha of the five constructs is higher than 0.8, which is higher than the advocated standard of 0.6, showing high reliability, indicating that the measurement results of the questionnaire are reliable.



Note: "Linking Strategy" to "Inherited Culture" are marked as "AA" to "EE" respectively.

Figure 5-12 Factor analysis plot

Table 5-7 Reliability and validity verification of items in the measurement model

Construct/Factor	Factor loading	Item
Linking strategy (Factor A) <i>Cronbach's alpha</i> = 0.866, <i>CR</i> = 0.870, <i>AVE</i> = 0.629		
A 1	0.607	Corporate universities can be invited to participate in high-level seminars on corporate management topics
A 2	0.798	Corporate universities are able to plan research and provide high-quality reports or solutions in line with the company's high-level concerns

A3	0.902	Corporate University has the solution design and operational capabilities that drive companies to implement strategy through execution
A4	0.836	Corporate universities can closely follow the development of the industry, plan action themes according to the company's strategy, and convert them into landing projects
Contacting business (Factor B) <i>Cronbach's alpha</i> = 0.848, <i>CR</i> = 0.849, <i>AVE</i> = 0.589		
B1	0.696	Corporate University can improve employees' thinking about problem solving from customer value
B2	0.646	Corporate university can improve employees' skills in customer experience management and service
B3	0.871	Corporate University can help business units identify and gain insight into market and customer changes
B4	0.833	Corporate University helps business units innovate and iterate on businesses, products and services
(Fostering team (Factor C) <i>Cronbach's alpha</i> = 0.892, <i>CR</i> = 0.899, <i>AVE</i> = 0.690		
C1	0.78	Help the company develop employees who can actively undertake and perform high-quality work
C2	0.894	Helping companies develop employees who better meet customer needs than their competitors
C3	0.874	Help companies develop team leadership that balances performance and people orientation
C4	0.768	Formed effective programs and practices in nurturing and stimulating team vitality
Shaping organization (Factor D) <i>Cronbach's alpha</i> = 0.916, <i>CR</i> = 0.919, <i>AVE</i> = 0.586		
D1	0.692	Clearly define company organizational capabilities and be able to diagnose and nurture them
D2	0.755	Help companies promote smooth information communication and cross-team collaboration
D3	0.85	Help companies optimize organizational structure and resource allocation to improve overall performance
D4	0.746	Drives the organization and scheduling of tasks throughout the company around customer needs
D5	0.781	Drive rapid response to market and customer changes throughout the company
D6	0.722	Promote the company's reasonable authorization to make employees the protagonists in the work
D7	0.778	Promote the company to optimize the management process to make the work more concise and efficient
D8	0.789	Help the company to establish an incentive and restraint mechanism with clear guidance and standards
Inheriting culture (Factor E) <i>Cronbach's alpha</i> = 0.896, <i>CR</i> = 0.900, <i>AVE</i> = 0.752		
E 1	0.925	Help employees understand the meaning of work from the company's mission
E 2	0.906	Help employees inspire work commitment from company goals and vision
E 3	0.761	Helping companies gain employee buy-in to core values

Note: Each path is significant at the 95% level.

This study uses AMOS 25.0 for confirmatory factor analysis, and measures construct validity, convergent validity and discriminant validity respectively: (1) Construct validity is to test the rationality of the model in the empirical study, and the fit index shows the data and

hypothesis The fit of the model is good (χ^2/df is 1.969, less than 3). (2) As shown in Table 5-8, the combined reliability (CR) is higher than the standard of 0.7; the average variance extraction value (AVE) value is higher than 0.7, indicating that the relationship between the index and the item is statistically study meaning. (3) This study uses the square root of the AVE value to estimate the discriminant validity. The square root of the AVE is higher than the absolute value of the Person correlation coefficient (as shown in the Table 5-8) between variables, indicating that the discriminant validity of this study conforms to the theoretical construction. requirements. Moreover, the variables among the five constructs are basically correlated, reflecting that the path models in this study are closely related and have a certain degree of discrimination.

Table 5-8 Correlation analysis of constructs in measurement models

	A.	B.	C.	D.	E.
Linking Strategy (A)	0.793				
Contacting Business (B)	0.669 **	0.767			
Cultivating Team (C)	0.588 **	0.746 **	0.831		
Shaping Organization (D)	0.615 **	0.666 **	0.719 **	0.766	
Inheriting Culture (E)	0.415 **	0.541 **	0.722 **	0.591 **	0.867

Note: (1) The diagonal line is the square root of AVE; **. At the 0.01 level (two-tailed), the correlation is significant.

In conclusion, the analysis results of the survey questionnaire show that it can better fit the items constructed by the existing coding analysis. Therefore, the five-factor theoretical model constructed in this study has a high degree of fitting and can pass the empirical test of exploratory factor analysis. This study responds to the suggestion in qualitative research to adopt a large sample empirical test for coding results (Jia Xudong and Heng Liang, 2020), making the research results more general and credible.

5.4 Chapter Summary

In this chapter, this study provides an in-depth analysis of the enabling pathways of corporate universities using a mixed qualitative and quantitative approach. First, this research constructs the preliminary theoretical model of this research with the help of three-level coding layer by layer, which is divided into five dimensions: linking strategy, close business connection, cultivating team, shaping organization and inheriting culture. Secondly, this study

also designed a questionnaire according to the categories constructed in the coding analysis, verified the theoretical model of this study through the questionnaire, and responded to the call of some scholars for the combination of qualitative and quantitative methods.

6 THEORETICAL MODEL CONSTRUCTION OF CORPORATE UNIVERSITIES ENABLE ENTERPRISE DEVELOPMENT IN THE CHINESE CONTEXT

Although this study verified the theoretical model established by coding with the help of questionnaire method, it still could not completely avoid the self-report problem caused by the data provided by enterprises themselves. If the enterprise only discloses its advantages and keeps silent about the disadvantages, it will bring certain errors to the identification of the corporate university empowerment path. In order to avoid corporate universities only sharing successful experience and not revealing deficiencies and weaknesses, the method of learning from existing studies is based on the forward-looking content (development direction and forward-looking thinking) in the enterprise data collection of the second round of case studies. This section aims to increase the trend research of the five dynamic paths of corporate university enabling enterprise development. This is the practice being carried out by corporate university, reflecting the dynamic trend of corporate university enabling enterprise development.

In Chapter 5, this research proposes a path model for corporate university to empower enterprise development (as shown in Figure 5-5), focusing on explaining how corporate university can link strategies, closely link businesses, cultivate teams, shape organizations and inherit culture through five paths to enable business development. Combining coding method and questionnaire survey method, this study demonstrates that the internal validity of the theoretical model is acceptable. In order to avoid corporate universities only sharing successful experiences, not revealing deficiencies and shortcomings, it draws on the practice of existing research, based on the forward-looking content in the second round of case data collection ("development direction" or "forward thinking" question item) . This section aims to increase the trend research of the five dynamic paths of corporate university empowerment of enterprise development - this is the ongoing practice of corporate university, which reflects the dynamic trend of corporate university empowerment of enterprise development.

As shown in Figure 6-1, this research constructs a *trend model of corporate university empowerment of enterprise development*. The five trends correspond to the extension of the five paths of corporate university empowerment of enterprise development, namely: (1) "Path of linking strategy " to "Path of enabling strategy", (2) "Path of contacting business" to "Path of

enabling business", (3) "Path of fostering team" to "Path of enabling team", (4) "Path of shaping organization" to "Path of enabling organization", and (5) "Path of inheriting culture" to "Path of enabling culture". Among them, the inner pentagon is a refinement of the path model in Figure 5-5, that is, corporate university empowers enterprise development through strategy, business, team, organization and culture; Able to practice activities still fall within one or more aspects of strategy, business, team, organization and culture. The two nested pentagons from the inside to the outside constitute the trend model of the corporate university empowering enterprise development, and the trend model can effectively supplement the original path model, making the research conclusion more objective and comprehensive.

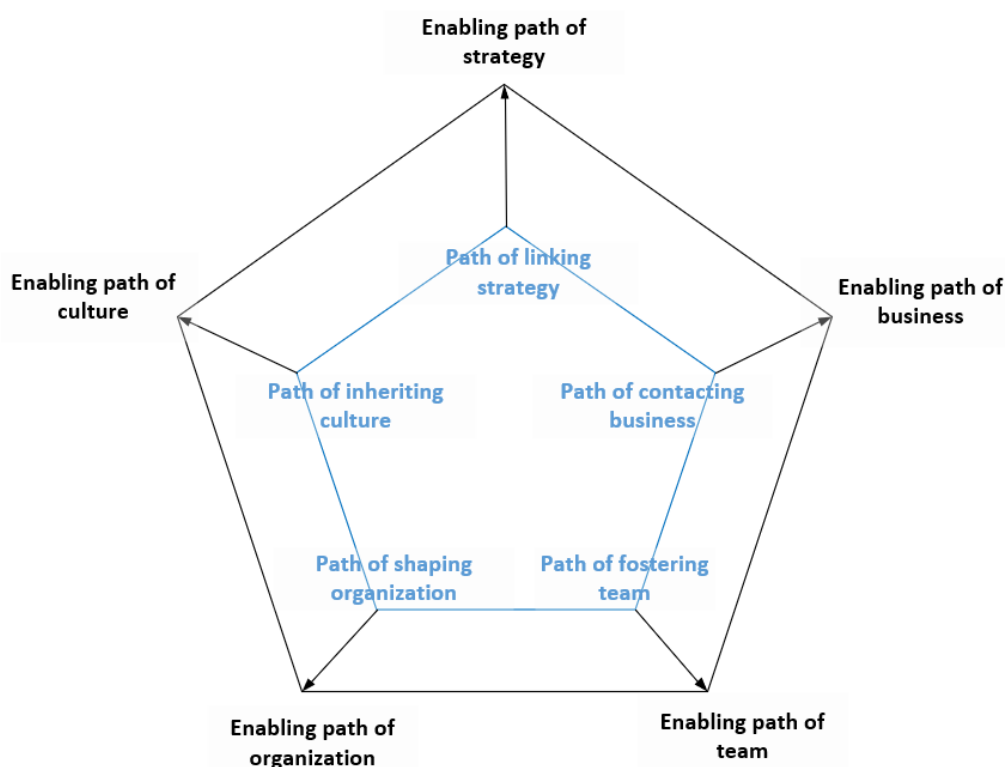


Figure 6-1 The trend model of corporate university empowering enterprise development

6.1 "Path of Linking Strategy" to "Path of Enabling Strategy"

There are two typical aspects in the deepening path from "*Path of linking strategy*" to "*Path of enabling strategy*", namely, building a digital learning context and enabling enterprises to develop the ecology where universities are located. system.

First, corporate universities can deepen and leverage digital learning to build new learning scenarios. For example, New Hope Business School will use the means of digital mobile

learning to publicize strategic intentions with rich "micro-lectures", aiming to enable employees of the company to deeply understand and recognize the company's strategic development direction. By building digital learning scenarios, corporate universities are creating an online and offline learning environment, and provide employees with strategic direction training through online learning platforms, mobile learning platforms, and corporate offline strategic course training. For example, New Hope Business School pointed out: "What we need to do is to keep the communication strategy from going astray, so that the company's employees can deeply understand the company's development direction and why it is developing in this way (see Table 6-1). In addition, New Hope Business School, as a corporate university, clearly pointed out that it is currently using digital mobile learning to continuously publicize the company's strategic intentions with a rich and diverse portfolio of micro-courses and activities.

Second, corporate universities are not only supporting their internal strategic development but are also extending strategic support to stakeholders within their supply chain and distribution network ecosystems. However, the approaches to ecosystem development vary among different corporate universities. For instance, Geely University emphasizes that competition exists not only between individual companies but also between groups of companies. Therefore, constructing ecosystems has become a strategic direction adopted by many companies to face future competition. Geely University has employed strategic cooperation by collaborating with suppliers and distribution partners, actively engaging in knowledge exchange to enhance the management and technical levels of their business alliance, thereby improving the overall competitiveness of their ecosystem (refer to Table 6-1).

On the other hand, Bank of China University, affiliated with a state-owned bank, adopts a strategic approach called "industry-education integration," focusing on creating a core ecosystem involving various partners. Furthermore, corporate universities of different attributes are striving to build empowering ecosystems as their core strategic direction. For example, Dongfeng Nissan University is expanding its empowerment scope to upstream and downstream partners, and Sany Enterprise University is attempting to establish the "Sany Ecosystem," providing training services for various stakeholders within the Sany ecosystem, including enterprises, industries, government, other social organizations, and individuals.

Regardless of their ownership type and industry, these corporate universities are actively creating core ecosystems involving diverse partners to enhance their roles within their respective ecosystems. Thus, this study suggests that the role of corporate universities extends beyond internal boundaries, making them essential supporters and enablers within their ecosystems. However, differences in the dominant company's attributes result in variations in the corporate university's ecosystems. These differences in empowerment strategies might lead to unique partner selections, strategic implementations, and outcomes among different corporate universities, further highlighting the impact of corporate universities on biodiversity within the entire business ecosystem.

6.2 "Path of Contacting Business" to "Path of Enabling Business"

As the degree of closeness between corporate universities and business continues to increase, simply maintaining close contact with business can no longer meet the development needs of corporate universities. Combined with the development trend of case corporate universities, facing future business needs will definitely become the mainstream choice. At present, there are already case corporate universities that have gradually embedded their functions into business units. The deepening of "*Path of contacting business*" to "*Path of enabling business*" is the most extensive development idea in the case corporate university, which reflects the implementation of the training and learning function of the corporate university. It is still necessary to rely on enabling business development.

First, corporate universities should help businesses establish a professional knowledge management system when empowering businesses in the future. For example, RSM Management School will help businesses establish professional learning maps and knowledge management, and for future businesses, knowledge will be reserved in advance and learning maps will be iteratively updated, which not only improves the basic capabilities of the business team (team coordination, The ability to solve problems), and improve the knowledge management ability of the business team to deal with the rapid iteration of multiple scenarios. Another New Hope Business School insists that the primary task of a corporate university is knowledge management, that is, to be able to master the core experience related to the efficiency of the company's operation and management. No matter whether the personnel change or not,

the core experience will always be in the hands of the organization. Therefore, the focus of future empowerment business is to fully excavate and refine the core experience of the organization, and to speed up the flow of experience and the dissemination of knowledge (see Table 6-1).

Second, corporate universities should fully listen to business needs when empowering businesses in the future, and at the same time focus on core resources to enhance business advantages. For example, RSM Management School will cooperate with new retail to carry out online and offline interactive learning to ensure the continuous advancement of new retail business. The corporate university can focus on sorting out the business knowledge required by employees, and provide targeted training for key employees (e.g., employees in charge of home improvement can master and decorate and maintain knowledge) to ensure the smooth development of the business line of the enterprise.

6.3 "Path of Fostering Team" to "Path of Enabling Team"

"Path of fostering team" to *"Path of enabling team"*, mainly through concepts, actions, structures and capabilities, through a variety of knowledge management capabilities to achieve enterprise-university. It means further empowerment of the organization.

Corporate universities extend from the path of fostering teams to empower teams in two aspects: increasing the driving force of enabling teams and reducing resistance: First, corporate universities can increase the driving force of building organizations. For example, as shown in the typical behavior of enterprises in Table 6-1, CCB University will increase the driving force of empowering teams in the future from three aspects: (1) Corporate universities can continue to pay attention to the quality of training, and change the past one-sided pursuit of training scale and single training. The business model pays more attention to employee needs, curriculum design, and training effects. (2) Corporate universities can continue to focus on ability improvement, customize products for specific audiences, continuously improve student experience and operation management, and continuously improve employee capabilities and work quality and efficiency; Provide support for the whole chain management of "selection-use-cultivation-retention-removal" of enterprise human resources. Therefore, corporate

universities can play a more important supporting and promoting role in empowering team training and management.

Second, corporate universities can reduce the resistance of existing mechanisms to enabling teams. For example, in terms of team empowerment, New Hope Business School will consider creating a culture and mechanism in the enterprise that promotes work with teams and can reduce obstacles to teamwork, such as a work culture of struggle and responsibility, while focusing on the practice of learning, Help the team to continuously improve their ability and methods of doing things through learning and experience, design learning projects that the team participates in, and help the team grow and solve practical problems.

6.4 "Path of Shaping Organization" to "Path of Enabling Organization"

From "shaping an organization" to "empowering an organization", corporate university mainly includes four aspects: concept, action, structure and capability. First, from the perspective of concept, the company should be committed to putting the new concept of corporate university into operation, into the team, and into the hearts of employees. Second, from the action level, give full play to the advantages of corporate universities in knowledge management, cultivate more action learning coaches, cultivate seeds, and fall into the organization to solve problems. Enable managers to use the methodology and tools of action learning to integrate into daily problem solving; thirdly, from the perspective of structure, based on the optimization of the existing organizational structure by corporate universities, through a broad understanding of corporate management and employees For different demands at different levels, build different architecture systems for different business sectors; fourth, from the perspective of ability, integrate internal and external resources of the organization, penetrate into the work of the organization through the learning and training methods of corporate universities, and build management With the professional sequence learning path map and knowledge map, it can continuously summarize and iterate best practices, form the management of organizational experience and knowledge, empower every member of the organization, and realize the common growth of individuals and organizations. The most typical example is CCB University's insistence that "all pain points are opportunities". From the concept, it first breaks the ice and reshapes the organization, and empowers the organization in terms of action,

structure and capability. For another example, as shown in the typical behaviors of enterprises in Table 6-1, Robam University will insist on improving the organizational system thinking ability through four aspects of knowledge management in the future, including team learning (management tools, knowledge productization, and reviewing normality), standardization, case standardization), mental model (unified language, template, method, implementation), self-transcendence (understanding the principle and comprehend the essence, mastering the tools and strictly implementing, combining with the actual practice, long-term persistence to form a habit), common vision (problem culture, Benchmarking culture, excellent culture, review culture).

6.5 "Path of Inheriting Culture" to "Path of Enabling Culture"

"Path of inheriting culture" to *"Path of enabling culture"* in corporate universities mainly includes vertical empowerment for sustainable development, horizontal empowerment for the coordinated development of ecosystems, and multi-channel empowerment. Three aspects of empowering corporate culture:

First, corporate universities can empower corporate culture vertically and promote the sustainable development of corporate culture. For example, as shown in the typical behavior of enterprises in Table 6-1, Robam University insists on building the Institute of Cultural Power, the Institute of Organizational Power, the Institute of Professional Power and the Institute of Leadership. By creating a carrier institution for corporate culture, corporate universities can maintain the strategy of empowering culture for many years, which is conducive to promoting corporate culture and implementing relevant strategies of corporate culture, and promoting the evergreen development of corporate culture.

Second, corporate universities can horizontally empower corporate culture and create a strategic cultural alliance that includes members of the ecosystem. For example, Red Star Macalline School of Management is relying on the Red Star system, closely surrounding the Red Star culture, and further extending and expanding the Luban culture, Red Star corporate culture and brand culture to the upstream and downstream of the industry through the combination of production, education and research.

Third, corporate universities can rely on existing cultural construction methods to achieve cultural development that empowers future organizations through various new communication methods. For example, New Hope Business School is digging out more interesting and informative benchmarking stories and disseminating them in various forms; the second is to integrate cultural self-reflection and mutual mirroring into training. Another example is CCB University, which has begun to deeply apply digital technology to cultural construction, and has been awarded the honors of "Advanced Corporate university" for talent training by state-owned enterprises, "2019 National Excellent Financial Media Works", "2020 China Internet Education Technology Innovation Enterprise Award", and The "Jinzhihuimin" project won the "2020 China Internet Education' Outstanding Contribution Award for 'Suspending Classes without Learning'", and CCB University's online platform also won the "2020 China Internet Education Quality Platform Product Award" (see Table 6-1).

In summary, the trend model of corporate university empowerment mechanisms depicted in Figure 6-1 is built upon the foundation of the path model shown in Figure 5-5. This study proposes that the trend model serves as a further enhancement of the path model, providing an explanation for the empowerment activities that corporate universities are currently engaged in. The study refines a five-dimensional theoretical framework for corporate university empowerment of organizational development, namely, 'Strategy-Business-Team-Organization-Culture.' This framework serves to guide corporate universities in operating within these dimensions effectively, thereby promoting the sustained development of organizations.

Table 6-1 Typical evidence citing table of corporate universities empowering enterprises to develop ideas in the future

Paths	Typical evidence cited
Path of Enabling Strategy	Strategic cognition and strategic understanding: What we need to do is to keep the communication strategy in the right direction, so that the employees of the company can deeply understand the development direction of the company and why it is developing in this way. Let all staff be in the same pace, up and down the same desire, left and right synchronization. In the future, New Hope University will use the means of digital mobile learning to continuously publicize its strategic intentions with a rich and diverse combination of micro-courses and activities (New Hope Business School).
	The competition between enterprises is not the competition of a single enterprise, but the competition between enterprise groups. The construction of "ecosystem" has become the strategic direction chosen by more and more enterprises to face future competition. In addition to supporting specific strategic topics, corporate universities In addition to the promotion of the enterprise alliance, it must also assume the role of the development and cooperation center of the enterprise alliance. Taking training as the starting point, it will cooperate with suppliers and distributors to exchange talents and knowledge, promote cooperation, and jointly improve the management and technical level of the enterprise alliance. Competitiveness of the entire business group (Geely University).
Path of Enabling Business	In the next step, Robam University will focus on the company's ability to build the Cultural Power Research Institute, the Organizational Power Research Institute, the Professional Power Research Institute, and the Leadership Research Institute (Robam University).
	Cooperate with new retail to carry out online and offline interactive learning to ensure the continuous advancement of new retail business; at the same time, focus on advantageous resources, focus on sorting out home improvement business knowledge, focus on cultivating and building home improvement talents, and provide corresponding decoration, maintenance and talent management related to home improvement. Training and coaching to ensure the smooth and full development of the home improvement business (Red Star Macalline School of Management).
	In terms of knowledge management, as a corporate university, the primary task is knowledge management, that is, to be able to master the core experience related to the efficiency of the company's operation and management. No matter whether the personnel change or not, the core experience will always be in the hands of the organization. Let the experience be fully excavated, refined, flowed and disseminated, making learning easier and anytime, anywhere (New Hope Business School).
Path of Enabling Team	In terms of empowering teams, in addition to providing some corresponding team courses and tools and methods for team division and cooperation, corporate universities can also consider creating a culture and mechanism in the enterprise that promotes work with teams and can reduce barriers to teamwork, such as The work culture of striving and responsibility, while focusing on the practice of learning, helps the team to continuously improve their ability and methods of doing things through learning and experience, design learning projects for team participation, and help the team grow and solve practical problems (New Hope Business School).

	In terms of employee growth, it is necessary to look beyond the business to look at the business, to jump out of the training to do training, and to reflect the value creation at different levels. The first layer is concerned with the quality of training. Change the previous mode of one-sided pursuit of training scale and satisfaction with completing training affairs, and pay more attention to employee needs, curriculum design, and training effects. The second layer, the ability to focus is improved. Customized development of products for specific audiences, continuous improvement of student experience and operation management, and continuous improvement of employee capabilities and work quality and efficiency. The third layer is service talent management. It is necessary to provide support for the whole-chain management of "selection, cultivation, retention and elimination" of human resources, and make training play a more important supporting and promoting role in terms of training and recruitment, training and employee motivation, and training and cadre training. The fourth layer is the corporate strategy of Milian. Standing at the strategic height of the CCB Group, it earnestly fulfills the mission of serving employees and strategies, and accurately conveys the spirit and strategic intention of the Party Committee of the Head Office, making training a powerful tool to promote the implementation of the strategy (CCB University).
Path of Enabling Organizations	- Improve organizational system thinking through four aspects of knowledge management, namely: team learning (management tools, knowledge productization, normalization of review, and case standardization), mental models (unified language, templates, methods, execution), Self-transcendence (understanding the principle and comprehend the essence, mastering the tools and strictly implementing, combining with the actual practice, and long-term persistence to form a habit), common vision (problem culture, benchmarking culture, excellence culture, review culture) (Robam University). - Talent empowerment, based on the difficulties and pain points in the process of strategy implementation, is guided by the front line of business, breaks through traditions, focuses on innovation, and establishes a talent empowerment mechanism that closely follows the 4+N business body. Talent experience, based on employees' opinions and needs, criticism and suggestions, and focusing on the key dimensions of "satisfaction and talent thickness", to further improve employees' perceived experience (Geely University). "When you use a new financial logic and thinking to think about this issue, you will find that nothing has nothing to do with us. All pain points are opportunities." Therefore, CCB University first breaks the ice and rebuilds the organization in terms of concept. Shape, empower the organization in terms of concept, action, structure, and capability (CCB University, interview content).
Path of Enabling Culture	- In the future, Red Star Macalline School of Management will continue to rely on the Red Star system, closely focus on the Red Star culture, and further extend and expand the Luban culture, Red Star corporate culture and brand culture to the upstream and downstream of the industry through the combination of production, education and research. Through the Luban University Leadership Academy, university alliances, and cross-value chain interaction, we will further deliver talents to the entire industry, and promote the further development and growth of Red Star Culture (Red Star Macalline School of Management). "The future cultural training is to let everyone discover the original culture of New Hope through experiential and guided classes. Explore more interesting and informative benchmark stories and make them into various forms to spread. Cultural self-reflection and mutual reflection are integrated into the training. Our training logos, energy words, posters, and promotional material designs must be moisturizing and silently infiltrating the cultural content." (New Hope Business School, interview content).

	• Publish the "Culture Outline" with the main culture of the group as the core, deeply refine the business culture (such as R&D culture, IT culture), and build a complete cultural system; formulate the "Corporate Culture Evaluation and Incentive Measures", and carry out the inspection system of the four major cultural landing mechanisms; Formulate the "Sparkling Prairie SOP Manual" to help disadvantaged units in the group to improve their cultural construction; at the same time, it regularly conducts cultural training and cultural integration and exchange activities for foreign employees, organizes cultural research and diagnosis activities for overseas companies, and guides overseas companies to output the "Cultural Outline", which will The "Cultural Ambassador" series of activities are promoted to overseas companies to achieve the same cultural cognition at home and abroad (Geely University).
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6.6 Chapter Summary

This study constructs an integrated analysis path model based on the Chinese context of corporate university empowerment of enterprise development, constructs a local theoretical system from the bottom up, and attempts to answer the question of how this research in the Chinese context compares with the Western academic research on corporate universities. "Where is the special", "why is it special", and "what phenomena exist to show that it is really special, these three questions reveal whether the research really shows China Context-specific and unique theoretical contributions.

First, regards the first question of "where is the special", Since 2021, the Chinese government has undergone significant policy changes in the education sector, particularly in lifelong education and vocational training. This marks a new era for vocational education in China. The government now actively promotes quality education and vocational training, categorizing adult continuing education into academic and non-academic streams. Enterprises of various kinds are encouraged to participate in legally sanctioned vocational education programs, enriching and enhancing the vocational education system. Report shows a continual growth in China's vocational education market, with a substantial portion dedicated to non-academic vocational education (Deloitte, 2021).

This unique policy environment creates ample opportunities for the development of corporate universities. In this context, Chinese corporate universities have become integral components of lifelong, vocational, and continuing education. They emphasize the importance of models such as 'integration of industry and education' and 'school-enterprise cooperation.' Consequently, this study asserts that Chinese corporate universities are not merely educational institutions; they serve as vital links connecting businesses and education. This synergy fosters the emergence of an entirely new educational ecosystem.

Considering the interconnected nature of lifelong education, vocational training, and continuing education, the key to developing vocational education within the context of Chinese corporate universities lies in establishing innovative cooperative models. These models, notably represented by the 'integration of industry and education' and 'school-enterprise cooperation'

approaches, are paramount for the evolution of vocational education in this unique Chinese context

The second question is "why is it special? The uniqueness of Chinese corporate universities lies not only in government policy support but also in their exceptional innovative capabilities. Existing research suggests a close connection between an organization's innovation capacity and its intellectual capital or ability to leverage knowledge resources (Subramaniam & Youndt, 2005). With the advent of the VUCA era, traditional management methods have gradually lost their inherent advantages, leading to delayed organizational responses to change (Whiteman, 1998). Effective collaboration between teams has become challenging, and the inner drive of team members often remains underutilized. Empowerment, on the other hand, emphasizes igniting the interest and motivation of creative individuals by presenting challenges.

China's digital transformation and mobile internet technologies have gained exceptional prominence globally (Song et al., 2021; Wang et al., 2021), creating opportunities for enterprise digital development. Simultaneously, these advancements have provided an advantageous external environment for the digital evolution of corporate universities. Consequently, in the VUCA era, Chinese corporate universities have achieved efficient internal and external collaboration through widespread application of digital transformation and mobile internet technologies. This digital environment has provided corporate universities with abundant innovation opportunities, sparking employees' creativity and driving their motivation for innovation. The concept of empowerment finds expression in Chinese corporate universities, encouraging employees to challenge themselves and pursue excellence. In this positive atmosphere, corporate universities are not just places for knowledge dissemination; they serve as incubators that inspire innovative ideas, fostering both technological and managerial innovation within organizations."

Third, what phenomena demonstrate its uniqueness? The distinctiveness of Chinese corporate universities is reflected in their developmental trajectory and mission. In comparison to corporate universities in Western contexts, Chinese corporate universities have emerged prominently within a unique institutional environment. They are not merely places for nurturing talent but also emphasize organizational empowerment. The role of corporate universities extends beyond imparting knowledge to encompass building organizational culture, unearthing

employee potential, and facilitating industry connections. This multidimensional development has turned corporate universities into vital platforms for creating value within organizations. In China, corporate universities are no longer merely adapting to market demands passively; instead, they are actively shaping market trends, driving industrial upgrades, and championing innovation and creativity. This unique phenomenon illustrates the distinctive position and role of Chinese corporate universities within the entire educational system. They serve as fountains of knowledge dissemination and engines for organizational development, providing robust support for continuous innovation and sustainable development in Chinese enterprises.

Therefore, based on the three-level coding, this study constructs a path model for corporate universities empowering organizational development. Integrating quantitative empirical results and qualitative trend analysis, this research combines the path model and trend model of corporate university empowerment for organizational development, proposing an integrated analytical model for corporate university empowerment in the Chinese context (as illustrated in Figure 6-2).

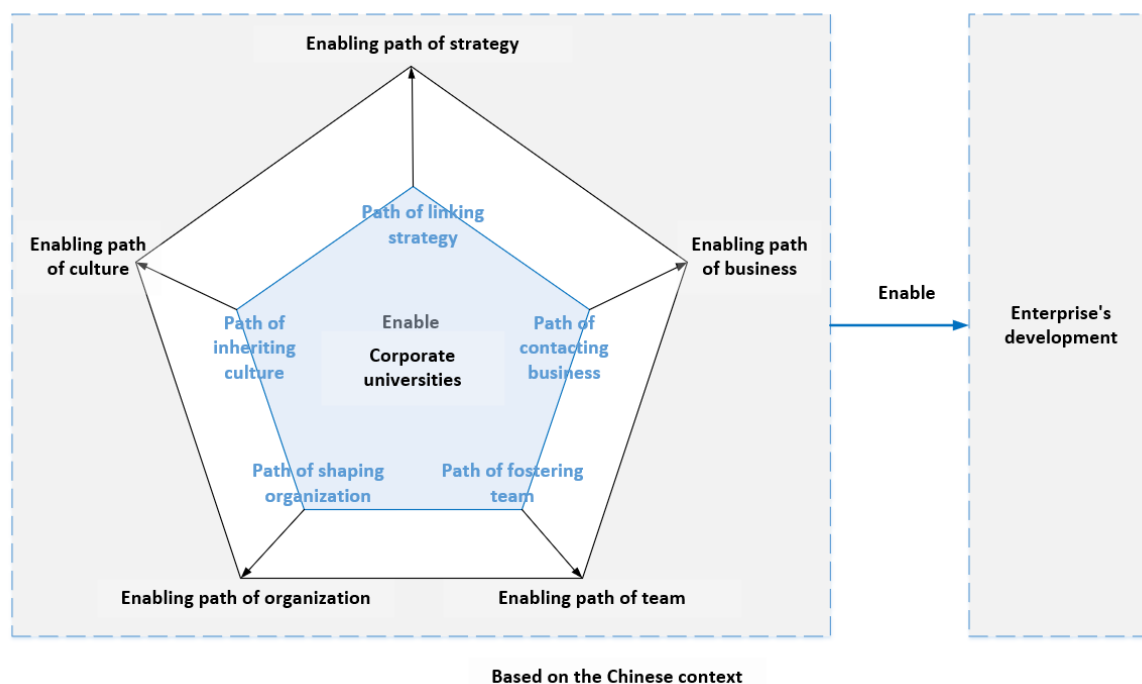


Figure 6-2 An integrated analysis model of corporate university empowering enterprise development in Chinese Context

In this model, there is an expanded path model showcasing the unique developmental trajectory of corporate university empowerment in the Chinese context. Specifically, corporate

universities empower organizational development through five expanded pathways: strategy, business, teams, organizations, and culture:

(1) *The trend path of linking strategy → enabling strategy*: in the Chinese context, corporate universities establish close strategic connections, aligning the long-term development goals of the enterprise with the educational mission of the corporate university. This strategic alignment enables corporate universities to formulate specific educational strategies, ensuring that training and education align with the company's strategic objectives.

(2) *The trend path of contacting business integration → enabling business*: By deeply integrating with the company's operations, corporate universities gain a better understanding of the enterprise's actual needs and challenges. This close business relationship enables corporate universities to develop training programs and courses tailored to practical requirements, providing employees with practical knowledge and skills.

(3) *The trend path of fostering teams → enabling teams*: By cultivating and enhancing the skills and leadership of internal teams, corporate universities strengthen employees' teamwork and innovative thinking. This empowerment of teams enhances internal collaboration, fostering improved innovation and problem-solving capabilities.

(4) *The trend path of shaping organizations → enabling organizations*: Corporate universities intervene at the organizational level, assisting enterprises in building more flexible and adaptable organizational structures. This organizational empowerment, achieved through training and guidance, helps companies better adapt to market changes, thereby enhancing competitiveness.

(5) *The trend path of inheriting of culture → enabling culture*: By preserving and promoting corporate culture, corporate universities help employees better understand and internalize the company's core values and cultural characteristics. This cultural empowerment enhances employee cohesion, increases their sense of belonging, thereby improving employee loyalty and job satisfaction.

It is important to note that due to this study's focus on corporate universities in the Chinese context, the theoretical model has clear boundaries of interpretation—it is specifically applicable to empowering practices within corporate universities in China. Its applicability to corporate universities in developed countries is challenging to assess, and whether this

theoretical model can be extended to other emerging economies remains a topic for further discussion. On one hand, this study has constructed an integrated model based on the practices of Chinese corporate universities, uniquely developed within China's specific cultural, political, and economic context. China's corporate university empowerment practices are influenced not only by the local culture but also deeply shaped by China's distinctive social values and business environment. This uniqueness makes it challenging for this model to be directly applicable to corporate university practices in other developed countries. Cultural differences, variations in educational systems, and career development paths in developed countries could all pose challenges, making the model difficult to apply in cross-cultural contexts.

On the other hand, even within emerging economies, cultural disparities and diverse educational systems between different countries present challenges. Cultural differences might lead to communication barriers, mismatched training philosophies, and incompatible leadership styles in corporate university empowerment practices. These differences could potentially impact the effectiveness of corporate university empowerment practices. Therefore, future research aiming to extend this model to other emerging economies will need to conduct in-depth studies on the cultural characteristics, educational systems, and career development models of the target countries. This in-depth analysis will facilitate necessary adjustments and customization in the practical application of the model.

7 CONCLUSION AND PROSPECTS

This chapter first summarizes the main conclusions of the study and proposes the practical contribution of this study to guiding corporate university enabling behaviors. Finally, this study identifies the limitations of the research, and on this basis puts forward suggestions for future research directions.

7.1 Discussion and Conclusions

With the diversification of corporate business sectors, organizational mergers and acquisitions or restructuring, the rapid expansion of revenue scale and the total number of employees, as well as the influence of industry leadership, industrial integration policies, and market orientation, corporate universities have also been endowed with more new missions and functions by the company, to support the overall strategic development of the enterprise. This study uses a mixed research method to construct a path model for the corporate universities to empower enterprise development and uses a questionnaire survey method to verify the conclusion. During the important transition period when the state regulates corporate universities and encourages the integration of production and education, this research pays attention to the changes and trends of this group promptly.

This study clarifies the role of corporate universities in empowering organizations. Based on organizational learning theory, organizational capital theory, and knowledge-based view. The research method conducts an in-depth analysis of the path of corporate university empowerment of enterprise development under the Chinese context, and the research conclusions are as follows: First, this study establishes a path model of corporate university empowerment enterprise development through the three-level coding method, and identifies the current Chinese context. Enterprises plan the mission of corporate university and the core path of corporate university empowering enterprise development, including "linking strategy", "closely connecting business", "cultivating team", "shaping organization" and "inheriting culture" paths, which are the paths of most corporate universities. The core value of being and having an impact. Thus, the path model of corporate university empowerment organizations established in this study shows how corporate university deploys and implements strategies at the employee, team, and organizational levels, which further enriches the existing literature on

corporate university and empowerment research. This research regards the development process of corporate university as an enabling process, and corporate universities can interact with enterprises to exert an enabling effect. During this process, this study found that the core of corporate university empowerment of enterprise development lies in connecting multiple subjects (such as internal employees, management teams, external suppliers, and strategic alliance partners) to create value for customers. In the final analysis, this connection is an enabling capability, which requires the integration of organizational elements (business, structure, process, and mechanism) to drive the formation of organizational vitality and organizational capabilities, thereby improving organizational performance and creating sustainable competitive advantages.

Second, this study analyzes the knowledge management function that corporate universities empower enterprises to develop. With the continuous emergence of knowledge management research in corporate universities (Antonelli et al., 2013; Li et al., 2015; Singh et al., 2019), the academic community is paying more and more attention to how corporate university knowledge management can help corporate universities. Since most of the existing research in the field of corporate universities is mainly descriptive research, more discussions are devoted to the construction of the operational support system of corporate universities. Theoretical construction, which is in stark contrast to the thriving development of corporate universities in reality. In addition, although some studies have analyzed the content of the relationship between corporate university and organizational capacity, more is to analyze the resources required for the development of corporate universities from the perspective of the corporate university itself. For example, some studies believe that the faculty of corporate universities is an effective carrier to support the business system, and the business operation of corporate universities requires hardware equipment, the latest learning technology, and information-based system platform support (Kimiloglu et al., 2017).

However, there is still a lack of research on corporate universities and organizational development. Some scholars have called for more attention to the construction of organizational capital and capacity systems in corporate universities (Chen Yunqi and Xu Yusen, 2018). Therefore, this study goes further based on the existing research and studies the multiple paths of corporate universities empowering enterprise development. In addition, this study also refers

to the view of business success in the "Yang's Triangle Theory of Organizational Development", that is, a successful organization should have advantages in thinking, ability, and governance. Based on this, this study proposes that corporate universities can effectively enhance strategies, develop collaboratively with business departments, stimulate team creativity, promote value co-creation among departments at all levels, and enhance corporate cultural identity in terms of corporate thinking, capabilities, and governance, which enriches the research status of corporate university capability system.

Third, this study expands the path model of corporate university empowering enterprise development and incorporates future trend analysis into the theoretical model. Based on the characteristics and changing trends of the external environment of organizational development, this study points out that the current growing demand for organizational agility and flexibility in China puts forward new requirements for organizational development theory, and uses the trend model to show the visible trend of corporate universities empowering organizations in the context of China, providing a reference perspective from practice for opening the cognitive "black box" of research on excellent corporate universities and organizational capabilities.

7.2 Practical Contributions

This research has implications for the future practice of policymakers, corporations, and corporate university administrators. For policymakers: The research conclusions provide a reference for government departments to formulate top-level plans for corporate universities. On August 26, 2021, the Shanghai Municipal Government issued the "Opinions on Promoting the Innovative Development of Continuing Education for Employees in the New Era", proposing that by 2025, Shanghai needs a distinctive "dual system" continuing education model for employees, which includes both academic education, but also vocational education (e.g., training provided by corporate universities, career development planning, study coaching). The guiding opinions of the Shanghai government have pointed out the direction for the innovative development of continuing education for employees in the new era.

Therefore, the practice of how corporate universities can empower the development of enterprises must consider the local environment of Chinese enterprises, to be truly applicable to the improvement of Chinese enterprise management and provide practical suggestions for

managers. This research is a policy for the future government to encourage enterprises to participate in the lifelong education system. The formulation provides a practical reference. In addition, although the naming method of "corporate university" is controversial in China, when the functional complexity of corporate training and talent development work rises from employee skills training to hierarchical and classified talent training, professional team training, and emphasis on individual When the improvement of organizational performance is required to promote the development of the company's strategy, business, and culture, such a corporate training center, corporate learning and development center or corporate business school has played a strategic role as a "corporate university". As a regular form of "corporate university", "corporate university" is not important whether the name is "university", but whether the core functions are closely matched with the strategy (such as business team, talent development, organizational culture) is the key.

For the managers of enterprises and corporate universities: First, the conclusions of the study inspire corporate university managers to do a good job in the construction of corporate universities from the perspective of organizational development. In prior research and practice, from the perspective of corporate universities, human resources work planning and training work planning are carried out according to the planning and needs of organizational development. Corporate universities play a supporting role in the entire organizational system. With the increasingly mature development of corporate universities, excellent corporate universities have completed self-management and self-development and formed a self-evolving ecosystem. It is necessary to further refine and summarize the methodology, and start the practice from "the organization wants me to do it" to "I want to do it for the organization", from "passive activity" to "active action". Second, the research conclusion inspires that corporate managers need to pay full attention to the corporate university of knowledge management carrier. Top managers of enterprises need to think about how to establish a three-dimensional empowerment system within the enterprise so that the enterprise can become an enabling organization - this is also one of the directions of enterprise self-evolution (Frederick, 2017).

When an organization can evolve itself and grasp the key to empowerment, it can accelerate the improvement of organizational performance, acquire the ability of healthy and rapid development of the organization, and finally achieve a "win-win" between the enterprise

and the corporate university. Third, the research conclusions inspire local managers to formulate corporate university empowerment strategies based on the reality of China. The analysis framework of the corporate university empowerment enterprise path proposed in this study is beneficial to help enterprise managers explore the boundaries of empowerment, improve the effectiveness of leadership management behaviors, talent training, and performance management empowerment methods, and will also serve as a platform for more Chinese enterprises and global enterprises. Enterprise empowerment management provides practical reference value.

7.3 Limitations and Prospects

This study has limitations in sample selection and data collection. First, in the qualitative stage, 13 corporate universities were chosen as research samples due to their outstanding performance in empowering businesses. Consequently, all cases were unable to provide information and data on failures. Although the selection of cases was carefully considered for their representativeness and relevance to the research questions, efforts are needed to enhance the generalizability of the study results. Future research will consider including "failed corporate universities" to supplement the existing empowerment pathway model, focusing on analyzing the phenomena of corporate university transformation failure and corporate withdrawal from corporate universities.

Second, in the quantitative analysis stage, a survey questionnaire regarding corporate university empowerment pathways was distributed, with 119 valid responses collected. Cross-sectional data may not effectively reveal causal relationships. Therefore, future research could employ secondary data to construct longitudinal panel data, further validating the conclusions drawn in this study.

Third, as the core analysis in this study originates from the corporate university empowerment practices in China, the 'Corporate University Empowerment of Business Development Integrated Analysis Model' established using the coding method is applicable to corporate universities in China in terms of empowering practices in strategy, business, teams, organizations, and culture. The research conclusions may not apply to corporate university empowerment practices in developed countries with earlier and more refined practices. Further

discussions are needed to determine whether the theoretical model can guide corporate university empowerment practices in other emerging economies.

This study has focused on the pathways and trends of corporate universities empowering business development in the context of China. It also provides valuable directions for future research. Firstly, researchers can continue exploring new phenomena, concepts, and practices within corporate university empowerment in future studies. While this study identified innovative practices in case analysis and coding, such as the *"Production-Education Integration" emphasized by Bank of China University, clearly stating that "Bank of China University is not only a vocational education platform for internal staff but also an important platform for providing vocational education and training services to the society and achieving Production-Education Integration."* Additionally, corporate universities led by private enterprises, such as Dongfeng Nissan University, have proposed the concept of *"building a shared ecosystem for supply chain partners,"* emphasizing that the future business and functions of corporate universities should not only empower internal users and employees but also cover partners in the supply chain, industry chain, and even value chain. This trend is also reflected in other corporate universities: Red Star Macalline Group Management College's practices include focusing on the industrial chain ecosystem, and Geely Corporate University has similar practices focusing on the ecosystem, such as *"building a comprehensive empowering learning ecosystem."* Therefore, future research can explore the trend of "Production-Education Integration" in corporate universities, the transformation of digital learning platforms, and new practices like internal innovation incubators. Researchers can particularly focus on the interaction between innovation and corporate university empowerment, investigating which innovation resources and capabilities can enhance the efficiency of corporate universities in empowering business development.

Second, researchers can analyze the obstacles in the pathways of corporate university empowerment, examining factors that may lead to inadequate support from corporate universities to businesses. Besides, future researchers can analyze the relationship between corporate university empowerment pathways and organizational success using the integrated Path-Trend Model established in this study. Utilizing the Qualitative Comparative Analysis (QCA) method, researchers can further refine typical configurations.

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APPENDIXES

Appendix 1: Literature Visualization Results

1. Research Status

The input data collected in our study is generated by a topic search query from Web of Science Core Collection databases. There are two reasons why we choose the 1985-2020 period to conduct this survey: on one hand, the first paper on the “corporate university” appeared in 1985, which was the beginning of research in this field; On the other hand, our search period is up to September 2021 and incomplete years are not considered. The extraction of the sample took place for all “Articles” and “Review” types, excluding the other types of documents in the search: 120 articles whose topics (titles, keywords, and abstracts) contain “corporate university” were obtained. Data format transformations were undertaken using Citespace 5.7.R5 SE.

For these publications, we selected all information relevant to the research to analyze the following two terms: Literature overview: for instance, the number of articles, geographical and periodical distributions, leading authors and institutions, academic collaboration, article citations, and national comprehensive strength were investigated. Together with the terms, the latest research status of the corporate university was evaluated.

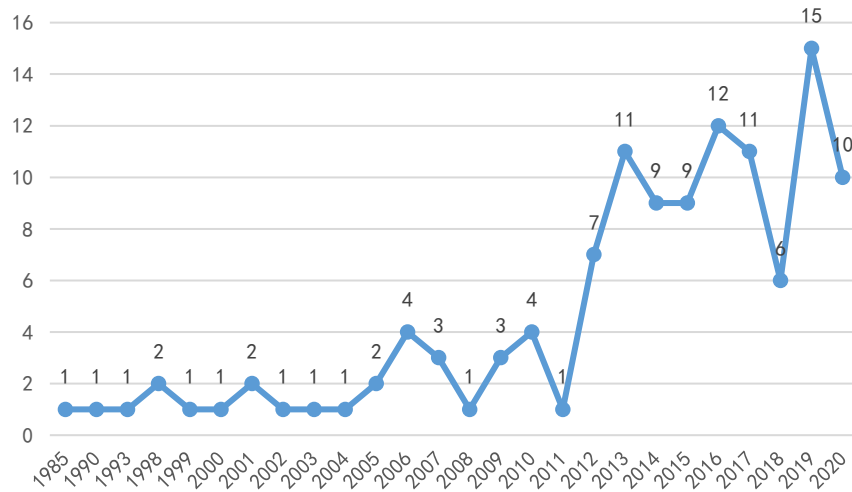


Figure A1. The number of papers annually in the field of “corporate universities”

Table A1. Top10 authors in the field of “corporate university”

Author	Publications	Percentage
Rademakers MF	4	3.333
Buchbinder H	3	2.5
Gardoni M	3	2.5
Rheume L	3	2.5
Rolfe G	3	2.5
Cappiello G	2	1.667
Lytovchenko I	2	1.667
Newson J	2	1.667
Payne PG	2	1.667
Pedrini G	2	1.667

Table A2. Top10 institutions in the field of “corporate university”

Institutions	Publications	Percentage
Erasmus University Rotterdam	5	4.17%
York University Canada	5	4.17%
Ctr Strategy Leadership	4	3.33%
Dutch Fdn Corp Univ	4	3.33%
University of Quebec	4	3.33%
Ecole De Technologie Superieure Canada	3	2.50%
Monash University	3	2.50%
State University System of Florida	3	2.50%
Swansea University	3	2.50%
University of Toronto	3	2.50%

Table A3. Top10 countries/regions in the field of “corporate university”

Country/Region	Publications	Percentage
USA	33	27.50%
Canada	19	15.83%

Australia	17	14.17%
Brazil	9	7.50%
England	7	5.83%
Netherlands	5	4.17%
Peoples R China	5	4.17%
Italy	4	3.33%
Poland	3	2.50%
South Africa	3	2.50%

Table A4. Top10 categories of WOS database in the field of corporate university

Categories of WOS database	Publications	Percentage
Education Educational Research	38	31.67%
Management	18	15.00%
Business	7	5.83%
Social Sciences Interdisciplinary	7	5.83%
Information Science Library Science	6	5.00%
Sociology	6	5.00%
Cultural Studies	5	4.17%
Literature	5	4.17%
Environmental Studies	4	3.33%
Anthropology	3	2.50%

2. Research hotspots

Research hotspots: author keywords contain the core information about which authors are most concerned, therefore co-citation analysis, as well as frequency analysis of keywords has been widely used to investigate the development trends and research hotspots in a lot of previous studies (Xie, Zhang et al., 2008; Li, Ding et al., 2009; Du, Li et al., 2014). In this sense, this study applies Citespace 5.7.R5 SE to visual analysis of the latest research hotspots.

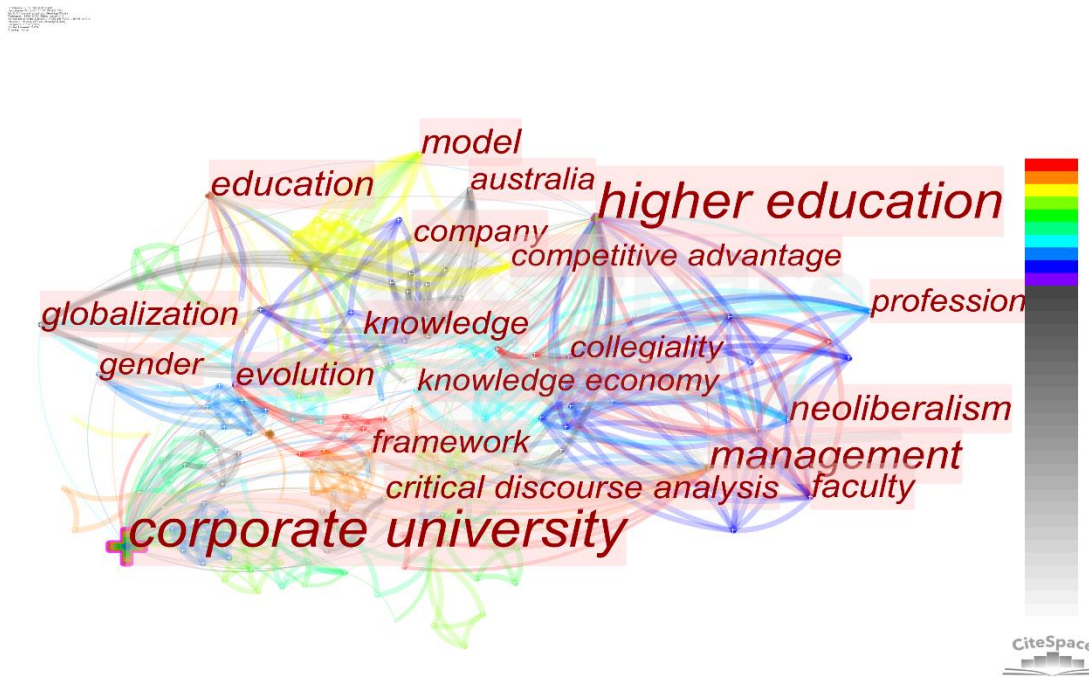


Figure A2. Core co-occurrence keywords in the field of “corporate university”

We set the time slice to one year and selected the top words from each slice to citation-view the data with a modified g-index method. Thus, we obtain 260 nodes and 774 links between them. It can be seen that “higher education”, “knowledge economy”, “competitive advantage”, and “management” are the most popular research areas.

Table A5. Keywords with top10 co-occurrence frequency

Frequency	Centricity	Year	Keywords
32	0.65	2003	corporate university
17	0.54	2005	higher education
8	0.1	2013	management
8	0.1	2012	education
4	0.01	2015	corporate training
3	0	2015	organizational learning
3	0.03	2005	learning
3	0.02	2019	knowledge management
3	0.02	2004	knowledge
3	0.04	2008	corporate education

Next, we perform cluster analysis for co-occurrence keywords. According to visual analysis and statistical analysis, the corporate university can be primarily divided into the following fields¹:

¹ The clustering with too few members is automatically eliminated by the algorithm.

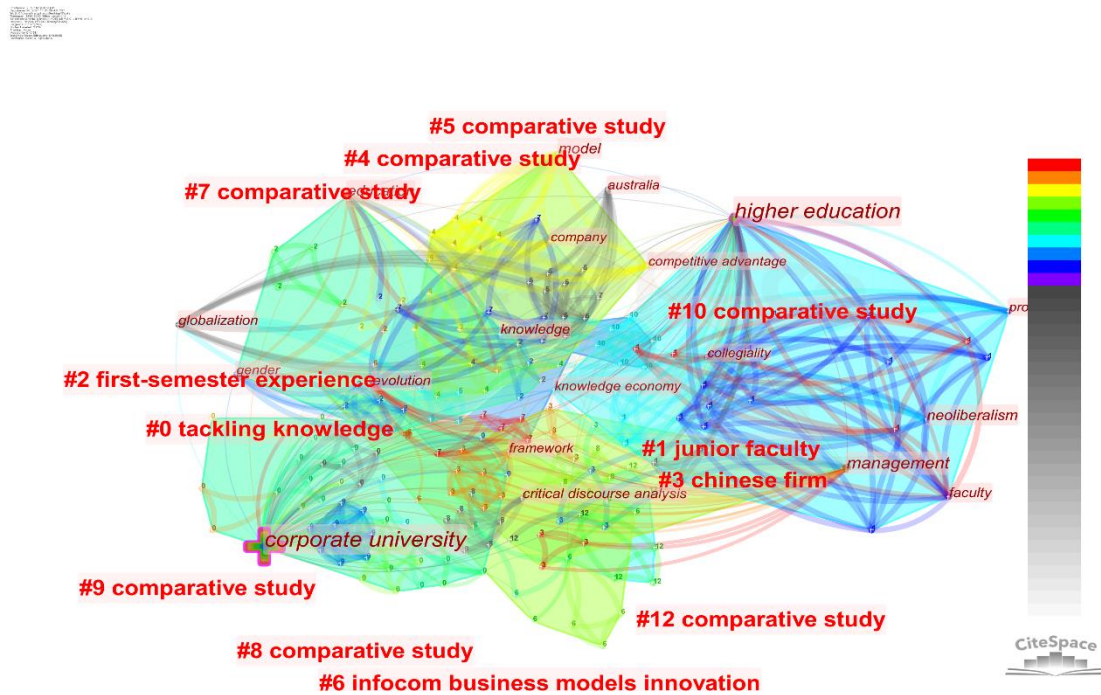


Figure A3. The main clustering of co-occurrence keywords

For example, #0 tackling knowledge, the members of the cluster whose citation frequency is higher than 2 are: “academic hiring”, “corporate training”, “corporate education”, “corporate university”, and “organizational learning”.

Table A6. Members of cluster #0 with a frequency greater than two

Cluster	Frequency	Degree	Centricity	Members of the cluster
#0	2	6	0.02	academic hiring
#0	4	8	0.01	corporate training
#0	2	2	0.01	corporate education
#0	29	87	0.65	corporate university
#0	3	8	0	organizational learning

Appendix 2: Interview Outline

1. How to establish the goal, mission, and vision of a corporate university based on the three cores of strategy, organization, and talent? How to keep the vision and goals of the corporate university in line with the strategy in the process of operation?
2. How to develop a feasible talent empowerment strategy based on corporate university positioning and talent management practice? How to grasp the internal mechanism of value creation of a corporate university to formulate the path of establishing and upgrading a corporate university, to realize the sound development of a corporate university?
3. How to establish the position of a corporate university in organization management? How to ensure that the corporate university is integrated into the business unit and management process? How to realize vertical integration and horizontal collaboration between enterprise and university?
4. How can corporate universities form a systematic enabling strategy based on ability and performance? What are some ways to cover the whole process of employee career development? How can enterprises make the corporate university business solution scientific?
5. How to construct a corporate university management process that covers many business units? How can corporate universities support the diverse needs of their employees?
6. How can corporate universities enhance their radiation and attraction? What practices do corporate universities currently have that reach into their supply chains and ecosystems?

Appendix 3: Summary of Vision, Mission, and Values of Corporate Universities

Corporate university name	Basis information	Functional positioning	Mission purpose	Development Vision
Dongfeng Nissan University	Enterprise: Dongfeng Motor Co., Ltd. Dongfeng Nissan Passenger Vehicle Company Type of enterprise: Three foreign-funded enterprises (Sino-foreign joint ventures, Sino-foreign cooperative enterprises, and foreign-funded enterprises) Industry type: Manufacturing Year established: 2013 Location: Guangdong Province	Dongfeng Nissan University undertakes the four functional positioning of the enterprise talent development learning center, the enterprise management innovation practice center, the enterprise culture brand communication center, and the enterprise alliance development cooperation center.	Dongfeng Nissan University focuses on the five major missions of improving employees' professional ability, improving organizational competitiveness, disseminating and inheriting corporate culture, enhancing the overall strength of the industry value chain, undertaking social responsibilities, providing strong talent guarantee for the sustainable and healthy development of DFN's business, and China's automobile industry. contribute to its development.	With the vision of learning igniting passion, Dongfeng Nissan University encourages all employees to learn better by creating a learning atmosphere, creating learning opportunities, creating learning motivation, and adopting various methods to improve employees' enthusiasm for learning.
China Resources Gas College	Enterprise: China Resources Gas (Group) Co., Ltd. Type of enterprise: Central/state-owned enterprise Industry type: Electricity, heat, gas Year established: 2012 Location: Guangdong Province	Positioning: the cradle of talent training in key positions, the center for summarizing and disseminating management practices, the platform for managers to inherit cultural experience, and the window for external exchanges of innovative practices	Talent cultivation of elites in the gas industry, actively disseminating the company's strategic thinking and corporate culture, carrying out clean government education, and building party members' ideological and work styles; summarizing management practices, inheriting excellent culture, creating a smart ecological complex, promote industrial upgrading, and contribute to China Resources Gas' rapid and sustainable development. Sustained development provides intellectual support.	To become a leading professional management school in China

Red Star Macalline Group School of Management	Enterprise: Red Star Macalline Home Furnishing Group Co., Ltd. Type of enterprise: Private Enterprise Industry type: Wholesale and retail trade Year established: 2008 Location: Shanghai	Focus on the industrial chain ecosystem	The Whampoa Military Academy in the Home Furnishing Industry	Gather industry wisdom and lead talent development
CCB University	Enterprise: China Construction Bank Group Type of enterprise: Central/state-owned enterprise Industry type: Bank, Finance Year established: 2018 Location: Beijing	Cultural Auditorium, cultivating the excellent culture of CCB; Thought Academy, creates and shares management wisdom; Empower the workshop and impart practical knowledge and skills. Empowering people, empowering organizations, empowering society, and empowering national education.	Mission: Serving Society, Serving Strategy, Serving Employees Serve society and contribute financial solutions Service strategy to promote steady and innovative development Serving employees and improving professional competence	Vision: New Era, New Finance, New Ecological Corporate university
Geely Corporate university	Enterprise: Zhejiang Geely Holding Group Co., Ltd. Type of enterprise: Private Enterprise Industry type: Manufacturing Year established: 2015 Location: Zhejiang Province	Talent delivery, cultural empowerment, organizational development	Empowering businesses and achieving every Geely fellow	In the future, Geely Corporate university will further cooperate with upstream and downstream industry chains, industry institutions, global universities, and other partners to create an ecological chain of talent training throughout the career cycle and help Geely's global business continue to develop.
Robam University	Enterprise: Hangzhou Robam Electric Co., Ltd. Type of enterprise: Central/state-owned enterprise Industry type: Home appliance manufacturing Year established: 2014	Improve employee capabilities and promote career development; help improves business capabilities and focus on comprehensive talent management; inherit corporate culture and assist in organizational transformation; promote ecological	Cultivate outstanding employee, cultivate our leading talents, and strive to contribute to the sustainable development of small and medium-sized enterprises in China, and become a consultant for employee growth, business	enabler of the company's organizational capacity building, and continuously create value for the sustainable development of China's small and medium-sized entities

	Location: Zhejiang Province	collaboration and cultivate industry talents.	development partner, organizational culture inheritance, and practical value platform.	
Sany Corporate University	Enterprise: Sany Group Co., Ltd. Type of enterprise: Private Enterprise Industry type: Manufacturing Year established: 2003 Location: Hunan Province	The sole purpose of a corporate university is to serve industries and businesses well. According to the development model of "one body and two wings", the college takes the development of the service industry as the main body and takes academic education and vocational training as the two wings. "Based on Sany, facing the society, deeply integrating into the industrial chain, combining engineering and learning, improving quality, and connecting the front and rear markets of high-end equipment manufacturing", cultivate sustainable development of innovative and applied talents. With the concept of "creating first-class talents", cultivate and develop talents that meet the strategic development needs of Sany Group, and build a learning organization; undertake external training services from the industry, government, and other social organizations, export Sany management and culture, and assume social responsibilities.	Mission and purpose: The college takes whole-person education and "humanities + science + specialties" as its guiding ideology, aims at empowering learners with the abilities they need for lifelong development, and realizes a diversified, personalized, and customized talent training model.	Building a new type of university facing the future To run a future-oriented university: selective education, suitable sex education, precision sex education, high-efficiency sex education, open school, and smart campus. Run a university with the most distinctive features: differentiated development, not the first, but the only. To run a university that returns to the essence of education: general ability + professional ability, education that guides and supports people's lifelong growth. To run a university that integrates technology into education: work efficiency, teaching efficiency, learning efficiency, life efficiency, high degree of freedom, personalization, and customization. To run a university that empowers learners and employees with the ability to grow continuously: the ability to dream, the ability to realize dreams, and the ability to hold dreams.

				Run a university dedicated to promoting social progress and cultural development: exporting qualified talents, doing things first, promoting positive energy and excellent culture, and creating value.
Taikang Business School	Enterprise: Taikang Insurance Group Co., Ltd. Type of enterprise: Private Enterprise Industry type: Financial Industry Year established: 2016 Location: Beijing	The home of Taikang people's spirit is the Taikang Spirit Inheritance Ceremony, Taikang Cultural Experience Base, and Taikang Strategic Communication Center; Taikang Business School is the cradle of Taikang talent training, focusing on building international leaders, professional cadres, and professionalism Taikang Business School is a new starting point for Taikang people's career, a high-end platform for building industry-university-research integration and cultivating a large health industry, focusing on strategic advantages to cultivate core talents, and continuously linking external first-class resources.	Taikang people's "spiritual home, the cradle of talents, and new starting point of career" is Taikang's corporate university.	Taikang Business School adheres closely to the company's strategy and builds Taikang's characteristic education soft power around the "Taikang Plan for Longevity Times". Focus on the training of core cadres, empower organizations, teams, and employees, and start the new three-year construction.
New Hope Business School	Enterprise: New Hope Co., Ltd. Type of enterprise: Private Enterprise Industry type: Agriculture, forestry, animal husbandry, and fisheries Year established: 1995 Location: Sichuan Province	Committed to "undertaking the company's strategy, promoting performance improvement, and helping talent development", integrating action learning, case teaching, blended learning, cross-border research, coaching, and other diverse learning methods to form a unique talent training and Develop	Mission and purpose activate the learning energy of employees and industry chain partners, create a learning environment that stimulates innovative thinking and continuously improves problem-solving ability, shape and spread professionalism and business demeanor, improve the skills of	Goal and Vision: Integrate Eastern and Western wisdom, adhere to the spirit of positivity, pragmatic, and innovation, promote the global development of New Hope Liuhe, and provide a high-value, continuous innovation and learning environment for the agriculture, animal husbandry, and food industry.

		models and embed learning into organizational workflows and practices.	employees and customers, and promote the company's sustainable development.	
China General Nuclear University	Enterprise: China General Nuclear Power Corporation Type of enterprise: central/state-owned enterprise Industry type: Electricity, heat, gas Year established: 2003 Location: Guangdong Province	Positioning of the CGN Party School: the main channel of training rotation, the melting pot of party spirit training, the main position for theoretical study, the decision-making consultation think tank Positioning of CGN University: Origin of management change, group strategy boost, talent doubling incubation, business development partner, corporate culture dissemination, organizational knowledge accumulation	The origin of management change, corporate strategy boost, talent doubling incubation, business development partner, corporate culture dissemination, organizational knowledge precipitation.	Become a benchmark for enterprise talent training.
China Railway Fourth Bureau Group Corporate university (Party School)	Enterprise: China Railway Fourth Bureau Group Co., Ltd. Type of enterprise: Central/state-owned enterprise Industry type: Construction industry Year established: 2016 Location: Anhui Province	Cultivating talents, promoting change, managing knowledge, promoting culture, sharing communication	Serve the development strategy of the enterprise as the center, and become the business booster, strategy driver, performance facilitator, and industry coordinator of the enterprise	To be a leader in talent training and organizational development in the field of infrastructure

China Mobile Academy	Enterprise: China Mobile Communications Group Co., Ltd. Type of enterprise: Central/state-owned enterprise Industry type: Information transmission, software, and information technology services Year established: 2000 Location: Beijing	With the goal of "striving to be a leader in first-class education and training institutions and serving as a leader in education and training of the whole group", the college has continuously built its core competitiveness of "trustworthy, reliable, usable, inseparable, and irreplaceable". The functions of the college are positioned in the four dimensions of party school education, management training, online learning, and business support. Enhance the value of the teaching platform, support the development of business work in an all-around way, and promote the implementation of the company's strategy and the sustainable and healthy development of the company.	As a corporate university of China Mobile, China Mobile College shoulders the mission of cultivating Chinese-level management talents, professional technical talents, and business backbones for China Mobile.	China Mobile Academy is committed to becoming a talent training base and creating a platform for ideological interaction.
ZTE College	Enterprise: ZTE Corporation Type of enterprise: Central/state-owned enterprise Industry type: Manufacturing Year established: 2003 Location: Guangdong Province	Learning consultant for employee growth, business unit partner, an engine for learning organization construction	Learning consultant for employee growth, business unit partner, an engine for learning organization construction	ZTE College is committed to building an employee capability development system that is "combat-oriented and connected from top to bottom", and supports the company's operations with the model of "mainly focused on business and focused on professional development", delivering customer value and leading employee growth.

Note: The rankings of corporate universities in this table are in no particular order, only in alphabetical order in Chinese.

Appendix 4: Expert Evaluation Form of Some Chinese Corporate Universities

Corporate university	Expert evaluation
Dongfeng Nissan University	Under the background of the Industry 4.0 era, Dongfeng Nissan University shoulders the five missions of promoting the symbiotic development of enterprises, industries, and society, with the concept of "whole person education", supplemented by systematic construction methods, agile learning technology, and unique "Dongfeng Nissan University". Learning methods", is committed to accelerating organizational transformation, closing business linkage, and helping employees grow "happily".
Robam University	Robam University creatively inherits the cultural genes of its affiliated enterprise "engraving the brand with ingenuity", and constantly innovates the concept of talent training, explores the learning mode, updates the learning content, and optimizes the learning technology according to the growing talent development needs of the internal and external industrial value chain of the enterprise. , strive to build a corporate university learning development and operating system with the characteristics of this enterprise and this industry, use the "soul of ingenuity" of education to boost enterprises and industries, and compose the dream of the ingenuity of Chinese brands!
Red Star Macalline Group School of Management	Since its establishment in 2008, Red Star Macalline School of Management has accumulated rich experience in running schools, teachers, and knowledge accumulation through "joint school running, internal training, curriculum development" and other methods, and constantly integrates upstream and downstream resources in the industry, from "talent training", industry research, trend guidance, cultural dissemination" and other aspects, created the first "Whampoa Military Academy" in the real sense of the home furnishing industry, and promoted the innovative development of corporate universities in practical education.
Geely Corporate university	Adhering to the development orientation of "talent delivery, cultural empowerment, and organizational development", Geely University has established a comprehensive coverage of enterprises, including A "comprehensive empowerment learning ecosystem" for high-level and grass-roots leadership training, professional ability training throughout the value chain, and general ability training for all employees. Through the "customized, scenario-based, and personalized" training plan, employees at all levels can learn and grow by benchmarking, and business partners can develop and win together. Actively advocate the learning concept of "everyone is a teacher, and everyone is a student", emphasizing that "everyone is a talent, and everyone can become a talent", providing a broad platform for the growth and development of employees. At the same time, combined with the actual needs of the industry, strengthen the carrier of skill competitions and project actual combat, "rehearse with competition", "combination of training and combat", and test employees' true abilities based on real business scenarios. Geely Corporate university has fully supported the business development of Geely Group from the four aspects of "cultural integration, talent delivery, knowledge sharing, and organizational pain-point tackling"!

CCB University	As a corporate university of a state-owned bank, CCB University always adheres to the principle of "serving the society, serving the strategy, and serving the employees", actively contributes financial solutions, promotes the steady and innovative development of enterprises, and enhances the professional ability of employees as its responsibility, and actively explores the "future bank". "The way of empowerment. At the same time, using a new generation of information technology, combining the advantageous resources of all parties, actively explore and make bold attempts at technological innovation in the field of talent development and training. Based on years of data accumulation, it pioneered and designed the IPI infrastructure construction index in the industry, established the CCB-Nankai systemic risk index system, promoted the continuous monitoring of the systemic risk index, and played a pioneering and leading role as a large bank.
Taikang Business School	Taikang Business School always focuses on the company's strategy and the need for cadre training and strives to become the "spiritual home, the cradle of talents, and a new starting point for the career" of Taikang people. Firmly build the soft power of Taikang education and training, continue to operate steadily, and build a business school for "Taikang people". Constantly "going out" and "inviting in", establishing extensive resource connections, providing "Chinese sample Taikang solutions" for Taikang's big health industry ecosystem construction and the era of longevity, and for realizing the combination of insurance and physical medical care. positive contribution.
Sennics School of Management	Scenics School of Management is committed to cultivating all kinds of talents needed for enterprise development, and to providing continuous talent power and cultural power for the sustainable development of "Sennics". Sennics School of Management has a complete system, continuous innovation, and remarkable results. The college organically combines project practice and training growth and has established a complete talent team training model, advocating the TBP model for training business partners, serving business, and effectively promoting organizational performance. Sennics School of Management not only laid the foundation for the development of the company but also used the "Amoeba" business philosophy to innovate and practice the model of the college's value and mutual benefit, becoming a successful case in the practice of corporate university.
New Hope Business School	Incubate long-term value and help achieve performance. New Hope Business School, as a strategic engine that empowers organizational development and helps talent cultivation, has always adhered to the talent cultivation concept of "combining combat training and learning war in war", closely following the company's "strengthening feed, growing pigs, In 2020, overcoming the impact of ASF and COVID-19, resolutely optimizing the talent training model, digging deep into business needs, interconnecting online and offline, with the help of the digitalization of New Hope University The learning platform, relying on multi-dimensional methodologies such as experience extraction and action learning, focuses on performance improvement. In addition, to ensure the sustainable development of the company, New Hope Business School has linked the organization department to open up the hierarchical

	talent supply chain of "talent identification-cultivation-development" to ensure the benign flow and supply of talent. New Hope Business School's diversified talent training and development model, digital exploration and practice, knowledge management, and other aspects have high reference and promotion value and can become a learning benchmark for Chinese corporate universities.
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Note: The rankings of corporate universities in this table are in no particular order, only in alphabetical order in Chinese.

Appendix 5: The First Round of Qualitative Data Collection (2019)

Fill in by way of proof, that is, present the content points, key practices, or quantitative data that can prove the actual situation of the enterprise; unless there is a clear time limit, the time limit for filling in the content is within two years (2018-2019). Practices and results; except for a clear word count requirement, there are no other word count requirements, and the standard is clear content, clear practice, and credible results; each column has clear filling requirements, please try to follow the details of each column as much as possible. If required, do not omit; if there is no relevant content, you can fill in none.

All content collected in this survey will be used as a reference for the selection of the best corporate university rankings. With your consent, the content will also be used in the research of the academic projects of the University of Otago and Shanghai Jiaotong University, which are related to Chinese companies. development of the university. However, there is no commercial interest involved, and we will be grateful if you finally decide to participate, and if you decide not to participate, there will never be any adverse impact on you. Thank you for your cooperation!

1. Overall overview

Company Profile	(1) Company name	1
	(2) Date of establishment	2
	(3) The nature of the company	3
	(4) Company scale: number of enterprises; annual operating income	4
	(5) Industry	5
Introduction to Corporate University	(1) Name of the corporate university	6
	(2) Date of establishment	7
	(3) Organizational structure and full-time staff	8
	(4) Management Mechanism	9
	(5) Development System	10
Value proposition	(1) Mission and purpose of the corporate university	11
	(2) Functional orientation of corporate university	12
	(3) Training objects of corporate universities	13
	(4) Development Vision of Corporate University	14
Development path	A landmark event since the establishment and operation of the corporate university	15

Innovative features	The value characteristics and innovative practices of corporate universities in the process of promoting organizational development	16
Direction of development	The stage summary of the corporate university's development process	17
	The focus and action direction of corporate university in the next stage	18

2. Strategic Alignment

Planning to lead	(1) The practice and key content of the connection between the corporate university and the company's strategic theme	19
	(2) The corporate university's grasp of the characteristics and growth laws of the talents of the training objects	20
	(3) The learning map covering the training objects or the key learning projects formed by the corporate university	21
	(4) Important practices of corporate universities in promoting the digital transformation strategy of enterprises (if any)	22
Business close connection	(1) Training themes planned in 2019 to connect with the needs of business and management departments	2
	(2) Introduction to the training and learning activities for business and management problem-solving implemented in 2019	24
	(3) A business training system that effectively promotes organizational performance improvement	25
Ability match	(1) Introduction to the learning project for the development and implementation of high-level or enterprise key talents	26
	(2) Professional support for high-level decision-making or prospective research	27
	(3) Successful practice of corporate university in corporate internal consulting	28
Cross-border integration	The integration and application of learning resources within and outside the enterprise by corporate university	29

3. Organizational coordination

Vertical integration	(1) The positioning of the corporate university in the corporate organizational structure	30
	(2) Introduce the company's training management system structure and management mode	31
	(3) Powers and responsibilities of corporate universities in the training management system	32
Horizontal collaboration	(1) Collaboration between corporate universities and business units and functional departments	33

	(2) The company's process or mechanism for the connection between talent training and talent management	34
	(3) The practice of corporate university to enhance employee cohesion and sense of belonging	35
	(4) Implementation methods of corporate universities in the process of promoting corporate cultural identity	36
Organization is effective	(1) Function setting and staffing of corporate university	37
	(2) The corporate university development concept, talent training concept, business construction, and operation concept that the corporate university operation management team upholds	38
	(3) The number of projects per capita and the number of students trained per capita in 2019 by the operation and management team of the corporate university	39

4. Complete business

Content docking	(1) The connection and operation of training programs and talent development standards (vocational qualifications, competency models, performance standards, and career development channels)	40
	(2) The connection and operation of training projects and talent development plans or talent echelon construction	41
	(3) The connection and operation of training projects and talent evaluation (ability and performance)	42
Unification of elements	(1) The key success factors of corporate university training programs and their application and promotion	43
Advanced method	(1) Training methods or learning techniques for different types and levels of training programs (including mobile internet learning techniques and methods)	44
	(2) The characteristic cultivation theory, model, method, or technology formed by precipitation	45

5. Knowledge Integration

Knowledge management	(1) Standards and templates for knowledge extraction, precipitation, dissemination, and application	46
	(2) The process and mechanism of knowledge extraction, precipitation, dissemination, and application	47
	(3) Construction of an information-based knowledge management platform	48
Knowledge innovation	Knowledge innovation achievements connected with talent training, including but not limited to courses, cases, teaching materials	49
	(1) Application of knowledge innovation achievements	50

Knowledge operation	(2) Application of information-based knowledge management platform (coverage, traffic, quantity of knowledge resources, platform operation results)	51
	(3) Successful cases of corporate universities in external training and corporate university consulting	52

6. Resource matching

Strong teachers	(1) The process and mechanism of the construction and management of the teaching staff	53
	(2) The number and proportion of existing internal full-time and part-time teachers, the number and proportion of teachers at all levels, and the number and proportion of teachers in various professional categories	54
	(3) The list and performance of influential teachers within the company and in the industry	55
Perfect service	(1) Construction and use of training facilities or infrastructure	56
	(2) Typical practices of student service management	57

7. Lean operation

Exquisite development	(1) Precipitate and form fruitful project development models, mechanisms, and tools	58
	(2) Brand courses and benchmarking projects of enterprises or industries formed by precipitation	59
Fine implementation	(1) High-efficiency, high-quality, and well-experienced project implementation management models, mechanisms, and methods	60
	(2) Formed innovative project effect evaluation methods and tools	61
Efficient support	(1) The practice and application of training project management using network technology, mobile Internet, and data technology.	62
	(2) Effectively implemented learning management system	63

8. Value Demonstration

Talent growth	(1) Construction of a talent training system	64
	(2) Excellent practices in the growth cycle of training objects, the ability of key talents and the advancement of positions, the competency of key talents based on talent inventory, or the optimization of talent structure.	65
Performance contribution	(1) Quantitative and qualitative impact of corporate universities on stakeholders (executives, business units, functional departments, students.)	66
	(2) Contribution of corporate universities to the promotion of annual strategic themes and the improvement of organizational capabilities	67
	(3) Input and output of learning and training projects	68
Cultural push	(1) Corporate learning culture performance	69

	(2) The attractiveness of corporate universities to students and teachers	70
	(3) Effectiveness of corporate culture communication	71
Brand influence	(1) The output of training standards, projects, courses, teaching materials, and training theories, models, methods, and tools outside the enterprise	72
	(2) Awards of corporate universities	73
	(3) The award or recognition of the trainees	74
	(4) The status of the corporate university being recognized by the government, industry organizations, or serving in society	75
Social responsibility	(1) Industry research and cross-border research practice conducted by corporate universities in operation	76
	(2) Knowledge output in the operation of corporate universities	77
	(3) The practice of industry-university cooperation in corporate universities	78

Appendix 6: The Second Round of Qualitative Data Collection (2020)

Filling out the application materials for the selection of corporate universities requires considerable effort, but this is exactly the value of the selection of the best corporate universities in China - the process of filling in the application is a process of value precipitation and focus improvement. At the same time, it has also become an important basis for selection. The specific instructions are as follows:

Filling in the materials includes four aspects:

- Basic information: basic information about companies and corporate universities;
- Systematic: the solidity of the construction and operation of the corporate university system;
- Innovative: Corporate University's leading practice in cultivating talents and empowering enterprises;
- Ideological: Forward-looking thinking and unique propositions of corporate university empowerment.

Present key content that realistically demonstrates best practices, including evidence.

Follow the template prompts to fill in the text (Song type, No. 5 font), form, and picture.

There is no word count requirement for filling in the content, and the standard is to have a complete outline, clear practices, and credible results.

In addition to the selection of the best corporate university rankings, all the content collected by the survey will also be used as a reference for academic research on corporate universities, but it will never involve any commercial interests. Thank you for your cooperation!

1. Basic information

Basic information about the company			
Full name of the company			
Established		Enterprise nature	
Industry		Is it listed	
Revenue scale		Staff size	
Development Summary			

Corporate University Basic Information			
Full name of the corporate university			
Date of establishment		Institutional attributes	Profit Center/Cost Center
Number of full-time staff		Number of teachers	
Service object		Principal's name	
Contact		Contact details	
Development path			

2. Systematic (100 points)

1. About corporate university value proposition (10 points)	
Definition of mission and purpose	(According to the actual situation of the company)
functional positioning	(According to the actual situation of the company)
goal vision definition	(According to the actual situation of the company)
Team evaluation (dimension example)	● There is a clear definition of why it exists and it is accepted at the top ● Where it works is clearly defined and reflected ● A clear picture of what success looks like and a clear action design and validation
2. About the corporate university project system (15 points)	
Project Architecture	(A brief introduction to the basic logic and core features of learning project construction)
boutique project	(Outline introduction of benchmarking projects formed around core talents and key issues)
Group evaluation (dimension example)	Grasp the law of talent growth, environment, and state to build a project system in place Build a training program that connects growth channels, competencies, and performance promotion Have a talent training project that is followed by high-level attention and praised by students and continues to iterate
3. About the corporate university curriculum system (15 points)	
Curriculum Structure	(A brief introduction to the basic logic and core features of the subject setting)
Excellent course	(Outline introduction to quality course content around key competency development)

Group evaluation (dimension example)	Have a job competency learning map in place and implement competency certification Extract key position practical experience methods and convert them into the course content Rich online and offline learning courses attract employees to learn actively
4. About the corporate university faculty system (10 points)	
Faculty	(A brief introduction to the basic logic, quantity, and structure of the teaching staff)
Brand teachers	(Outline introduction of influential lecturers and their expertise from their own full-time and part-time faculty)
Team evaluation (dimension example)	Set up full-time and part-time teachers covering various business fields and talent echelons Established useful standards and procedures for teacher selection, development, and certification The company's senior management and high-end professional and technical personnel will take the initiative to teach Internal part-time teachers are attractive to students
5. About corporate university technology system (10 points)	
technical reference	(Outline introduction of new technologies and methods cited in the field of talent development and organizational development)
Technological innovation	(An overview of models, tools, and methodologies specific to the field of talent development and organizational development)
Team evaluation (dimension example)	Problem-oriented introduction of new technologies and effective transformation and application Developed its proprietary and practical models, and tools, and iterated continuously Proprietary methods, products, or projects are actively referenced by internal and external business units
6. About the corporate university support system (15 points)	
Information system	(A brief introduction to the characteristics of using information technology to support the efficient operation of corporate universities)
resource capacity	(A brief introduction to the integration and utilization of corporate university resources and the capabilities and characteristics of the operating team)
Team evaluation (dimension example)	Use information technology to build a learning platform covering diverse needs There are in place practices for integrating internal and external resources to promote high-quality development The professional team has the independent ability to research, consult, develop, and operation
7. About corporate university management system (10 points)	
Architecture settings	(Outline introduction of the organizational structure of the corporate university and the characteristics of its functions)
operating organization	(A brief introduction to the organization and main features of the business operation of the corporate university)

8. About the corporate-university collaboration system (15 points)	
vertical synergy	(Outline introduction of the mechanism and characteristics of collaboration in the field of personnel training and management)
Horizontal collaboration	(Overview of the synergy mechanism between corporate universities, business units, and functional departments)
Team evaluation (dimension example)	Have a working mechanism that drives management to effectively invest in talent development Establishing a Measuring Mechanism for Measuring the Value Creation Capability of Corporate university Have a mechanism to drive each function and unit to invest in talent training Effectively build a working mechanism that closely links talent training and talent management

3. Innovation and Thought (100 points)

1. About the empowerment strategy (20 points)	
innovative approach	(Summary of successful practices that enable company strategy formulation and execution)
forward-thinking	(Summary presents the idea and thinking that corporate university can better empower the company's strategy)
Team evaluation (example of dimensions)	Follow the theme of industrial development, company strategic planning, and action and transform it into a landing project Possess the ability to design and operate solutions that drive the company's strategy through the execution Plan research and provide high-quality reports or solutions in line with the company's high-level concerns Can be invited to participate in high-level seminars on company management topics
2. About enabling business (20 points)	
innovative approach	(Summary of successful practices that empower the company's business development)
forward-thinking	(Summary presents the idea and thinking that corporate university can better empower the company's business development)
Team evaluation (example of dimensions)	Improve employees' thinking about problem-solving from the perspective of customer value Improve staff skills in customer experience management and service Help business units identify and gain insight into market and customer changes Help business units innovate and iterate on businesses, products, and services
3. About enabling culture (20 points)	

innovative approach	(Outline of practices that empower company culture)
forward-thinking	(Summary presents the idea and thinking that corporate university can better empower company culture)
Team evaluation (example of dimensions)	Help employees understand the meaning of work from the company's mission Help employees inspire work commitment to company goals and vision Help the company gain employee recognition of the core values
4. About empowering organizations (20 points)	
innovative approach	(Outline presenting practices to improve the company's organizational capabilities)
forward-thinking	(Summary presents the assumptions and thinking that corporate university can better improve the company's organizational capabilities)
Team evaluation (example of dimensions)	Clearly define company organizational capabilities and enable diagnosis and training Drives the organization and scheduling of tasks throughout the company around customer needs Drive rapid response to market and customer changes throughout the company Help companies optimize organizational structure and resource allocation to improve overall performance Promote the company to optimize the management process to make the work more concise and efficient Help companies promote smooth information communication and cross-team collaboration Promote the company's reasonable authorization to make employees the protagonists in the work Help the company to establish an incentive and restraint mechanism with clear guidance and standards
5. About empowering teams (20 points)	
innovative approach	(Outline of practices for empowering team competence and vitality)
Forward-thinking	(Summary presents the idea and thinking that corporate university can better empower team ability and vitality)
Team evaluation (example of dimensions)	Formed effective programs and practices for nurturing and stimulating team vitality Help companies develop team leadership that balances performance and people orientation Helping companies develop employees who better meet customer needs than their competitors Help the company develop employees who can actively undertake and perform high-quality work

Appendix 7: Questionnaire for Corporate Universities Enabling Enterprises (2021)

Dear Participants:

Thank you very much for taking the time to fill out this questionnaire, which is a survey on corporate universities empowering business development. The internal empowerment platform of the enterprise serves the corporate strategy, and is or has been named as "XX University, College, School of Management, Business School, Research Institute, Training Institute, Empowerment Center, School", excluding traditional Enterprise training center/training department; and at least have the function of serving the talent training of the enterprise. The purpose of this survey is to verify the enabling mechanism of corporate universities and to provide a reference for management decision-making. The information obtained is for academic research purposes only, not for commercial purposes. We will keep your personal, team, and company information strictly confidential, and no personal information about you will be disclosed or appear in the research! Your valuable comments will be of great contribution and value to this research, thank you again for your help and support!

Completion Instructions:

1. This questionnaire consists of two parts, the first part is basic information, and the second part is answering questions.
2. The questionnaires are all multiple-choice and scaled questions. Please tick “√” or fill in the blanks before the options you think are most suitable.
3. Please fill in the questionnaire truthfully, we will keep your personal information and answers confidential.

To thank you for your support of this research project, the research participants who fill in the questionnaire carefully will receive a red envelope reward after submitting the questionnaire. In addition, if you would like to know more about the results of the study, or if you have any questions or suggestions about this study, please contact us at icur@sjtu-oec.com

Corporate University Empowerment Research Team

August 2021

The first part basic information

1. Your gender: ☐ Male ☐ Female

2. Your age group: ☐ Less than 20 years old ☐ 0~30 years old ☐ 30~40 years old ☐ 40~50 years old ☐ Over 50 years old
3. Your education level: ☐ high school and below ☐ College ☐ Undergraduate ☐ Master students ☐ PhD degree and above
4. Your working years in the current company: ☐ Less than 1 year ☐ 1-3 years ☐ 3-5 years ☐ 5-10 years ☐ 10 years or more
5. The type of enterprise you are in is: ☐ Central Enterprise ☐ Local state-owned enterprises ☐ Foreign-funded enterprises ☐ Private enterprise ☐ Other
6. The type of your department: ☐ Production department ☐ Technical department (R&D, testing.) ☐ Functional departments (finance, human resources, public relations, administration, legal affairs.) ☐ Marketing department (marketing, sales) ☐ Other
7. The size of your team: ☐ 5 people or less ☐ 5-10 people ☐ 10-20 people ☐ 20 people or more
8. Your identity in the team: ☐ leader ☐ Members
9. Does your company have a corporate university (college/research institute): ☐ Yes ☐ No
10. Do you think the corporate university has met your learning, training, and growth needs (1-10 is an increasing degree of satisfaction, and 10 is a full score): _ _ _ _

Part two answers questions

As a corporate employee, based on your observation and participation in the actual work and output of your company's corporate university (corporate college/research institute), do you agree with the view that "corporate university can play the following driving forces in the following aspects"? (1-5 are: "disagree", "somewhat disagree", "general", "agree", "agree")

1、 In my opinion, corporate universities, in terms of enabling strategies, can:

- ✧ A1 was invited to participate in high-level seminars on the topic of the company operation and management (scoring ____)
- ✧ A2 Plan research and provide high-quality reports or programs closely related to the company's high-level concerns (scoring ____)
- ✧ A3 can design and manage solutions that drive the company's strategy through execution (scoring ____)

- ✧ A4 Keeping close to the development of the industry, planning the action theme according to the company's strategy and converting it into a landing project (scoring ____)

2、In my opinion, corporate universities, in terms of enabling businesses, can:

- ✧ B1 Improve employees' thinking about problem-solving from the perspective of customer value (scoring ____)
- ✧ B2 Improve employees' skills in customer experience management and service (scoring ____)
- ✧ B3 Help business units identify and gain insight into market and customer changes (scoring ____)
- ✧ B4 Help business units innovate and iterate businesses, products, and services (scoring ____)

3、In my opinion, corporate universities can do the following in terms of empowering teams:

- ✧ C1 helps the company develop employees who can actively undertake high-quality work (scoring ____)
- ✧ C2 helps the company develop employees who better meet customer needs than its competitors (scoring ____)
- ✧ C3 Helps companies develop team leadership that balances performance and people orientation (scoring ____)
- ✧ C4 has formed effective programs and practices in nurturing and stimulating team vitality (scoring ____)

4、In my opinion, corporate universities can do, in terms of empowering organizations:

- ✧ D1 Clearly define the company's organizational capabilities and can diagnose and nurture (scoring ____)
- ✧ D2 helps companies promote smooth information communication and cross-team collaboration (scoring ____)
- ✧ D3 helps companies optimize organizational structure and resource allocation to improve overall performance (scoring ____)
- ✧ D4 Push the company to organize and arrange tasks around customer needs (scoring ____)

- ✧ D5 drives the company to respond quickly to market and customer changes (scoring ____)
- ✧ D6 Promote the company's reasonable authorization to make employees the protagonists in the work (scoring ____)
- ✧ D7 Promote the company to optimize the management process to make the work more concise and efficient (score ____)
- ✧ D8 helps the company to establish an incentive and restraint mechanism with clear guidance and standards (scoring ____)

5、 In my opinion, corporate universities can do the, in terms of enabling culture:

- ✧ E1 Help employees understand the meaning of work from the company's mission (scoring ____)
- ✧ E2 helps employees motivate work engagement from company goals and vision (scoring ____)
- ✧ E3 helps the company to gain employee recognition of core values (scoring ____)

In the questionnaire to be filled out in this research, it is specially marked: "All the content collected by this survey will be used for the selection of the best corporate university rankings, if you agree, the content will also be used for the University of Otago and in the academic project research of Shanghai Jiaotong University, the research project is related to the development of Chinese corporate universities. But it does not involve any commercial interests. If you finally decide to participate, we will be very grateful, and if you decide not to participate, we will never cause any adverse effect."